



one point   4 lines   12 neighbours   27 far points

# Forty Points

*One qutrit, the exceptional algebras, and a light cone*

An exact and honest account of the  $W(3,3)$  programme,  
written so that anyone can follow it

The  $W(3,3)$  Theory Programme

repository [wilcompute/W33-Theory](https://github.com/wilcompute/W33-Theory) · 26 September 2026

Every mathematical statement in this paper carries a status badge and, where it is a computation, the path of the committed script and certificate that proves it. Interpretations are labelled as interpretations. Things that were believed and later shown false are kept on the page, with the measurement that overturned them.

## Abstract

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**For everyone.** Take the smallest quantum system that has three settings instead of two — a *qutrit* — and let it act on itself. The ways it can be transformed, and the ways those transformations do or do not interfere with one another, form a finite geometric object with exactly forty points. This paper follows that single object as far as exact mathematics can take it. It turns out to contain a tiny version of space-time with a light cone; four interlocking “clocks”; the largest exceptional symmetry in mathematics,  $E_8$ ; the octonionic algebra of  $3 \times 3$  matrices that physicists have long suspected sits behind particle physics; and, inside that algebra, a genuine ten-dimensional Lorentz group whose spinors are the matter particles of one generation. We also say, just as clearly, what the object does *not* do: it does not by itself produce the masses, the forces, or the dynamics of our universe, and it provably cannot decide whether nature is left-handed or right-handed. Finally we settle, by an exhaustive exact computation, the most important open question about the string-theory models built on it: in their supersymmetric vacua the proton is not protected, because a field with the quantum numbers of a right-handed sneutrino is forced to condense — and that field carries exactly the charge that switches the forbidden couplings back on. We follow every escape route we could find to its end: other discrete symmetries, parities built from rotations of the hidden dimensions, and very heavy superpartners. Each fails — one only after we found that the selection rule it relied on is incomplete — and the last would need a top quark lighter than the one we measure.

**For specialists.** The symplectic generalized quadrangle  $W(3,3)$  is the projective commutation geometry of the two-sided Weyl action on one qutrit’s Hilbert–Schmidt space; its Bell line is the unitary-channel diagonal. Fixing that line, the 27 transverse Lagrangians form  $\text{Sym}_2(\mathbb{F}_3)$  with the finite Minkowski form  $t^2 - x^2 - y^2$ , whose null relation is context intersection; the four null directions are  $\mathbb{P}^1(\mathbb{F}_3)$ , the tetracode is their evaluation code, and the same four labels index the Hesse striations and an  $A_2^4 \subset E_8$ . On the committed executable  $E_8$  Chevalley basis, a Hesse tick line induces the contact grading  $1 + 56 + 134 + 56 + 1$ ; its 56 is exactly the Freudenthal triple system of the split Albert algebra whose norm is the repository’s signed 45-triad  $E_6$  cubic,  $Q = (\alpha\beta + T)^2 - 4\beta N(X) - 4\alpha N(Y) + 4T(X^\#, Y^\#)$ , and the 57-dimensional Heisenberg extension carries the quasiconformal cone  $N = Q/4 - \tau^2$  with the Günaydin–Koepsell–Nicolai inversion law. The Killing form obeys  $K(e_\alpha, e_{-\alpha}) = -60(-1)^{\text{ht}\alpha}$ ; the three tick lines give commuting  $E_{8(-24)}$  involutions with  $E_8 = 80 + 56 + 56 + 56$ , restricting to  $E_{7(-25)}$ , whose Hermitian Jordan triple yields a Euclidean Albert algebra with Peirce split  $1 + 16 + 10$ , a Lorentzian  $(1,9)$  Peirce slice,  $\text{Der} = F_{4(-52)}$ ,  $\text{str}_0 = E_{6(-26)}$ , and an  $\mathfrak{so}(1,9)$  acting on a Majorana–Weyl 16. The repository’s matter-parity charge is  $6L_c - 2$ , and matter parity is the  $2\pi$  rotation of  $\text{Spin}(9)$ . The committed split real form, by contrast, is the height twist  $\exp(i\pi\rho^\vee)$  and gives a  $(5,5)$  slice: signature is a real-form datum, invisible over  $\mathbb{F}_3$ . For the 215 heterotic  $\mathbb{Z}_6$  Standard Models of the  $W(3,3)$  twist we recover and freeze the exact spectra and prove, over the full rational  $B - L$  freedom and over all 1024  $\mathbb{Z}_2$  characters odd on quarks and leptons, that no FI-cancelling  $D$ -flat singlet vacuum preserves matter parity: the never-even singlets have  $B - L = \pm 1$  and every  $D$ -flat vacuum condenses one. The obstruction is one  $E_6$ -shaped  $U(1)$ ,  $4Q_\psi - \frac{4}{5}Q_\chi$ ; it extends to every element of the gauge torus times the space group (Smith normal forms), the forced field obeys  $Q(n) = -Q(u^c d^c d^c) = -Q(q l l^c) = -Q(l l e^c)$  and regenerates all three couplings at quartic order, and baryon triality is excluded by  $Q(u^c) + Q(e^c) = 2Q(q)$ . Plane-rotation

$R$ -symmetries appear to reopen parity in one model, but only through an  $R$ -rule that is incomplete on non-prime planes; with the rules derived from the worldsheet instantons (Bizet et al.) the  $\mathbb{Z}_6$ -II class closes (0/128), as do the  $\mathbb{Z}_2 \times \mathbb{Z}_6$  and  $\mathbb{Z}_3 \times \mathbb{Z}_6$  families of the same twist, and  $\mathbb{Z}_{12}$ -I once the  $E_6$  lattice fixes its admissible rule; in the prime orbifold  $\mathbb{Z}_3 \times \mathbb{Z}_3$ , where every rule is established, the parity survives and the vacua are supersymmetric — the superpotential is a  $\mathbb{Z}_3$ -graded function of one or two composites — but a hidden-representation index leaves six massless charge- $\frac{1}{3}$  states in every one, and breaking the hidden group to cure it leaves fractional charges protected by an unbroken discrete 3-group; and two-loop running of the Higgs quartic caps any superpartner scale that could tame the regenerated couplings unless  $m_t \leq 171$  GeV.

*The forty points are forty views of one thing. Looked at through the right lens, the same finite object is a quantum measurement geometry, a finite space-time, a clock, a lattice, an exceptional Lie algebra, and a Lorentzian light cone — and each identification in this paper is an exact computation, not an analogy.*

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## How to read this paper

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This paper is written for two readers at once. One has never studied algebra beyond school and wants to understand what has actually been found. The other wants every statement to be precise enough to check. We have tried not to cheat either of them. The mathematics is stated in full; next to it, in gold-edged boxes, the same idea is explained in ordinary words, and in teal-edged boxes we say what the idea means physically. Rose-edged boxes mark the edge of what is known.

### The five badges

Every claim of substance carries one of five labels. They are the most important typographical device in the paper, because the whole value of a speculative subject lies in being able to tell the proved from the hoped-for.

#### THEOREM

A mathematical theorem: proved by an argument in the text, or a standard result with a reference.

#### COMPUTATION

An exact finite computation, carried out by a committed script whose output certificate is named. “Exact” means integer or rational arithmetic, or an exhaustive enumeration — never a numerical approximation, unless we say so.

#### BRIDGE

A bridge: a mathematically true statement *plus* an interpretation of it (for example, calling a mathematical light cone a light cone of time). The mathematics is certain; the interpretation is a proposal.

#### OPEN

An open question, stated as sharply as we can.

#### REFUTED

Something that was once believed, written down, and then shown false. We keep these, with the reason.

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### A map of the journey

The paper follows one chain of exact identifications. Figure 1 is the whole argument in one picture; each box is a section.

### What you will need

Nothing beyond patience. We introduce every structure before we use it, and the glossary in Appendix B collects the vocabulary. Readers who want only the story can read the plain-words boxes and the key-idea panels and skip the displayed mathematics; nothing essential will be lost. Readers who want only the mathematics can do the reverse.



**Figure 1.** The chain of exact identifications followed in this paper. Shaded boxes are the three places where the story changes scale: from a qubit-sized object to a finite geometry, from the geometry to  $E_8$ , and from  $E_8$  to a Lorentzian algebra.

## Where the material comes from

Everything here is drawn from the public repository `wilcompute/W33-Theory`, a record of several thousand numbered “passes” by two independent research tracks, and from its three long manuscripts — the  $W(3,3)$  paper, the photonic Holonet paper and the Holonet machine blueprint — some of whose pages are adapted here. Passes 10949 and 10950, which supply Sections 6–8, were carried out for this synthesis. A reader who wants to check a statement can find its script and certificate in the ledger of Appendix A.

## 1 Prologue: a shape made of measurements

### 1.1 Why look at something finite?

Physics describes the world with continuous quantities: positions, times, field strengths, each a real number that can be divided forever. Yet the deepest facts we know about nature are *discrete*. An electron’s spin, measured along any axis, gives one of exactly two answers. Quarks come in exactly three colours. There are exactly three generations of matter. The symmetry groups that organise the forces —  $SU(3)$ ,  $SU(2)$ ,  $U(1)$  — are continuous, but their representations, which decide what particles can exist, are labelled by whole numbers.

This programme starts from the discrete end. It asks: is there a single finite object, small enough to write down completely, whose structure already contains the patterns physics keeps finding? The candidate studied here is a configuration of forty points called  $W(3,3)$ . It is not new — geometers have known it for a century — but it has an unusual habit: every time it is examined through a different lens, it turns out to be a well-known structure from a different part of mathematics or physics. This paper is an account of those lenses, and of exactly how far each one reaches.

**In plain words**

A useful picture:  $W(3,3)$  is like a crystal. A crystal is finite in its unit cell, but the unit cell determines the whole crystal's behaviour — how it breaks, how light passes through it, what sounds it makes. The hope of this programme is that  $W(3,3)$  is a unit cell for something larger. The honest answer, as you will see, is that it is a remarkable unit cell for the *symmetries* of physics, and a much weaker one for its *dynamics*.

**1.2 Bits, trits and qutrits**

Every computer you have used stores information as bits — switches that are either off or on. A *trit* is a three-way switch: 0, 1, or 2. On its own that is not interesting; two bits can imitate a trit. A *qutrit* is different. It is a physical system with three distinguishable states, obeying quantum mechanics, which means it can be in a blend of all three settings at once, with relative phases. In symbols, a qutrit state is a unit vector

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle + a_2|2\rangle \in \mathbb{C}^3,$$

and the phases of the complex numbers  $a_j$  matter.

Two operations generate everything one can do to the “shape” of a qutrit. The *shift*  $X$  moves each setting to the next, and the *clock*  $Z$  stamps each setting with a phase:

$$X|j\rangle = |j+1 \bmod 3\rangle, \quad Z|j\rangle = \omega^j|j\rangle, \quad \omega = e^{2\pi i/3}.$$

They do not commute:  $ZX = \omega XZ$ . That single twist — a phase of one third of a turn picked up when the order of two operations is swapped — is the seed of everything in this paper.

**What it means physically**

$X$  and  $Z$  are the finite versions of the two most important operations in quantum mechanics: translation in position and translation in momentum. The relation  $ZX = \omega XZ$  is the three-state version of Heisenberg's  $[x, p] = i\hbar$ . Physicists call the group generated by  $X, Z$  and the phases the *Heisenberg–Weyl group*; its elements are the qutrit *Pauli operators*. They label the possible measurements one can make on a qutrit that are “as simple as possible” — the ones every other measurement is built from.

**1.3 Arithmetic with three numbers**

Because  $X^3 = Z^3 = 1$ , exponents only matter modulo 3. We therefore work in the *field with three elements*,  $\mathbb{F}_3 = \{0, 1, 2\}$ , where addition and multiplication are done modulo 3 — clock arithmetic on a three-hour clock. It is a genuine number system: every nonzero element has an inverse ( $1 \cdot 1 = 1$ ,  $2 \cdot 2 = 4 = 1$ ). In particular  $2 = -1$  and  $\frac{1}{2} = 2$ , facts we will use repeatedly.

A qutrit Pauli operator is named by two numbers,  $X^a Z^b$  with  $(a, b) \in \mathbb{F}_3^2$ . Two qutrits need four:  $(a_1, b_1, a_2, b_2) \in \mathbb{F}_3^4$ , a space with  $3^4 = 81$  points.

## 1.4 The forty in one breath

Here is the whole construction, which Sections 2–3 unpack slowly. Label the 81 two-qutrit Pauli operators by vectors  $x \in \mathbb{F}_3^4$ . Remove the identity, leaving 80. An operator and its inverse describe the same measurement (one is the other run backwards), and  $x$  and  $-x = 2x$  label that pair, so the distinct measurements number

$$\frac{3^4 - 1}{3 - 1} = \frac{80}{2} = 40.$$

Two of these forty measurements can be performed together exactly when their operators commute. Recording which pairs commute gives the forty-point geometry  $W(3, 3)$ .

*The forty points of  $W(3, 3)$  are the forty independent directions along which a pair of qutrits — or, as Section 3 shows, one qutrit acting on itself — can be measured. The lines of the geometry are the complete sets of measurements that can be made at the same time.*

## 1.5 What this paper claims, and what it does not

A “theory of everything” should, at a minimum, reproduce the particles and forces of the Standard Model, the geometry of space-time and gravity, and the numbers — masses, couplings, mixing angles — that we measure. It is important to say at the start that nothing in this paper achieves that. What it does achieve is a tight web of *exact* connections between a single finite object and the structures that any such theory seems to need:

- the qutrit Heisenberg group and its Clifford symmetries (Sections 2–3);
- a finite space-time with a light cone, finite clocks, and a rigorous distinction between reversible time and an arrow of time (Section 4);
- the exceptional Lie algebra  $E_8$  and its subalgebras  $E_7$  and  $E_6$  (Section 5);
- the Freudenthal triple system behind four-dimensional black-hole charges, and a 57-dimensional light cone on which  $E_8$  acts by conformal-like maps (Section 6);
- the octonionic Albert algebra, the Lorentz algebra  $\mathfrak{so}(1, 9)$  and its chiral spinor, which carries exactly the quantum numbers of one generation of matter (Sections 7–8).

Section 9 is devoted to the physics that has been attempted on top of this — string compactifications that realise the Standard Model on the same structure — including the negative results, and Section 10 to a quantum computer architecture read off the same geometry. The Epilogue lists what remains open.

### Where this stops

The finite geometry supplies *symmetry and structure*. It does not, by itself, supply a Hamiltonian, an energy scale, a value of Newton’s constant, or a reason for any particular mass. Where this paper uses words like “time”, “light cone” or “matter”, it is naming a mathematical structure that behaves like the physical one; whether nature uses it is a separate, empirical question.

## 2 The object: the symplectic quadrangle $W(3,3)$

### 2.1 Definition

Let  $V = \mathbb{F}_3^4$  with coordinates  $x = (x_1, x_2, x_3, x_4)$  and the *symplectic form*

$$\langle x, y \rangle = x_1y_3 + x_2y_4 - x_3y_1 - x_4y_2 \pmod{3}.$$

It is bilinear, alternating ( $\langle x, x \rangle = 0$ ) and nondegenerate (only  $x = 0$  pairs to zero with everything).

**Definition 2.1.** The *points* of  $W(3,3)$  are the one-dimensional subspaces of  $V$  (40 of them). Two distinct points  $\langle x \rangle, \langle y \rangle$  are *collinear* if  $\langle x, y \rangle = 0$ . The *lines* of  $W(3,3)$  are the two-dimensional subspaces  $L$  on which the form vanishes identically (*totally isotropic* or *Lagrangian planes*); each contains  $(9 - 1)/2 = 4$  points.

#### Proposition 2.1

#### THEOREM

$W(3,3)$  has 40 points and 40 lines. Every line has 4 points and every point lies on 4 lines. For a point  $p$  and a line  $L$  not containing it, exactly one point of  $L$  is collinear with  $p$ .

*Proof.* Points:  $(3^4 - 1)/(3 - 1) = 40$ . For a point  $p = \langle x \rangle$ , the set  $x^\perp = \{y : \langle x, y \rangle = 0\}$  is a 3-dimensional subspace containing  $x$ ; the lines through  $p$  are the 2-dimensional subspaces of  $x^\perp$  that contain  $x$ , and these correspond to the points of  $x^\perp / \langle x \rangle \cong \mathbb{F}_3^2$ , of which there are  $(9 - 1)/2 = 4$ . Double counting point–line incidences,  $40 \cdot 4 = (\# \text{lines}) \cdot 4$ , so there are 40 lines. Finally, if  $p \notin L$  then  $x^\perp \cap L$  is a subspace of  $L$  of dimension 1 (it cannot be all of  $L$ , since then  $\langle x \rangle + L$  would be a totally isotropic 3-space, impossible in a nondegenerate 4-space), so exactly one point of  $L$  is collinear with  $p$ .  $\square$

The last property is the defining axiom of a *generalized quadrangle*: there are no triangles made of three lines, but from any point you can always reach any line in one step, in exactly one way. With four points per line and four lines per point,  $W(3,3)$  is the generalized quadrangle of order  $(3,3)$ , usually written  $\text{GQ}(3,3)$  [41].

### 2.2 The collinearity graph and its spectrum

Join two points by an edge when they are collinear. Each point has  $4 \times 3 = 12$  neighbours, so the graph has  $40 \cdot 12/2 = 240$  edges. Two collinear points share exactly 2 further neighbours (the other two points of their line), and two non-collinear points share exactly 4 (the points of  $x^\perp \cap y^\perp$ , a nondegenerate plane). A graph in which these three numbers are constant is called *strongly regular*; this one is  $\text{SRG}(40, 12, 2, 4)$  [7].

#### Proposition 2.2

#### THEOREM

The adjacency matrix  $A$  of  $W(3,3)$  has eigenvalues 12 (once), 2 (24 times) and  $-4$  (15 times).



**Figure 2.** Left: the view of  $W(3,3)$  from one point. Right: the quadrangle axiom.

*Proof.* For  $\text{SRG}(v, k, \lambda, \mu)$  one has  $A^2 = kI + \lambda A + \mu(J - I - A)$ , where  $J$  is the all-ones matrix. On vectors orthogonal to the all-ones vector,  $J$  acts as zero, so every other eigenvalue  $\theta$  satisfies  $\theta^2 = (\lambda - \mu)\theta + (k - \mu)$ , i.e.  $\theta^2 + 2\theta - 8 = 0$ , giving  $\theta \in \{2, -4\}$ . Multiplicities  $f, g$  satisfy  $f + g = 39$  and  $\text{tr } A = 12 + 2f - 4g = 0$ , so  $f = 24, g = 15$ .  $\square$

Since  $|-4| < 2\sqrt{11} \approx 6.63$ , the graph is *Ramanujan*: it is as good an expander as a 12-regular graph can be. This is why the machine of Section 10 can route between any two of its forty nodes in two hops, and why the graph's minimum bisection is large (100 of the 240 edges, a computed value).

#### In plain words

“Strongly regular” means that the geometry looks exactly the same from every point, and from every pair of points. There is no privileged measurement. The three eigenvalues are the geometry’s “vibration frequencies”; that there are only three is the algebraic signature of this perfect evenness.

### 2.3 What the points and lines are physically

Let  $W(x) = X^{x_1}Z^{x_3} \otimes X^{x_2}Z^{x_4}$  be the two-qutrit Pauli operator with label  $x \in \mathbb{F}_3^4$ . A direct calculation from  $ZX = \omega XZ$  gives

$$W(x)W(y) = \omega^{\langle x, y \rangle} W(y)W(x). \quad (1)$$

So the symplectic form is not an arbitrary choice: it *is* the commutation phase of two-qutrit measurements.

#### What it means physically: contexts

- A **point** is a measurement direction: the cyclic group generated by one Pauli operator, which is the same whether it is run forwards or backwards.
- Two points are **collinear** when their operators commute and can be measured at the same time without disturbing each other.



**Figure 3.** A portrait of  $W(3,3)$ , computed from the symplectic form and drawn from the point of view of one point  $p$  (centre). Its 12 neighbours lie on four lines (four colours, three points each); the 27 far points sit on the outer ring, ordered by which neighbour they see on each of the four lines — by the quadrangle axiom every far point sees exactly one. The edge counts  $12 + 12 + 108 + 108 = 240$  are forced by  $\text{SRG}(40, 12, 2, 4)$ .

- A **line** is a maximal set of mutually compatible measurements: an abelian group of 9 Pauli operators whose common eigenvectors form a basis of the two-qutrit space  $\mathbb{C}^9$ . Physicists call this a *stabilizer group*, and each of its 9 joint eigenstates a *stabilizer state*. There are 40 lines and so  $40 \times 9 = 360$  stabilizer states — the standard count for two qutrits.
- The **quadrangle axiom** says: given any measurement and any complete measurement context not containing it, exactly one member of the context is compatible with it.

The quantum operations that map Pauli operators to Pauli operators (up to phase) form the two-qutrit *Clifford group* [17]. Modulo the Paulis themselves and phases, it acts on the labels  $x$  by exactly the linear maps preserving  $\langle, \rangle$ : the symplectic group  $\text{Sp}(4, 3)$ . So the symmetries of the geometry are, literally, the “cheap” quantum gates.

## 2.4 Two groups of order 51,840, which are not the same group

The symmetry groups that appear around  $W(3, 3)$  are easy to confuse, and confusing them has cost this programme real work (Section 5.4). Adapting the blueprint’s account:

$$\text{PSp}(4, 3) \cong U_4(2) \text{ (order 25 920)}, \quad \text{Sp}(4, 3) \cong 2.U_4(2), \quad \text{PGSp}(4, 3) \cong U_4(2):2 \cong W(E_6).$$

The last two both have order 51 840 but are *not isomorphic*:  $\text{Sp}(4, 3)$  has a centre of order two (the map  $x \mapsto -x$ ), while  $\text{PGSp}(4, 3)$  — the full automorphism group of  $W(3, 3)$  — has trivial centre and is the Weyl group of the exceptional root system  $E_6$ . **THEOREM**

### In plain words

The two groups have the same size but different “shapes”. One of them contains a special element that commutes with everything — a global sign flip. Whether that sign is visible or invisible turns out to decide whether the structure can tell left from right. This is why the same forty points can *host* a handedness without being able to *choose* one (Section 9.5).

## 2.5 The dual picture and a firewall

Swap the roles of points and lines: the 40 lines of  $W(3, 3)$ , with two lines “adjacent” when they meet, form the dual quadrangle, which is the parabolic quadric  $Q(4, 3)$ . For odd  $q$  the quadrangle  $W(3, q)$  is *not* isomorphic to its dual, although both have collinearity graph parameters  $\text{SRG}(40, 12, 2, 4)$ . This distinction matters several times below: the finite space-time of Section 4 lives on the *line* side, while the qutrit operators live on the *point* side. Repeated coincidences of counts between the two sides (for instance 27 vertices, 108 edges, 36 triangles) do not make them the same object — the repository calls this the *point/line firewall*, and it has caught several tempting mistakes. **COMPUTATION** (`w33_20260924_history_pointline_cospectral_firewall.py`)

## 2.6 Why three?

A sceptical reader should ask whether “three” was chosen because it works. The repository records five equations in which  $q$  appears as an unknown integer and each has exactly one positive solution,  $q = 3$ . Two can be checked by hand:

- The number of edges of the collinearity graph of  $W(3, q)$  is  $q(q+1)^2(q^2+1)/2$ ; the number of elements of  $\mathbb{F}_{q^5}$  outside  $\mathbb{F}_q$  is  $q^5 - q$ . Setting them equal gives  $2(q-1) = q+1$ , so  $q = 3$ : both equal 240.
- The number of points not collinear with a given point is  $q^3$ ; it equals 27, the dimension of the fundamental representation of  $E_6$ , only for  $q = 3$ .

### Where this stops

These are coincidences of integers, not derivations. They do not prove that three is *necessary* for anything physical. They do show that the programme’s most unusual choice — three-level systems rather than two-level ones — is the one value at which several independent structures meet. **BRIDGE**

## 3 One qutrit, acting on itself

Section 2 built  $W(3, 3)$  from *two* qutrits. This section shows that one is enough — provided we look not at the qutrit’s states but at the things that can be done to it. That shift of viewpoint is the precise content of the phrase the Holonet papers use, *self-entanglement*, and it is worth getting exactly right, because the phrase can also be read in ways that are not true.

### 3.1 Operators are states

An operator on one qutrit is a  $3 \times 3$  matrix: nine complex numbers. So the space of operators  $\text{End}(\mathbb{C}^3)$  is itself a nine-dimensional complex vector space, and with the Hilbert–Schmidt inner product  $\langle A, B \rangle = \text{tr}(A^\dagger B)$  it is a Hilbert space. Writing  $|A\rangle\rangle$  for a matrix regarded as a vector,

$$\text{End}(\mathbb{C}^3) \cong \mathbb{C}^3 \otimes \mathbb{C}^{3*} \cong \mathbb{C}^9, \quad |i\rangle\langle j| \longleftrightarrow |i\rangle \otimes |j\rangle.$$

Under this identification the identity matrix becomes the maximally entangled “Bell” state of two qutrits,

$$|\Omega\rangle = \frac{1}{\sqrt{3}}|I\rangle\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |11\rangle + |22\rangle).$$

This is the Choi–Jamiołkowski correspondence, a standard tool of quantum information: every operation on a system is encoded by a state of two copies of it.

### In plain words

Imagine describing a machine not by its current setting but by its complete instruction table: “if the input is  $j$ , produce  $i$  with amplitude  $A_{ij}$ ”. That table has nine entries. The nine entries behave exactly like the state of two three-way systems — an “input” and an “output”. Nothing mystical is happening; it is the same information written in a

different shape. The “two qutrits” of Section 2 can be read as the before-and-after of a single qutrit.

### 3.2 The two-sided action and its geometry

Let  $P_u = X^{u_1} Z^{u_2}$  for  $u \in \mathbb{F}_3^2$ . A qutrit operator  $A$  can be multiplied by Pauli operators from the left and from the right:

$$\Phi_{u,v}(A) = P_u A P_v^\dagger, \quad (u, v) \in \mathbb{F}_3^2 \oplus \mathbb{F}_3^2.$$

#### Theorem 3.1

THEOREM

COMPUTATION

The 81 maps  $\Phi_{u,v}$  form an orthogonal basis of  $\text{End}(\text{End}(\mathbb{C}^3)) \cong M_9(\mathbb{C})$ ; the 80 non-identity maps span  $\mathfrak{sl}_9$ . They commute up to the phase

$$\Phi_{u,v} \Phi_{u',v'} = \omega^{[u,u'] - [v,v']} \Phi_{u',v'} \Phi_{u,v}, \quad [u, u'] = u_1 u'_2 - u_2 u'_1.$$

The form  $[u, u'] - [v, v']$  is symplectic on  $\mathbb{F}_3^4$ , and the resulting commutation geometry on the 40 projective classes is  $W(3,3)$ , with collinearity graph  $\text{SRG}(40, 12, 2, 4)$  and spectrum  $12^1 2^{24} (-4)^{15}$ . `w33_20260924_temporal_su9_e8_compiler.py`

*Proof of the phase.*  $\Phi_{u,v} \Phi_{u',v'}(A) = P_u P_{u'} A P_{v'}^\dagger P_v^\dagger$  and  $\Phi_{u',v'} \Phi_{u,v}(A) = P_{u'} P_u A P_v^\dagger P_{v'}^\dagger$ . Using  $P_u P_{u'} = \omega^{[u,u']} P_{u'} P_u$  and taking adjoints of  $P_v P_{v'} = \omega^{[v,v']} P_{v'} P_v$  gives the ratio  $\omega^{[u,u'] - [v,v']}$ .  $\square$

The minus sign is the fingerprint of right multiplication: acting from the right reverses the order of products, so the right-hand copy of the qutrit carries the *opposite* algebra. The two “qutrits” are not two particles, and not even two subsystems in the usual sense: they are the algebra of left actions and its commutant, the algebra of right actions, on one qutrit’s operator space.

### 3.3 The Bell line is the set of physical channels

Among all  $\Phi_{u,v}$ , which ones are genuine quantum operations? A linear map on operators is physically realisable as a quantum channel only if it is *completely positive*, which for  $\Phi_{u,v}$  means its Choi matrix  $|P_u\rangle\rangle\langle\langle P_v|$  is positive semidefinite. A rank-one matrix  $|a\rangle\langle b|$  is positive only when  $b$  is a positive multiple of  $a$ ; so up to phase,  $\Phi_{u,v}$  is completely positive exactly when  $u = v$ . The diagonal  $\Delta = \{(u, u)\}$  is totally isotropic ( $[u, u'] - [u, u'] = 0$ ) and its four projective points form one line of  $W(3,3)$ .

#### Proposition 3.2

THEOREM

COMPUTATION

The unitary channels  $A \mapsto P_u A P_u^\dagger$  form a distinguished line of  $W(3,3)$ , the *Bell line*; each fixes  $|\Omega\rangle$ . The left–right exchange  $(u, v) \mapsto (v, u)$  reverses the symplectic form; its eight fixed rays split into two Lagrangian lines, and the  $+$  line is exactly the unitary-

channel diagonal. `w33_20260924_liouville_swap_cp_outer_bridge.py`

#### What it means physically: a built-in time reversal

Exchanging left and right multiplication is the operator version of running a process backwards:  $A \mapsto P_u A P_v^\dagger$  becomes  $A \mapsto P_v A P_u^\dagger$ . Because it reverses the sign of the commutation form, it is an *anti*-symmetry of  $W(3,3)$  — the same kind of map as complex conjugation, which is how quantum mechanics implements time reversal. The repository shows that ordinary matrix transposition of the Pauli algebra induces exactly this class of outer symmetry, the one earlier work had identified abstractly (BT172). **COMPUTATION** (`w33_20260924_intrinsic_transpose_outer.py`, up to a labelling gauge.)

### 3.4 Three different things called “self-entanglement”

The phrase can mean three distinct things, and the Holonet programme has, at times, slid between them. It is worth separating them once and for all.

| Notion           | What is entangled                                                                                                | Status here                               |
|------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Intraparticle    | Two independent degrees of freedom of one photon (for example its arrival-time bin and its frequency bin).       | Standard and experimentally demonstrated. |
| Operator (Choi)  | The input and output legs of one operation, as in §3.1.                                                          | The exact algebraic home of $W(3,3)$ .    |
| Temporal process | Quantum correlations between interventions at different times, described by a multi-time <i>process tensor</i> . | Needs its own witness (§3.5).             |

#### A correction the papers needed

An earlier draft of the photonic paper described three optical time bins as containing “a past qutrit and a future qutrit”. Three orthogonal bins span only  $\mathbb{C}^3$ , whereas two qutrits need  $\mathbb{C}^9$ ; the two-bin subsets it named overlap and cannot be independent tensor factors. The architecture now uses two genuinely independent qutrit degrees of freedom of one photon — time bin  $\otimes$  frequency bin — and reads “past” and “future” as the input and output legs of a process, not as simultaneously existing copies. **REFUTED** (the three-bin reading) **THEOREM** (the corrected encoding).

### 3.5 Memory across a causal break

What does it take for a system to carry quantum information from one time to a later one? The repository’s cleanest test is a three-step process with a qutrit memory  $M$ : swap the system  $S$  into  $M$ ; discard  $S$  and replace it with anything at all (a *causal break*); swap back. If the original state reappears, the process had memory. But a classical memory that merely

records “0, 1 or 2” would pass that test too. The decisive witness entangles  $S$  with a reference  $R$  first.

### Theorem 3.3

COMPUTATION

With  $S$  initially maximally entangled with a reference qutrit  $R$ :

- a coherent memory returns the Bell state exactly (negativity 1, Bell fidelity 1);
- a classical label memory returns  $\frac{1}{3} \sum_j |jj\rangle\langle jj|$  (negativity 0, Bell fidelity  $\frac{1}{3}$ ), although it passes every basis-state retrieval test;
- a depolarizing memory  $\rho \mapsto (1-p)\rho + pI/3$  returns an isotropic state with Bell fidelity  $1 - \frac{8p}{9}$  and negativity  $\max(0, \frac{3-4p}{3})$ ; quantum memory survives exactly while  $p < \frac{3}{4}$ .

w33\_20260924\_temporal\_quantum\_memory\_witness.json

#### What it means physically

The threshold  $p = \frac{3}{4}$  is the standard point at which a qutrit depolarizing channel becomes entanglement-breaking: beyond it, the link between the earlier and the later system can still carry classical information but no longer quantum information. Retrieval after a break proves that the past is *remembered*; preserved entanglement with a reference proves that it is remembered *quantumly*. Neither involves any influence travelling backwards in time.

### 3.6 A state picks a clock

A faithful qutrit state  $\rho = \text{diag}(p_0, p_1, p_2)$  defines a natural flow on operators, the *modular flow* of Tomita–Takesaki theory, with generator  $K = -\log \rho \otimes 1 + 1 \otimes \log \rho$  acting on  $\text{End}(\mathbb{C}^3)$ . The matrix units are its eigenvectors:

$$K E_{ij} = \log \frac{p_j}{p_i} E_{ij}.$$

The three diagonal units have frequency zero; the six off-diagonal units  $E_{ij}$  ( $i \neq j$ ) — the six directed transitions between the three levels — come in three opposite pairs. Their pattern of differences is exactly the root system  $A_2$ . **COMPUTATION** (w33\_20260924\_modular\_a2\_event\_clock.py)

#### In plain words

A perfectly balanced qutrit ( $p_0 = p_1 = p_2$ ) has no preferred direction: every transition has zero frequency, and nothing distinguishes “forward” from “backward”. Any imbalance gives each of the six transitions a frequency, and pairs each with its reverse at the opposite frequency. This is the finite form of an old idea of Connes and Rovelli [8]: that a state of a system can itself define a flow of time. It orients change; it does not by itself make change irreversible. That distinction — between an orientation of time and an arrow of time — returns in Section 4.

### 3.7 Space and time on the same forty points

Section 2 built  $W(3,3)$  from two qutrits, with commutation form  $[a, a'] + [b, b']$ ; Theorem 3.1 built it from one qutrit acting on itself, with form  $[u, u'] - [v, v']$ . The two constructions live on the same forty projective points of  $\mathbb{F}_3^4$ , and they are not the same structure.

#### Theorem 3.4 (two quadrangles)

THEOREM

COMPUTATION

Both forms make the 40 points a generalized quadrangle  $GQ(3,3)$  with collinearity  $SRG(40,12,2,4)$ . The two quadrangles share exactly 16 lines — the product planes  $\ell_1 \oplus \ell_2$  — and each has 24 lines of its own: the spatial ones are the graphs of the 24 maps with determinant  $-1$ , carrying the 216 maximally entangled two-qutrit stabilizer states; the temporal ones are the graphs of  $SL(2,3)$ , the qutrit Clifford group modulo Paulis. Partial transposition exchanges the two sets of own lines, and on states it is a bijection: the partial transpose of each of the 216 maximally entangled stabilizer states is the two-time pseudo-density operator of one qutrit passing through one of the 216 Clifford gates (spectrum  $(+\frac{1}{3})^6(-\frac{1}{3})^3$ , temporal negativity 1). Exactly 2304 of the 103680 similitudes of the spatial form also preserve the temporal one: the local operations and the exchange of the parties. `w33_pass11143_space_time_two_quadrangles.py`

#### What it means physically: one qutrit across time is a Bell pair across space

Space and time agree on everything classical — the product lines — and disagree on exactly the entangled lines. A maximally entangled pair of qutrits and a single qutrit persisting through a gate are the same line of points, read with the two forms. This is the  $W(3,3)$  form of a general fact (“partial transpose as a space–time swap”, Fullwood and Li [45]; Marcovitch and Reznik [46]); the two-quadrangle structure, the 16/24 split and the state–gate bijection are specific to  $W(3,3)$ .

#### Proposition 3.5 (the spookiness budget)

COMPUTATION

For the qutrit Bell inequality of Collins, Gisin, Linden, Massar and Popescu [47] (local bound 2): using only  $W(3,3)$ ’s own resources — stabilizer states, Pauli measurements, Clifford gates — the maximum is exactly 2, both for two qutrits in space and for one qutrit measured at two times. Along the path from the Pauli bases to the optimal bases the value is  $2 + \frac{\pi^2}{6}t^2 + O(t^3)$ :  $W(3,3)$  sits exactly on the edge of the local polytope. One qutrit starting maximally mixed and measured twice gives exactly the Bell-pair value 2.8729; optimised over starts and bases it reaches 3.1628, above the best spatial value 2.9149 [48] — for qubits the two coincide [49]. A classical system with any memory reaches the algebraic 4 in time. `w33_pass11144_spookiness_budget.py`

#### Spooky action at a distance, in these terms

The correlation geometry of a Bell pair is exactly that of one qutrit correlated with itself across time. The temporal version needs only persistence; its larger values measure how much a measurement disturbs, not nonlocality. The spatial version is “spooky” precisely because it cannot be used to signal, and it needs a resource outside  $W(3,3)$  — magic — to exceed the local bound at all (the odd-dimensional discrete Wigner

function of stabilizer states is non-negative, Gross [50], and its phase space is the point space of  $W(3,3)$ . Nothing here travels backwards in time: what the two readings share is the geometry of the correlations, not a mechanism.

The same swap works in imaginary time. For a Gibbs state  $\rho = e^{-\beta K}/Z$  of one qutrit, the partial transpose of the thermofield double  $\sum_i \sqrt{p_i} |ii\rangle$  equals the Jamiołkowski matrix of the Euclidean half-evolution  $A \mapsto \rho^{1/2} A \rho^{1/2}$ , with negativity  $((\sum_i \sqrt{p_i})^2 - 1)/2$ : the temporal Bell line at  $\beta = 0$ , a product line as  $\beta \rightarrow \infty$ . The modular clock above is the flow whose Euclidean continuation prepares this entanglement. **COMPUTATION** (w33\_pass11146\_thermal\_reading.py)

### Proposition 3.6 (polygamy in time, and where the magic sits)

**COMPUTATION**

A qutrit evolving by the identity is maximally entangled with its past and with its future at once: every pair of times carries the temporal Bell line, so the pairwise temporal negativities of three times sum to 3. Three qutrits in space reach at most 1.1394 for the same sum, and  $N_{AB} + N_{AC} \leq 1.0328 = 4/\sqrt{15}$  (numerical optimum, identified to  $10^{-15}$ ). The temporal advantage of Proposition 3.5 is not special to qutrits: one qudit measured twice beats two qudits for every  $d \geq 3$  ( $d = 3, 4, 5$ : 3.163, 3.283, 3.377 against  $2.915 = 1 + \sqrt{11/3}$ , 2.973, 3.016), and only qubits coincide. Magic can sit in either place: with Pauli measurements only, the best state reaches  $2\sqrt{2} \cos(\frac{1}{3} \arccos \frac{1}{2\sqrt{2}}) = 2.6017$ , the largest root of  $x^3 - 6x - 2$  (mana 0.39), while the stabilizer state  $|\Omega\rangle$  with magic measurements reaches 2.8729. There is no threshold: the least mana that reaches  $\square$  with Pauli measurements grows linearly from zero, about  $\text{usr/bin/bash.65}$  per unit of violation. **COMPUTATION** (w33\_pass11147\_temporal\_cglmp\_by\_dimension.py, w33\_pass11148\_polygamy\_in\_time.py, w33\_pass11149\_magic\_for\_spookiness.py)

### In plain words

In space, a system can give its full entanglement to only one partner. Across time it gives it to its whole history at once — because it is not sharing anything, it is simply persisting. That is why the spatial kind is scarce and “spooky”, and the temporal kind is free.

### Theorem 3.7 (perfect space–time gates)

**THEOREM**
**COMPUTATION**

Write a two-qutrit Clifford gate as  $S = \begin{pmatrix} S_{AA} & S_{AB} \\ S_{BA} & S_{BB} \end{pmatrix} \in \text{Sp}(4,3)$ . Its Choi state on  $A_{\text{in}}, B_{\text{in}}, A_{\text{out}}, B_{\text{out}}$  is maximally entangled across time (always), carries entanglement rank  $S_{BA}$  across space and rank  $S_{AA}$  across the space–time diagonal  $\{A_{\text{in}} B_{\text{out}}\}$ , and

$$\det S_{AA} + \det S_{BA} = 1 \pmod{3},$$

so a gate can never be weak across both space and the diagonal. The gates that are maximal across all three cuts — whose Choi state is the absolutely maximally entangled state of four qutrits — are exactly those with  $\det S_{AA} = \det S_{BA} = -1$ : there are  $13824 = 24^3$  of them ( $4/15$  of  $\text{Sp}(4,3)$ ), forming a single orbit under local Clifford operations. For time-reversing Clifford operations the law reads  $\det S_{AA} + \det S_{BA} = -1$ ,

and the tetracode gate  $K \otimes I$ ,  $K = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ , is perfect; composed with a local time reversal it is the perfect unitary gate. `w33_pass11156_perfect_spacetime_gates.py`

### Entanglement that is spatial and temporal at once

A perfect gate hides everything in pairs: every two-qutrit marginal of its Choi state is maximally mixed, so  $A$ 's past is uncorrelated with  $A$ 's future and with  $B$ 's future separately, and entirely correlated with them jointly — maximal scrambling. No such gate exists for qubits; for qutrits it exists, and in  $W(3,3)$  it is the tetracode that sits at every point. That the four-qutrit absolutely maximally entangled state comes from the tetracode and is unique up to local unitaries is known [52]; the complementarity law and the space/time/diagonal reading are specific to  $W(3,3)$ .

Four refinements of Proposition 3.6. The Pauli-only Bell operator has characteristic polynomial  $(x^3 - 6x - 2)(x^3 - 3x + 1)^2$  exactly, its remaining eigenvalues being  $2 \cos(2\pi k/9)$ . In time the magic cannot sit in the start: with Pauli measurements a single qutrit gives exactly 2 whatever its initial state, whereas in space a magic state already violates. The temporal optimum keeps climbing with dimension (3.453, 3.517 at  $d = 6, 7$ , deficit from 4 falling like  $d^{-0.65}$ ) while the spatial one barely moves (3.050, 3.078). And the  $4/\sqrt{15}$  optimum is carried by a state whose invariants are all fifteenths: spectra  $(7, 4, 4)/15$  and  $(11, 2, 2)/15$ ,  $N_{AB} = N_{AC} = 2/\sqrt{15}$ ,  $N_{BC} = 1/15$ . **COMPUTATION** (`w33_pass11152_pauli_bell_charpoly.py`, `w33_pass11153_temporal_magic_budget.py`, `w33_pass11154_polygamy_optimum_structure.py`, `w33_pass11155_temporal_cglmp_trend.py`)

The loose ends close. The ninth roots come from one identity: with  $U = (XZ^2) \otimes (XZ^2)$ , the joint relabelling of both outcomes,  $B^3 - 3B + 1 = (B + 1)(I + U + U^2)$ , so on the charged sectors of  $U$  the Bell operator trisects the angle  $2\pi/3$  and on the neutral sector it gives  $x^3 - 6x - 2$ . The determinant law is general: for  $n$  qutrits every column and every row of blocks satisfies  $\sum_i \det S_{ij} = 1$ , and perfect three-qutrit gates exist (about 3.1% of  $\text{Sp}(6, 3)$ ), each column carrying determinants  $(1, 1, -1)$ . The value  $4/\sqrt{15}$  is derived: on the family  $\sqrt{x}|000\rangle + \sqrt{(1-x)/4} \sum_{a=1,2} |a\rangle(|0a\rangle + |a0\rangle)$  the negativity is  $\sqrt{(1+15x)(1-x)/4}$ , maximal at  $x = 7/15$ . The temporal optimum keeps climbing: 3.576, 3.621, 3.658 at  $d = 8, 9, 10$  against 3.101, 3.122, 3.140 in space, its deficit from 4 falling like  $d^{-0.75}$ . **COMPUTATION** (`w33_pass11157_ninth_roots_explained.py`, `w33_pass11158_perfect_three_qutrit_gates.py`, `w33_pass11159_explicit_polygamy_state.py`, `w33_pass11160_temporal_cglmp_high_d.py`)

Three qutrits can be counted exactly. A symplectic matrix is an ordered symplectic basis, the determinant of a block of a column pair is the local symplectic form, and Jacobi's complementary-minor identity shows that invertibility of the nine blocks already makes the Choi state absolutely maximally entangled. The enumeration gives  $286\,654\,464 = 2^{17}3^7$  perfect three-qutrit gates, a fraction  $128/4095$  of  $\text{Sp}(6, 3)$  (the same method returns 13824 for two qutrits; the single two-qutrit orbit agrees with Pahari's classification of Clifford dual-unitary gates over finite fields [53]). The tetracode gate has no scalar analogue — no  $3 \times 3$  matrix over  $\mathbb{F}_3$  has all entries and all  $2 \times 2$  minors nonzero — but it has one over  $\mathbb{F}_9 = \mathbb{F}_3[i]$ . There  $\text{Im}\langle u, v \rangle$  is the symplectic form, so a unitary matrix over  $\mathbb{F}_9$  is a symplectic matrix, perfect exactly when all its entries are nonzero: 12288 of the 24192 elements of  $U(3, \mathbb{F}_9)$ . The simplest is  $K = P^2 + i\mathbf{1}$ , a cyclic shift plus  $i$  times the all-ones matrix, unitary because  $\mathbf{1}^2 = 3 \cdot \mathbf{1} = 0$

— the number of parties equals the characteristic. Finally,  $4/\sqrt{15}$  is a certified local maximum: the partial transposes at the optimal state have no eigenvalue within  $2/15$  of zero, and the Hessian has 32 strictly negative directions and exactly 20 flat ones, the local-unitary orbit; 96 further random restarts reach  $4/\sqrt{15}$  and nothing higher. Global optimality remains open. **COMPUTATION** (`w33_pass11162_polygamy_local_maximum.py`, `w33_pass11163_perfect_three_qutrit_count.py`, `w33_pass11164_three_qutrit_tetrcode.py`)

## 4 Time: a finite light cone and four clocks

Section 3 ended with a qutrit whose operators behave like the before-and-after of a process. This section asks what *time* looks like inside  $W(3,3)$ . The answer comes in three layers, and keeping them apart is the main lesson of the section:

1. a finite **present**: 27 “events” carrying an exact Minkowski-type form  $t^2 - x^2 - y^2$ , with a light cone and four null directions;
2. an unbounded **history**: the paths through those events, which never close up and which remember a winding number that the present forgets;
3. an **arrow**: the direction of time in the thermodynamic sense, which appears only when information is thrown away.

Everything in layers 1 and 2 is reversible. Layer 3 is where irreversibility enters, and we shall see exactly what it costs.

### 4.1 Choosing a “now”

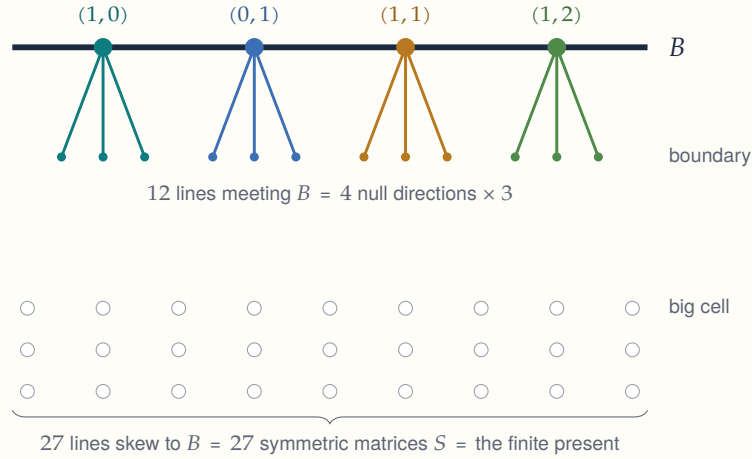
Fix one line  $B$  of  $W(3,3)$  — the Bell line of Section 3, the unitary channels. The other 39 lines fall into two kinds: 12 lines meet  $B$  (three through each of its four points), and 27 lines miss it entirely. Every line that misses  $B$  can be written uniquely as

$$L_S = \{(Sy, y) : y \in \mathbb{F}_3^2\}, \quad S = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a, b, c \in \mathbb{F}_3,$$

for a *symmetric*  $2 \times 2$  matrix  $S$ . There are  $3^3 = 27$  such matrices. So the lines of  $W(3,3)$  decompose as

$$40 = \underbrace{1}_B + \underbrace{12}_{\text{meet } B} + \underbrace{27}_{\text{histories } S}.$$

The 13 boundary lines are not an arbitrary leftover. In Plücker coordinates the 40 lines of  $W(3,3)$  lie on the quadric  $x_0x_4 - x_1x_3 + x_2^2 = 0$  in  $\mathbb{P}^4(\mathbb{F}_3)$  — this is the dual quadrangle  $Q(4,3)$  of Section 2 — and the 27 histories are its “big cell”  $x_0 \neq 0$ , reached by  $S \mapsto [1 : a : b : c : ac - b^2]$ . The 13 points with  $x_0 = 0$  form the tangent cone of the quadric at  $B$ :  $B$  itself plus four three-point generators, one for each point of  $B$ . **COMPUTATION** (`w33_20260924_history_bigcell_q43_compactification.py`)



**Figure 4.** The forty lines of  $W(3,3)$  seen from the Bell line  $B$ . The four points of  $B$  are labelled by the four projective directions of  $\mathbb{F}_3^2$ ; they will turn out to be the four null directions of a finite light cone. The  $1 + 12 = 13$  lines touching  $B$  form the “boundary”; the 27 others are the events of the finite present.

## 4.2 The finite light cone

Two histories  $L_S$  and  $L_T$  are either disjoint or they meet in a point. A short calculation shows they meet exactly when

$$\det(S - T) = 0.$$

Write  $S = \begin{pmatrix} t+x & y \\ y & t-x \end{pmatrix}$  (possible because 2 is invertible mod 3). Then

$$\det S = t^2 - x^2 - y^2$$

— the Minkowski form of a space-time with one time and two space dimensions. Two events are *null-separated* (“light can travel between them”) exactly when their lines meet, i.e. when the two contexts share a measurement.

### Theorem 4.1

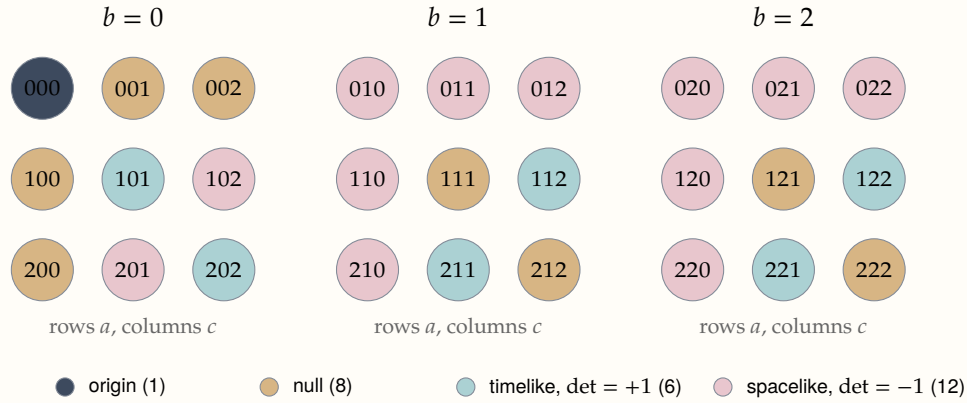
THEOREM

COMPUTATION

The 27 histories form the finite Minkowski space  $(\mathbb{F}_3^3, t^2 - x^2 - y^2)$ . Relative to any event, the other 26 split as 8 null, 6 timelike ( $\det = 1$ ) and 12 spacelike ( $\det = -1$ ). The null directions through an event are the four rank-one matrices  $vv^T$ ,  $v \in \mathbb{P}^1(\mathbb{F}_3)$ , and they correspond bijectively to the four points of  $B$ . [w33\\_20260924\\_history\\_bigcell\\_q43\\_compactification.py](#)

### In plain words

In ordinary relativity, two events are “null-separated” if a flash of light could connect them. Here there is no light and no distance — only three possible values of each coordinate — yet the same algebraic form  $t^2 - x^2 - y^2$  organises the 27 events, and “null” has a sharp operational meaning: two complete measurement contexts share exactly one measurement. The *celestial circle* — the set of directions light can travel — is in ordinary  $2 + 1$  dimensions a circle of directions; here it is the four points of the projective line over  $\mathbb{F}_3$ .



**Figure 5.** The 27 events  $S = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ , labelled  $abc$  and coloured by their separation from the origin 000. The counts  $1 + 8 + 6 + 12$  are those of a light cone over the field with three elements.

### 4.3 Four null pulses make one tick

The four null directions have integer representatives

$$D_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad D_\infty = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad D_+ = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad D_- = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

each positive semidefinite of rank one. Over  $\mathbb{F}_3$  they satisfy a single relation, and it is worth staring at:

#### Proposition 4.2 (the common mode)

THEOREM

COMPUTATION

The counter map  $\mathbb{F}_3^4 \rightarrow \text{Sym}_2(\mathbb{F}_3)$ ,  $(n_0, n_\infty, n_+, n_-) \mapsto \sum n_i D_i$ , is onto with kernel exactly  $\langle (1, 1, 1, 1) \rangle$ , so

$$\text{Sym}_2(\mathbb{F}_3) \cong \mathbb{F}_3^4 / \langle (1111) \rangle.$$

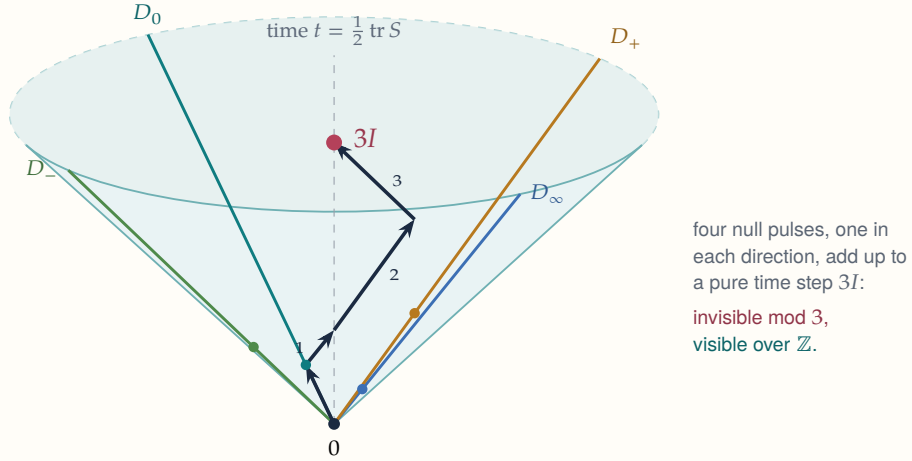
Over the integers, however,

$$D_0 + D_\infty + D_+ + D_- = 3I,$$

a strictly timelike matrix of trace 6. `w33_20260924_positive_null_counter_cover.py`

#### What it means physically: two kinds of time

Proposition 4.2 separates two notions that ordinary physics merges. The *state* of the finite present is a counter of null pulses modulo their common mode: 27 values. But a history that fires one pulse in every direction returns to the same present while advancing by a pure time step upstairs. The finite chart measures *where* you are; the lifted history measures *how long* it took. This is the structural shape of the Page–Wootters idea that time is a relation between a clock and a system, and of the distinction, in relativity, between coordinate position and proper time along a world line. **BRIDGE**



**Figure 6.** The real light cone of  $2 \times 2$  symmetric matrices, drawn with time  $t = \frac{1}{2} \text{tr } S$  upward. The four null rays  $D_0, D_\infty, D_\pm$  are the lifts of the four points of the Bell line. The black zig-zag adds one pulse in each direction: it ends on the time axis at  $3I$ . Modulo three this is the zero matrix — the finite present cannot see it — but the integer lift records it as elapsed proper time.

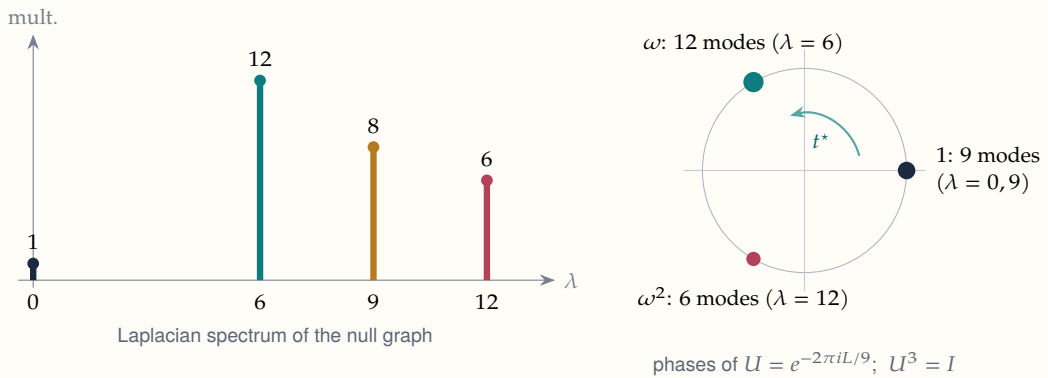
#### 4.4 Light-like steps: the null graph and its clock

Join two of the 27 events when they are null-separated. The resulting graph is 8-regular with 108 edges, and it can be diagonalised exactly.

##### Theorem 4.3 (spectral clock)

COMPUTATION

The null graph has adjacency spectrum  $8^1 2^{12} (-1)^8 (-4)^6$  and Laplacian spectrum  $0^1 6^{12} 9^8 12^6$ . At  $t^* = 2\pi/9$  the continuous-time walk  $U = e^{-it^*L}$  satisfies  $U^3 = I$ , with eigenvalue multiplicities  $1^9, \omega^{12}, (\omega^2)^6$ . The nine fixed modes are the constant mode and the eight nonzero dual-null vectors of  $\mathbb{F}_3^3$ . `w33_20260924_null_history_spectral_clock.py`



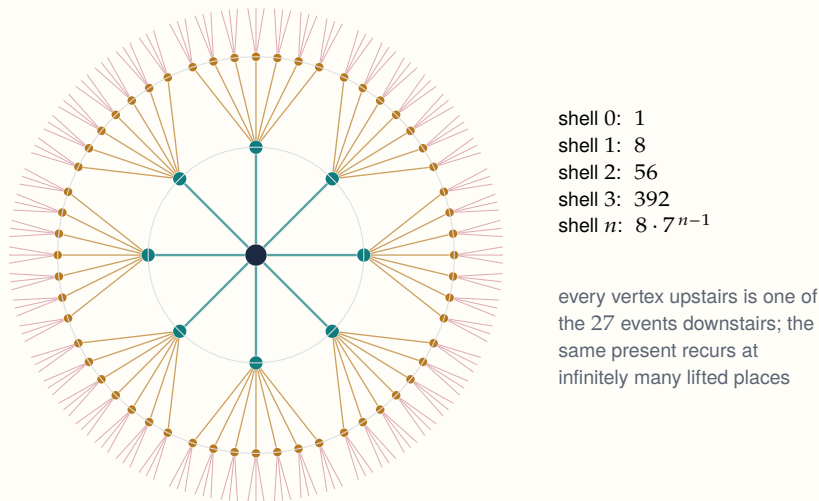
**Figure 7.** Left: the four Laplacian frequencies of the null graph. Right: running the graph as a quantum walk for time  $2\pi/9$  sends every mode to a cube root of unity — the 27-event present carries an exact qutrit-valued clock.

### A tempting identification that fails

The nine fixed modes invite the reading “9 = the nine operator histories of one qutrit” from Section 3. It is false. The symmetry group acting on the fixed nine is faithful of order 648, whereas the projective qutrit Clifford group has order 216, and the natural Bell-line  $S_4$  splits the eight null modes  $4 + 4$  while  $SL(2,3)$  is transitive on the eight nonidentity qutrit Pauli labels. The count matches; the module does not. **REFUTED** (as an identification)

## 4.5 History is a path, and paths do not close

The null graph is connected with 27 vertices and 108 edges, so its cycle space has rank  $108 - 27 + 1 = 82$ . Its fundamental group is the free group  $F_{82}$  and its universal cover is the infinite 8-regular tree.



**Figure 8.** The universal cover of the null graph (only three grandchildren of each vertex are drawn in the outer shell). The present is finite; the history is an unbounded tree.

### Theorem 4.4 (orientation and homology)

COMPUTATION

- (i) The 108 null edges are partitioned into 36 edge-disjoint triangles, one for each point of  $W(3,3)$  off the Bell line.
- (ii) A signed sum of the 36 triangle boundaries is a primitive cycle through all 108 edges with coefficients  $\pm 1$ . The index-2 subgroup of the Bell stabiliser (order 648) fixes it, and the other coset negates it: the cycle space splits objectwise as  $82 = 1_{\text{orientation}} + 81$ .
- (iii) Filling the 36 triangles leaves first Betti number 46, and the reduced 81-dimensional module splits as  $35 + 46$ . The 46 injects, by an explicit chain map, into  $H_1(W(3,3))$ , which is itself 81-dimensional and irreducible — the Steinberg module of  $\text{PSp}(4,3)$ .

w33\_20260924\_history\_invariant\_cycle\_orientation.py, w33\_20260924\_history\_to\_global\_h1\_intertwiner.py

So there is exactly one global orientation of the history graph, and the symmetry group of the “now” either preserves it or reverses it. This is a *direction* for time, but not yet an *arrow*: reversing it is a symmetry of the geometry.

| Temporal observable     | Lives on                                                           | Reversible?                              |
|-------------------------|--------------------------------------------------------------------|------------------------------------------|
| position in the present | one of 27 events                                                   | yes                                      |
| step count              | length of a path in the null graph                                 | yes                                      |
| reduced word            | element of the free group $F_{82}$                                 | yes                                      |
| winding                 | homology class in $\mathbb{Z}^{82}$ ; the common mode of Prop. 4.2 | yes                                      |
| orientation             | the sign of the unique invariant cycle                             | yes (up to a symmetry)                   |
| coherent phase          | the Weil phase $\mu_{12}$ below                                    | yes (complex conjugation)                |
| records                 | classical outcomes kept by an observer                             | <b>no</b> : discarding them is the arrow |

#### 4.6 A coherent twist: the twelfth roots of unity

For the putrit Fourier transform  $F$  and the chirp  $G = \text{diag}(1, \omega, \omega)$  one has  $(FG)^3 = iF^2$ . The metaplectic (Weil) representation that implements the symplectic symmetries on quantum states therefore carries phases in  $\langle i, \omega \rangle = \mu_{12}$ . Across the 27 quadratic forms of the present, the Weil phases occur with census  $1^{13}, i^4, (-1)^6, (-i)^4$ . Complex conjugation — the quantum-mechanical time reversal — sends  $i \mapsto -i$  and  $\omega \mapsto \omega^{-1}$ . This is a reversible, coherent orientation mechanism, distinct from entropy production. **COMPUTATION**

(w33\_20260924\_temporal\_maslov\_mul2.py)

#### 4.7 Four clocks: the Hesse striations and the tetracode

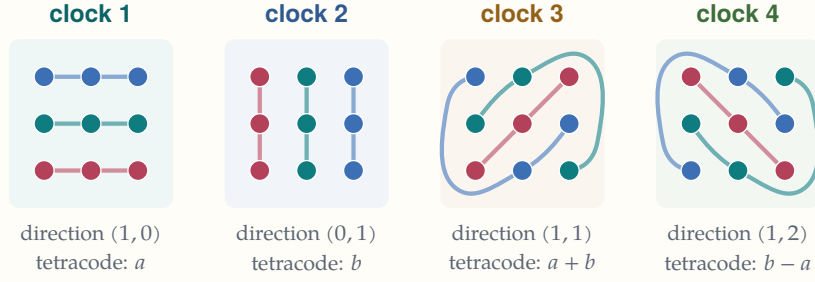
The four points of the Bell line — the four null directions — reappear in a second, seemingly unrelated place. Identify the nine operator histories  $|i\rangle\langle j|$  with the nine points of the affine plane  $AG(2, 3) = \mathbb{F}_3^2$ . Its 12 lines fall into four *parallel classes* of three lines each, one class for each direction  $v \in \mathbb{P}^1(\mathbb{F}_3)$ . Each class is a way of reading the nine cells as “three ticks of a clock”; we call the four classes the four *Hesse clocks*.

##### Theorem 4.5 (one $S_4$ for three structures)

**COMPUTATION**

The four projective null directions of the present, the four Hesse clocks, and (Section 5) four orthogonal  $A_2$  root systems inside  $E_8$  are all indexed by  $\mathbb{P}^1(\mathbb{F}_3) = \{(1, 0), (0, 1), (1, 1), (1, 2)\}$ . For every one of the 24 elements of  $PGL(2, 3) \cong S_4$ , the induced permutations of the three four-element sets are identical. The larger groups differ (the Bell stabiliser acts with kernel 27,  $AGL(2, 3)$  with kernel 18); the common object is the quotient  $S_4$  and its action. w33\_20260924\_null\_hesse\_4a2\_s4\_intertwiner.py

The four clocks also carry a code. Evaluate the linear “clock reading”  $\ell_{(a,b)}(x, y) = ax + by$



**Figure 9.** The four Hesse clocks. Each panel shows the nine operator histories of one qutrit as the affine plane  $\mathbb{F}_3^2$ , split into three parallel lines (the three “ticks”). Lines of the two diagonal classes wrap around, as on a torus. The four panels are labelled by the four points of  $\mathbb{P}^1(\mathbb{F}_3)$  — the same four labels as the null directions of Figure 4 — and by the coordinate of the tetracode they read.

at oriented representatives  $(1,0), (0,1), (1,1), (2,1)$  of the four directions. The result is

$$(a, b, a+b, b-a) \in \mathbb{F}_3^4,$$

and the nine such words are *exactly* the repository’s frozen *tetracode*, the  $[4,2,3]_3$  code at the heart of the ternary Golay code and of the  $E_8$  lattice’s tetracode construction — no coordinate permutation needed. **COMPUTATION** (`w33_pass10946_clock_code_cone_objectwise.py`)

| $(a,b)$ | $(1,0)$ | $(0,1)$ | $(1,1)$ | $(2,1)$ |
|---------|---------|---------|---------|---------|
| $(0,0)$ | 0       | 0       | 0       | 0       |
| $(1,0)$ | 1       | 0       | 1       | 2       |
| $(2,0)$ | 2       | 0       | 2       | 1       |
| $(0,1)$ | 0       | 1       | 1       | 1       |
| $(1,1)$ | 1       | 1       | 2       | 0       |
| $(2,1)$ | 2       | 1       | 0       | 2       |
| $(0,2)$ | 0       | 2       | 2       | 2       |
| $(1,2)$ | 1       | 2       | 0       | 1       |
| $(2,2)$ | 2       | 2       | 1       | 0       |

nine clock readings  $\ell_{(a,b)}$  evaluated at the four oriented null directions:  
every nonzero word vanishes on exactly one direction (one clock is “stopped”),  
minimum distance 3: the tetracode  $[4,2,3]_3$ .

**Figure 10.** The tetracode as a table of clock readings (colour: 1 teal, 2 rose).

#### What it means physically: a frame and an orientation

A projective direction picks a clock *frame*: which three-way split of the nine histories counts as the ticks. Choosing a representative of the direction —  $(1,2)$  or its negative  $(2,1)$  — picks an *orientation* of that clock. The tetracode’s peculiar sign  $b-a$  (rather than  $a-b$ ) in its last coordinate is exactly this orientation choice. Each nonzero codeword stops exactly one of the four clocks; 4 stopped clocks  $\times$  2 orientations  $\times$  3 origins give 24 complete clock calibrations.

The other research track found a further layer (Pass 10948, which we cite rather than re-derive). Over  $\mathbb{F}_3$ ,  $x^4 + 1 = (x^2 + x - 1)(x^2 - x - 1)$ , and after flipping the orientation of one clock the tetracode becomes the ideal  $(x^2 \pm x - 1)$  in  $\mathbb{F}_3[x]/(x^4 + 1)$ : the cyclic shift of clock readings with a sign,  $N(c_0, c_1, c_2, c_3) = (-c_3, c_0, c_1, c_2)$ , preserves it and satisfies  $N^4 = -I$ ,  $N^8 = I$ . The two orientations give two transverse self-dual tetracodes forming a hyperbolic

pair; the negacyclic presentation of the tetracode was used in a cyclotomic  $E_8$  construction by Watanabe *et al.* [44]. **COMPUTATION** (`w33_pass10948_clock_tetracode_negacyclic_hyperbolic_bridge.py`) A clock whose fourth tick is a sign flip and whose eighth tick is the identity behaves exactly like a spinor under rotation, and this is literal: the tick lifts to an element of order eight in  $GL(2, 3)$  whose fourth power is the central  $2\pi$  spin sign, and in Section 8 the same sign is matter parity (Theorem 8.3).

#### 4.8 Clocks as relations: the Page–Wootters weld

Label a clock qudit's phase by  $c \in \mathbb{F}_3$  and a system's phase by  $p \in \mathbb{F}_3$ . The four directions of  $\mathbb{F}_3^2$  are  $c, p, c + p$  and  $c - p$ . For an order-three evolution  $U$  of the system, the two stationary “history states”

$$(X_{\text{clock}} \otimes U)|\Psi_+\rangle = |\Psi_+\rangle, \quad (X_{\text{clock}} \otimes U^{-1})|\Psi_-\rangle = |\Psi_-\rangle$$

are supported, in the clock's phase basis, exactly on  $c + p = 0$  and  $c - p = 0$ : the two correlated directions. The two remaining directions ( $c = 0, p = 0$ ) are the uncorrelated axes. On the committed  $E_8$  basis, pairs of axis-only backgrounds generate a subalgebra of dimension 24, whereas any graph-correlated background generates all 248 dimensions. **COMPUTATION** (`w33_20260924_page_wootters_diagonal_weld.py`)

##### In plain words

In 1983 Don Page and William Wootters proposed [40] that the universe as a whole might be static, with time emerging as a correlation between a “clock” part and the rest. The finite version above is exact: a state that is globally unchanging, yet in which the clock reading and the system's phase are locked together. Only the two *correlated* clock directions produce such states — and only they, in the  $E_8$  computation, generate the whole algebra.

#### 4.9 A second light cone: Hermitian $3 + 1$ , and a weld that fails

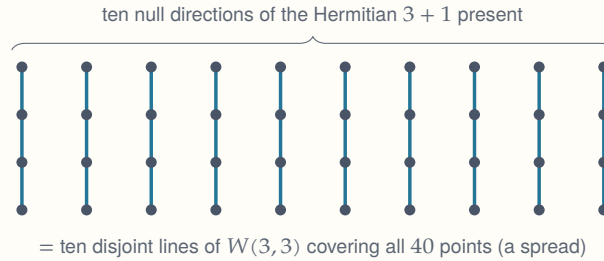
Replace real symmetric matrices by Hermitian ones over  $\mathbb{F}_9 = \mathbb{F}_3(u)$ ,  $u^2 = -1$ :  $X = \begin{pmatrix} a & z \\ \bar{z} & c \end{pmatrix}$  with  $z = x + uy$ . Then  $\det X = ac - x^2 - y^2$  is a form in four variables — a  $3 + 1$ -dimensional finite Minkowski space.

##### Theorem 4.6 (the Hermitian present is a spread)

**COMPUTATION**

The 81 Hermitian events split  $21 + 30 + 30$  by determinant, and the 40 projective directions split  $10 + 15 + 15$ . An explicit map sends the Hermitian quadric to the elliptic hyperplane section of  $Q(4, 3)$ , and under the line dictionary the ten null directions become ten pairwise disjoint lines of  $W(3, 3)$  covering all 40 points — a *spread*. The 36 elliptic sections are exactly the 36 spreads of  $W(3, 3)$ . `w33_20260924_hermitian_3pl_spread_bridge.py`

**The complex-structure weld does not exist**



**Figure 11.** The Hermitian 3 + 1 light cone is a partition of the forty points into ten complete measurement contexts — a complete “schedule” for measuring every two-qutrit direction exactly once.

One might hope to weld the Hermitian light cone to the symplectic geometry through a single complex structure  $J$  ( $J^2 = -I$ ) with  $\Omega(x, y) = B(Jx, y)$ . Exhausting all 729 candidates finds none. The light cone and the qutrit geometry meet through the point/-line duality of Section 2, not through one compatible  $J$ . **REFUTED** (for the direct weld)

#### 4.10 The arrow: forgetting two trits

Every structure so far is reversible. Where does irreversibility come from? There is a sharp answer.

##### Theorem 4.7 (the price of an arrow)

**THEOREM**

For every qutrit state  $\rho$ ,

$$\frac{1}{9} \sum_{u \in \mathbb{F}_3^2} P_u \rho P_u^\dagger = \text{tr}(\rho) \frac{I}{3}.$$

Each map  $\rho \mapsto P_u \rho P_u^\dagger$  is unitary and reversible; their average, obtained by forgetting which  $u$  occurred, is the completely depolarising channel, which erases every state to the same fixed point. The forgotten record is two trits, so by Landauer’s principle [32] making a qutrit clock irreversible in this way dissipates at least  $2 k_B T \ln 3$  per step.

*Proof.* The left side, as a function of  $\rho$ , commutes with conjugation by every  $P_v$  (which merely permutes the  $u$ ’s up to phases that cancel). Its image  $\Sigma$  therefore commutes with all nine Paulis, which span  $\text{End}(\mathbb{C}^3)$ , so  $\Sigma$  is a scalar; taking traces gives the scalar  $\text{tr}(\rho)/3$ .  $\square$

##### What it means physically

The finite present has a light cone; the history has an orientation; the Weil phase has a handedness. None of these is an arrow of time, because each can be reversed by a symmetry. The arrow appears exactly when a record is discarded. For a qutrit clock the minimal record is the two-trit label of the Pauli step — the photonic papers’ “two trits per cycle” (`bt823_the_closure.py`; see also `w33_20260924_adqc_record_arrow.py`) — and discarding it turns a reversible tick into the maximally irreversible channel. This is the finite form of the modern view that the thermodynamic arrow is information

loss, and it is *consistent* with the Connes–Rovelli idea of Section 3: a state can orient time, but only forgetting makes it flow one way. **BRIDGE**

The two-trit record need not be put in by hand. For a two-qutrit Clifford gate the tripartite information of its Choi state is  $I_3 = 2 - \text{rank } S_{AA} - \text{rank } S_{BA}$ , which the determinant law of Theorem 3.7 makes never positive: no gate concentrates information, and the perfect gates are the maximal scramblers ( $I_3 = -2$ ). Whenever  $S_{BA}$  is invertible, a tick with a partner in a maximally mixed state, followed by forgetting the partner and its purification — two trits — is exactly the completely depolarising channel of Theorem 4.7. The arrow appears as soon as a maximally entangling tick is followed by discarding what it entangled with. **COMPUTATION** (w33\_pass11161\_scrambling\_and\_arrow.py)

With three qutrits the record is shared rather than handed over. In a perfect three-qutrit tick each qutrit’s past is invisible to every proper subset of the three futures and fully present — two trits — in all three together. This is the secret-sharing property of absolutely maximally entangled states [54] (“exact local masking” [53]). So forgetting *any one* future erases the past completely, and the erased record is still two trits: the price of the arrow belongs to one qutrit, not to the number of partners. The spectral clock of Theorem 4.3 is not such a tick. Read on the three coordinates of  $\mathbb{F}_3^3$ , it is a Clifford gate,  $X_a \mapsto X_a$  and  $Z_b \mapsto Z_b X_{Mb}$  with  $M$  the Gram matrix of  $q$  — the free propagator  $e^{-iq(p)}$  of the Lorentzian form, coupling the two light-cone coordinates by one trit and acting locally on the transverse one. This is forced: a Clifford gate that commutes with all translations has off-diagonal blocks of rank at most one, so no translation-invariant clock is perfect. Conserved momentum caps scrambling at one trit per pair per tick, and forgetting a partner after a clock tick erases at most one trit; the two-trit arrow needs an interaction that breaks translation invariance. **COMPUTATION** (w33\_pass11165\_three\_qutrit\_arrow.py, w33\_pass11166\_clock\_tick\_clifford.py)

*Three layers of time, exactly separated: a finite present with a light cone (27 events,  $t^2 - x^2 - y^2$ ); an unbounded, oriented history (a free group, an 8-regular tree, a winding the present cannot see); and an arrow that costs exactly two trits of forgotten record per tick.*

#### Where this stops

None of this derives a continuum space-time, a speed of light, a metric that curves, or gravity. The light cone here is an algebraic form over a three-element field; “null” means “shares a measurement”. Whether the present, history and arrow of the finite model are the finite shadows of physical time is a bridge hypothesis, and the dynamics that would make it testable — a Hamiltonian, an energy scale — is not supplied by the geometry. **OPEN**

## 5 $E_8$ from nine histories

$E_8$  is the largest of the five exceptional simple Lie algebras: 248 dimensions, 240 roots, a symmetry so rigid that it appears, often unexpectedly, in string theory, in the densest packing of

spheres in eight dimensions, and in the physics of certain magnetic materials. This section shows that one qutrit's nine operator histories build  $E_8$  in a way in which every piece has a meaning in the finite geometry — and it records carefully one famous coincidence that turned out to be a trap.

### In plain words

A Lie algebra is the “infinitesimal” version of a continuous symmetry: the set of all directions in which you can nudge a system while keeping its rules intact, together with the rule (the *bracket*) for what happens when two nudges are combined in opposite orders. Its *roots* are the special directions that the symmetry's own “clock” (its maximal commuting part) rotates at definite rates.  $E_8$  has 240 of them.

## 5.1 Eighty plus eighty-four plus eighty-four

Recall from Section 3 that the 80 non-identity two-sided Weyl maps on one qutrit span  $\mathfrak{sl}_9$ , the Lie algebra of traceless  $9 \times 9$  matrices acting on the nine operator histories  $|i\rangle\langle j|$ . Now add *packets*: unordered triples of histories. There are  $\binom{9}{3} = 84$  of them, forming the representation  $\Lambda^3 \mathbb{C}^9$ ; their duals form  $\Lambda^6 \mathbb{C}^9$ , another 84. A classical construction [1, 3] makes

$$E_8 = \mathfrak{sl}_9 \oplus \Lambda^3 \mathbb{C}^9 \oplus \Lambda^6 \mathbb{C}^9, \quad 248 = 80 + 84 + 84,$$

into a Lie algebra. On the nine history cells the brackets are literal packet operations:  $E_{ij}$  replaces cell  $j$  by cell  $i$  inside a packet; two disjoint triples bracket to the complementary six-set; a triple and a six-set differing by one cell contract to an  $E_{ij}$ ; opposite packets return a Cartan direction. **COMPUTATION** (`w33_20260924_temporal_su9_e8_compiler.py`, `w33_20260924_temporal_a8_e8_process_bracket.py`)

## 5.2 Hesse lines are copies of $A_2$

Place the nine histories on the affine plane  $\mathbb{F}_3^2$  as in Figure 9. A triple  $S$  of cells labels the root

$$w_S = \sum_{i \in S} e_i - \frac{1}{3} \sum_{i=1}^9 e_i, \quad w_S \cdot w_T = |S \cap T| - 1.$$

(Indeed  $|w_S|^2 = 3 - 2 + 1 = 2$ .) The 84 triples fall into two orbits of the affine group  $AGL(2, 3)$  (order 432): the 12 *lines* of the plane and the 72 non-collinear *triangles*.

### Theorem 5.1 (four clocks, four $A_2$ 's)

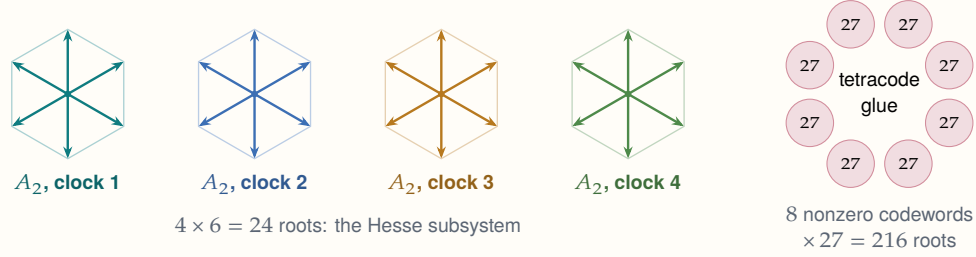
**THEOREM**
**COMPUTATION**

The three lines of one parallel class give three roots with pairwise products  $-1$  and sum  $0$ ; with their negatives they form a root system  $A_2$ . Lines of different classes meet in exactly one cell, so their roots are orthogonal. The 12 Hesse lines therefore span a reflection-closed subsystem  $4A_2 \subset E_8$ , one  $A_2$  per clock, and the root shell refines as

$$240 = \underbrace{72}_{A_8} + \underbrace{24}_{\text{Hesse } 4A_2} + \underbrace{144}_{\text{triangles}}.$$

`w33_20260924_temporal_hesse_4a2_a8_bridge.py`

*Proof.* Disjoint triples have  $|S \cap T| = 0$ , hence product  $-1$ ; the three lines of a class partition the nine cells, so  $\sum w_S = \sum_i e_i - 3 \cdot \frac{1}{3} \sum_i e_i = 0$ . Two non-parallel lines of an affine plane meet in one point, giving product  $1 - 1 = 0$ .  $\square$



**Figure 12.**  $E_8$  assembled from four clocks. Each Hesse clock contributes one  $A_2$  root system (left). The remaining 216 roots come in eight sectors of 27, one for each nonzero word of the tetracode (right):  $240 = 4 \cdot 6 + 8 \cdot 27$ .

### 5.3 The glue is the tetracode, and the tetracode is the clock

The lattice  $E_8$  contains the sublattice  $A_2^4$  with quotient  $\mathbb{F}_3^2$ , and the well-known construction of  $E_8$  from the tetracode [9] reads it as

$$E_8 = \bigcup_{c \in \mathcal{T}} (c + A_2^4), \quad 240 = 4 \cdot 6 + 8 \cdot 27,$$

where  $\mathcal{T} \subset \mathbb{F}_3^4$  is the tetracode: each of the eight nonzero codewords has weight three, and contributes  $3^3 = 27$  minimal vectors. Combined with Section 4, the four glue coordinates are the four clocks and the eight nonzero sectors are the eight oriented, nonzero linear clock readings. **THEOREM** (classical) **COMPUTATION** (the clock reading, `w33_pass10946_clock_code_cone_objectwise.py`)

### 5.4 A trap: $240 = 240$

The collinearity graph of  $W(3, 3)$  has 240 edges, and  $E_8$  has 240 roots. For years the repository looked for a natural bijection between them. The question has a precise and negative answer.

#### Theorem 5.2 (no equivariant bijection)

**COMPUTATION**

The two groups of order 51 840 of Section 2 each do only half the job:

- $\text{PGSp}(4, 3) \cong W(E_6)$  acts faithfully and transitively on the 240 edges of  $W(3, 3)$ , but *intransitively* on the 240 roots of  $E_8$ ;
- $\text{Sp}(4, 3)$  acts faithfully and transitively on the 240 roots, but its centre acts trivially on the edges, so its edge action is not faithful.

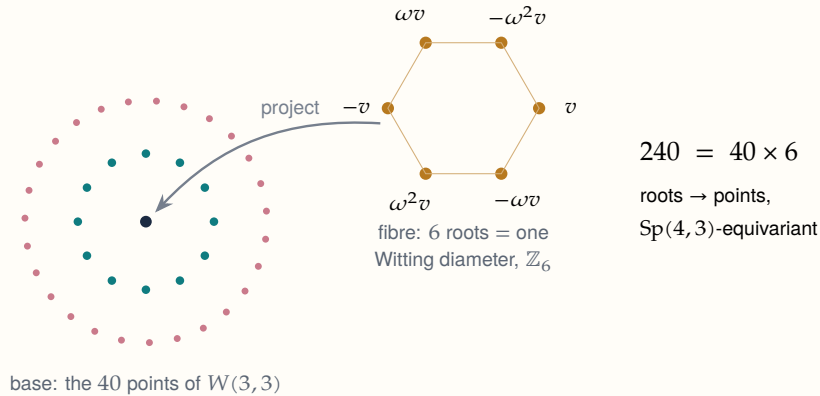
Hence no bijection between edges and roots is equivariant for a common group. `w33_pass1020_e8_transitive_51840.g`

What *does* exist is better. The Weyl group of  $E_8$  contains a regular element  $w$  of order three. By Springer's theory [42] its centraliser is the complex reflection group  $G_{32} \cong \mathbb{Z}_3 \times \text{Sp}(4, 3)$

of order 155 520 — the symmetry group of the *Witting polytope*, whose 240 vertices in  $\mathbb{C}^4$  are the  $E_8$  roots.

**Theorem 5.3 (the 6 : 1 fibration)**
**COMPUTATION**

The 240 roots of  $E_8$  fall into 40 complex lines  $\{\pm v, \pm\omega v, \pm\omega^2 v\}$ , each a copy of the six Eisenstein units.  $\text{Sp}(4, 3)$  permutes the 40 lines with subdegrees 1, 12, 27: the quotient is the collinearity graph of  $W(3, 3)$  — on its *points*, not its lines. The fibre group is  $\langle c^5 \rangle \cong \mathbb{Z}_6$  for a Coxeter element  $c$  of  $W(E_8)$ . `w33_pass1021_e8_fibration_over_forty.g`



**Figure 13.** The correct relation between  $E_8$  and  $W(3, 3)$ : not edges to roots, but a six-to-one map from roots to points, whose fibres are the Eisenstein unit hexagons.

**What it means physically: a photon's four-dimensional state space**

The 40 Witting diameters are 40 rays in  $\mathbb{C}^4$  — the state space of a photon's path and polarisation, or of two qubits. Two rays are orthogonal exactly when the corresponding points of  $W(3, 3)$  are collinear. So the same geometry is drawn twice: by *states* of a four-level system (the Witting rays) and by *operators* on two qutrits (the Pauli classes). The photonic Holonet papers call this operator–operand duality. **COMPUTATION** (`bt817_self_entangled_photon_atlas.py`, `bt821_operator_operand_duality.py`)

### 5.5 $E_6 \times A_2$ , the twenty-seven, and a cubic

Under its subalgebra  $E_6 \oplus A_2$ ,

$$E_8 = (78, 1) \oplus (1, 8) \oplus (27, 3) \oplus (\overline{27}, \overline{3}),$$

which is a  $\mathbb{Z}_3$ -grading  $248 = 86 + 81 + 81$ . The 27 of  $E_6$  carries a unique invariant cubic form  $N$ ; in the repository's committed basis it has exactly 45 signed terms (*triads*), each coordinate in 5 triads and each pair in at most one — precisely the incidence of the 27 lines and 45 tritangent planes of a smooth cubic surface, whose symmetry group is  $W(E_6)$ , the automorphism group of  $W(3, 3)$  [16]. **THEOREM** (classical) **COMPUTATION** (`canonical_su3_gauge_and_cubic.json`)

### What it means physically

In grand unified theories built on  $E_6$ , the **27** holds one generation of matter: under  $SO(10)$  it splits  $\mathbf{27} = \mathbf{16} + \mathbf{10} + \mathbf{1}$  — sixteen fermions (including a right-handed neutrino), a ten of Higgs-type fields, and a singlet. The cubic  $N$  is then the unique renormalisable Yukawa coupling  $\mathbf{27}^3$ . Section 8 shows this same split arising as scalar, vector and spinor of a ten-dimensional Lorentz group.

## 5.6 Clock gradings of $E_8$

A clock gives more than a  $\mathbb{Z}_3$ -grading: once an origin and an orientation are chosen it gives an *integer* grading. On the committed Chevalley basis:

### Theorem 5.4 (Hesse clocks grade $E_8$ )

COMPUTATION

(i) A single Hesse line (a “tick”) induces the *contact grading*

$$E_8 = \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2, \quad 1 + 56 + 134 + 56 + 1,$$

with  $\mathfrak{g}_0 = E_7 \oplus \mathbb{C}$ .

(ii) An oriented striation (three parallel ticks) induces the  $|3|$ -grading

$$2 + 27 + 54 + 82 + 54 + 27 + 2,$$

with Levi  $\mathfrak{g}_0 = E_6 \oplus A_1 \oplus \mathbb{C}$ ,  $\mathfrak{g}_{\pm 1} = \mathbf{27} \otimes \mathbf{2}$ ,  $\mathfrak{g}_{\pm 2} = \mathbf{27}$ ,  $\mathfrak{g}_{\pm 3} = \mathbf{2}$ .

(iii) The repository’s frozen  $\mathbb{Z}_3$ -grading  $86 + 81 + 81$  is exactly this integer grading reduced modulo three; all 8 347 nonzero committed structure constants obey the integer degree law.

w33\_20260925\_hesse\_clock\_e8\_gradings.py, w33\_20260925\_e8\_parabolic\_cubic\_clock\_lift.py

This parabolic spectrum was found independently by Kraft, Regeta and Zimmermann in a study of small varieties with group actions [28]. What is repository-specific is the identification with the Hesse clocks and the committed structure constants.

## 5.7 The cubic ladder: three steps to the top

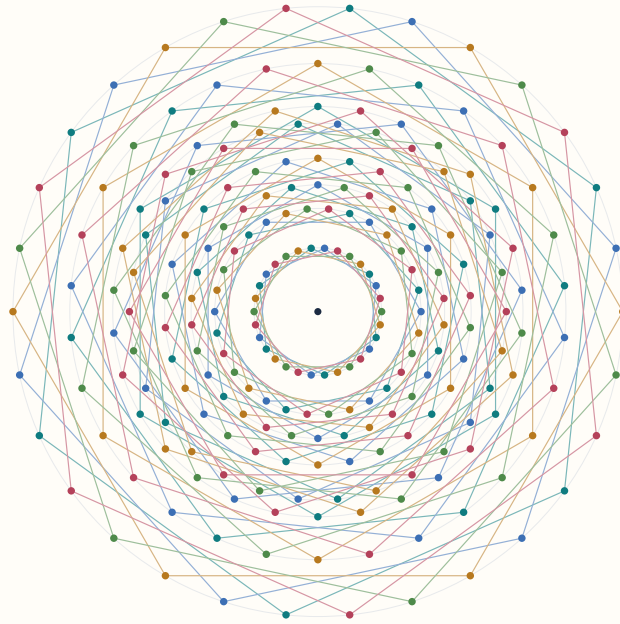
In the  $|3|$ -grading, positive degrees form a nilpotent algebra  $\mathfrak{g}_+ = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_3$  of dimension  $54 + 27 + 2 = 83$ . It is generated by  $\mathfrak{g}_1$ , and the way it fills up is governed by the  $E_6$  cubic.

### Theorem 5.5 (the bracket ladder)

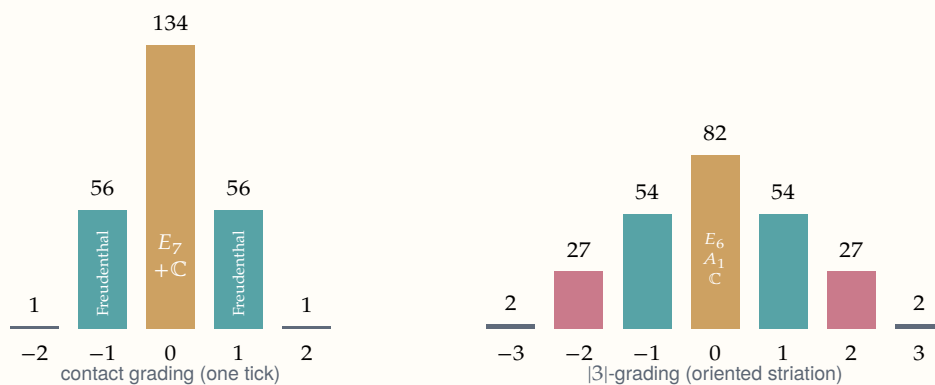
COMPUTATION

On the committed Chevalley bracket:

- (i)  $[\mathfrak{g}_1, \mathfrak{g}_1] = \mathfrak{g}_2$ : 270 nonzero pairs hit all 27 grade-two roots, each ten times;
- (ii)  $[\mathfrak{g}_1, \mathfrak{g}_2] = \mathfrak{g}_3$ : 54 nonzero pairs hit both top roots, each 27 times;

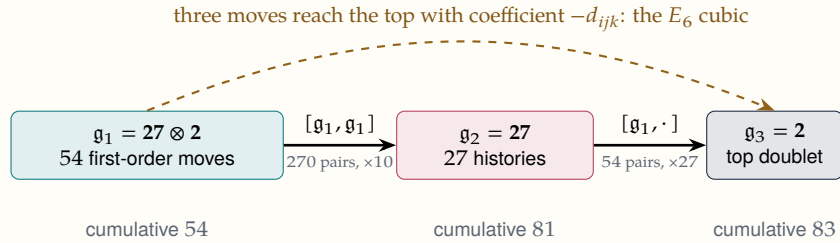


**Figure 14.** The 240 roots of  $E_8$  projected to the Coxeter plane of a Coxeter element  $c$  (order 30), which acts on the plane as a rotation by  $12^\circ$ . The roots lie on eight rings of thirty. The orbits of  $c^5$  — rotation by  $60^\circ$  — are regular hexagons, five per ring: these 40 hexagons are the 40 fibres of Theorem 5.3, one for each point of  $W(3,3)$ . Each hexagon is  $\{\pm v, \pm \omega v, \pm \omega^2 v\}$ , since  $c^{15} = -1$  and  $c^{10}$  has order three. All coordinates computed (`make_figures.py`).



**Figure 15.** Two ways a clock slices  $E_8$ . Heights are dimensions. On the left, the Freudenthal “56” of Section 6 appears at degree  $\pm 1$ ; on the right, the 27 of  $E_6$  appears doubled at degree  $\pm 1$  and single at degree  $\pm 2$ .

(iii) the nonzero nested triple brackets exhibit exactly the 45 triads of the cubic, each 12 times, and for every signed triad  $d_{ijk}$ ,  $[[e_{(i,0)}, e_{(j,1)}], e_{(k,a)}] = -d_{ijk} T_a$  for  $a \in \{0, 1\}$ . So the positive part grows  $54 \rightarrow 81 \rightarrow 83$ , and the committed cubic is exactly the rule by which three first-order moves reach the two-dimensional top. `w33_20260925_cubic_clock_bracket_ladder.py`



**Figure 16.** The cubic ladder. First-order moves combine in pairs to histories, and histories combine with one more move to reach the two-dimensional top. The only way three moves reach the top is through a triad of the cubic.

#### What it means physically: cubic time

Read degree as “order in time”. A first-order move is a state update; two moves compose into a second-order *history*; and three reach the top doublet, whose coefficient is the cubic volume  $N$  spanned by the three moves. In this reading the  $E_6$  cubic — in GUT language the Yukawa coupling — is the rule by which three independent updates produce a net displacement of the clock. The algebra is exact; the reading is a bridge, and it lacks what would make it physics: a Hamiltonian, a state and an energy scale. **BRIDGE**

*One qutrit’s nine histories build  $E_8$  as  $80 + 84 + 84$ . The four Hesse clocks are four orthogonal  $A_2$ ’s whose glue is the tetracode; a clock tick grades  $E_8$  as  $1 + 56 + 134 + 56 + 1$ , an oriented clock as  $2 + 27 + 54 + 82 + 54 + 27 + 2$ ; and  $E_8$ ’s roots fibre six-to-one over the forty points.*

## 6 The Freudenthal 56 and a 57-dimensional light cone

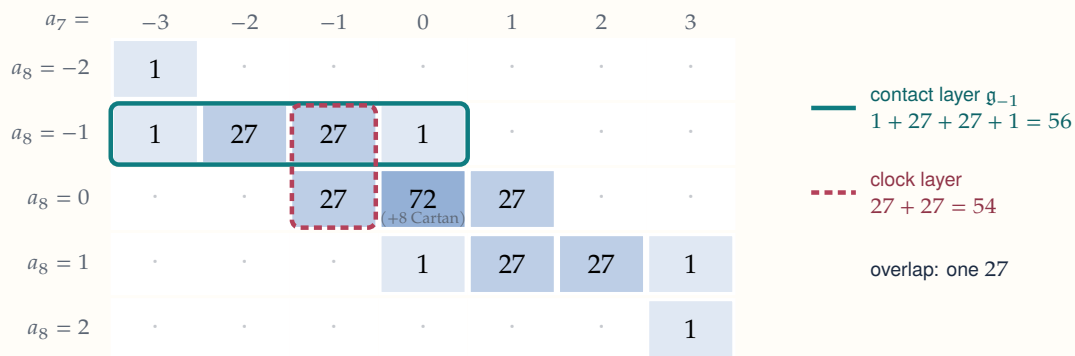
The contact grading of Theorem 5.4 has a 56-dimensional layer  $\mathfrak{g}_{-1}$ . In the 1950s Hans Freudenthal showed [15] that the 56-dimensional representation of  $E_7$  carries a remarkable structure — a symplectic form and a quartic form — now called a *Freudenthal triple system*. It has since turned up in the physics of black holes in supergravity and in the entanglement of qubits. This section computes it directly from the Hesse clock, on the committed  $E_8$  basis, and finds a light cone one dimension larger.

**In plain words**

Think of the 56 as the space of “charges” of a black hole in a theory with many electric and magnetic fields: 28 electric and 28 magnetic. The symplectic form pairs each electric charge with its magnetic partner. The quartic form is a single number built from all 56 charges; in string theory its square root gives the black hole’s entropy. What we show is that the forty-point geometry, through its clock, produces exactly this structure — including the right signs, with nothing adjusted by hand.

**6.1 Two clocks in one table**

Let  $\theta$  be the highest root of  $E_8$  and  $\alpha_8$  the simple root with  $\theta = \omega_8$ . The tick line’s contact grading is the  $\alpha_8$ -coefficient  $a_8$  of a root; the striation’s  $|3|$ -grading is its  $\alpha_7$ -coefficient  $a_7$ . Counting roots by both at once gives Figure 17.



**Figure 17.** The 240 roots of  $E_8$  counted by contact degree  $a_8$  (rows) and clock degree  $a_7$  (columns). The Freudenthal 56 (solid outline) and the clock’s first-order 54 (dashed outline) share exactly one 27.

**Theorem 6.1 (the 56 does not extend the 54)****COMPUTATION**

The contact layer  $a_8 = -1$  sits at clock degrees 0, -1, -2, -3 with sizes 1, 27, 27, 1; the clock layer  $a_7 = -1$  sits at contact degrees 0, -1 with 27 each. The two share exactly one 27. The second 27 of the Freudenthal system is the clock’s *second-order* layer, and its two scalar “poles” are  $\alpha = e_{-\alpha_8}$  (clock degree 0) and  $\beta = e_{-(\theta-\alpha_8)}$  (clock degree -3, a member of the clock’s top doublet). `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

**A count that misled**

A proposal circulating in the programme read the 56 as “the clock’s 54 plus the 2 top directions”. The dimensions agree ( $54 + 2 = 56$ ), but the objects do not: the 56 contains only half of the 54. **REFUTED** (as an identification)

## 6.2 The symplectic form and the Heisenberg algebra

Brackets of two elements of the 56 land in the one-dimensional  $\mathfrak{g}_{-2} = \mathbb{C}e_{-\theta}$ :  $[x_i, x_j] = \Omega_{ij}e_{-\theta}$ . On the root basis,  $\Omega$  is a signed perfect matching:  $\alpha \leftrightarrow \beta$ , and  $X_k \leftrightarrow Y_k$  where  $X_k$  and  $Y_k$  carry the *same* 27-label. Its determinant is  $\pm 1$ , so it is unimodular over  $\mathbb{Z}$ . Since  $e_{-\theta}$  is central in  $\mathfrak{g}_{-1} \oplus \mathfrak{g}_{-2}$ , the two together form a 57-dimensional *Heisenberg algebra* — the same algebra that, in quantum mechanics, is generated by positions, momenta and  $i\hbar$ . COMPUTATION

## 6.3 The quartic, in Freudenthal's normal form

Define  $Q(x) = \frac{1}{6}$  coefficient of  $e_{-\theta}$  in  $\text{ad}_x^4 e_\theta$  for  $x \in \mathfrak{g}_{-1}$ .

**Theorem 6.2 (the tick line is a Freudenthal triple system)**

COMPUTATION

$Q$  is an integral polynomial with 1036 monomials (primitive coefficients  $\{1 : 28, \pm 2 : 378, \pm 4 : 630\}$ ), exactly invariant under all 126 root generators of the Levi  $E_7$ . Writing  $x = (\alpha, X, Y, \beta)$  with  $X, Y$  in the two 27's,

$$Q = (\alpha\beta + T(X, Y))^2 - 4\beta N(X) - 4\alpha N(Y) + 4T(X^\#, Y^\#)$$

where  $N$  is the repository's committed 45-triad  $E_6$  cubic,  $X_k^\# = \partial N / \partial X_k$ , and  $T(X, Y) = \sum_k t_k X_k Y_k$  with  $t_k = \Omega(X_k, Y_k) \Omega(\alpha, \beta)$ . No sign has been adjusted by hand. Moreover  $(X^\#)^\# = N(X) X$  and  $X \cdot X^\# = 3N(X)$ , so  $N$  is the norm of a cubic Jordan algebra (a split Albert algebra) over  $\mathbb{Q}$  [29]. `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

The theorem makes one structural point vivid: the cubic “history volume”  $N(X)$  of Section 5 enters the quartic *only* multiplied by a clock pole  $\beta$ .

### Black holes and qubits

In  $\mathcal{N} = 8$  supergravity the 56 electric and magnetic charges of a black hole transform under  $E_{7(7)}$ , and the Bekenstein–Hawking entropy is  $S = \pi\sqrt{|Q|}$  with  $Q$  Cartan's quartic invariant [26]; in the “magic”  $\mathcal{N} = 2$  supergravity built on the octonionic Albert algebra the same formula holds with  $E_{7(-25)}$ . Duff and Ferrara showed that the same quartic measures the tripartite entanglement of seven qubits [11], and Borsten, Duff and collaborators built a dictionary between black holes and qubit entanglement on it [5]. Theorem 6.2 places that quartic, with its signs, inside the forty-point geometry's clock. The two real forms named here —  $E_{7(7)}$  and  $E_{7(-25)}$  — are exactly the two that appear, for two different reasons, in Section 7. BRIDGE

## 6.4 Jordan frames are the forty-five triads

A *frame* of a cubic Jordan algebra is a decomposition of the identity into three orthogonal idempotents. On the committed basis, in each of four 27-layers, the 45 triads of the cubic are exactly the 45 strongly orthogonal triples of roots: the coordinate frames. (This corrects an older reading of “27 minimal idempotents”: 27 counts the coordinate rank-one directions;

the frames are the 45 triads.) For every coordinate idempotent, the Peirce decomposition is  $27 = 1 + 16 + 10$ . **COMPUTATION**

## 6.5 The 57-dimensional light cone

Now use the 57-dimensional Heisenberg algebra as a space of points  $p = (X, \tau)$ , with  $X \in \mathfrak{g}_{-1}$  and  $\tau$  the central coordinate, and move the top vector  $e_\theta$  around:  $v_p = \exp(\text{ad}(X + \tau e_{-\theta})) e_\theta$ .

### Theorem 6.3 (quasiconformal light cone)

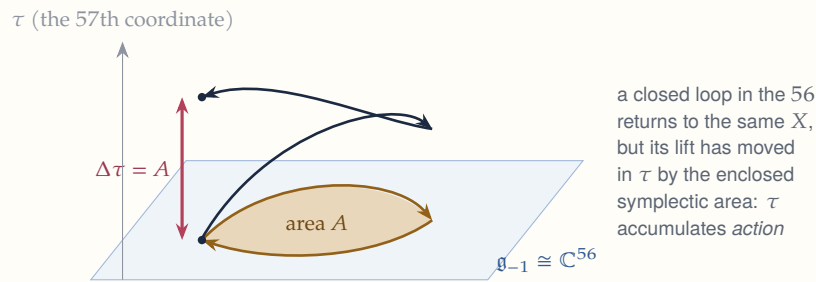
**COMPUTATION**

On the committed basis, symbolically:

- (i) the  $e_{-\theta}$ -component of  $v_p$  is  $\mathcal{N}(X, \tau) = \frac{1}{4}Q(X) - \tau^2$ ;
- (ii) the points compose by the Heisenberg law  $(X_1, \tau_1)(X_2, \tau_2) = (X_1 + X_2, \tau_1 + \tau_2 + \frac{1}{2}\Omega(X_1, X_2))$ ;
- (iii) the separation  $D(p, p') = \mathcal{N}(X - X', \tau - \tau' - \frac{1}{2}\Omega(X', X))$  equals  $B(v_p, v_{p'})/B(e_{-\theta}, e_\theta)$  for the Killing form  $B$ ;
- (iv) the Weyl reflection  $w$  exchanging  $e_\theta$  and  $-e_{-\theta}$  acts as an *inversion*  $\iota$ , and

$$D(\iota p, \iota p') = \frac{D(p, p')}{\mathcal{N}(p) \mathcal{N}(p')}.$$

This is the Günaydin–Koepsell–Nicolai quasiconformal realisation of  $E_8$  [19], now computed objectwise on the repository's basis. `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`



**Figure 18.** Why the 57th coordinate matters. In a Heisenberg group the central coordinate of a lifted path grows by the symplectic area its projection encloses. The quasiconformal separation  $D$  depends on  $\tau - \tau'$  only through this corrected difference  $\tau - \tau' - \frac{1}{2}\Omega(X', X)$ .

### What it means physically: a light cone, and its inversions

Ordinary Minkowski space has a *conformal* symmetry group,  $SO(4, 2)$ , larger than the Lorentz group; its most striking element is the inversion  $x \mapsto x/x^2$ , under which the interval between two events transforms as

$$(x' - y')^2 = \frac{(x - y)^2}{x^2 y^2}.$$

Theorem 6.3(iv) is exactly this law, with the Minkowski interval replaced by the quartic separation  $D$  on a 57-dimensional space. So  $E_8$  acts on the forty-point clock's Heisenberg algebra as a group of “conformal” symmetries of a quartic light cone. The central coordinate  $\tau$  plays the role of accumulated action: the same pattern met in Section 4, where the finite present forgot a common mode that the lifted history remembered. **BRIDGE** (the reading) **COMPUTATION** (the formulas)

*A single tick of a Hesse clock turns  $E_8$  into  $1+56+134+56+1$ . The 56 is Freudenthal's triple system, whose quartic — computed from the committed brackets with no sign adjusted — is the black-hole and seven-qubit invariant; and with one more coordinate it becomes a 57-dimensional light cone on which  $E_8$  acts by conformal-type inversions.*

#### Where this stops

The formulas are exact, but they are the formulas of the *complexified* (or split) structure. Whether the light cone is Lorentzian — one time direction, the rest space — depends on a choice of real form, which the finite field cannot see. That is the subject of Section 7. Nothing here supplies a black hole, a charge lattice of a physical theory, or dynamics.

**OPEN**

## 7 Real forms: where time's signature comes from

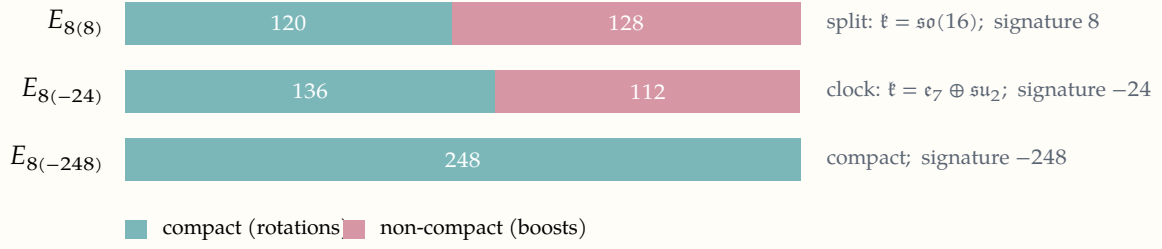
Every computation so far could have been done over the complex numbers — or over  $\mathbb{F}_3$ . But physics needs more: it needs to know which directions are *time-like* and which *space-like*. Mathematically, that is the choice of a *real form*.

#### In plain words

A complex Lie algebra is like a map drawn without saying which way is north. A real form chooses, for every direction, whether it behaves like a rotation (you come back to where you started — “compact”, like space) or like a boost (you never come back — “non-compact”, like the relation between observers moving at different speeds). The same complex  $E_8$  has three real forms, and they differ precisely in how many directions are rotation-like.

### 7.1 Three real forms of $E_8$

A real form is fixed by a *Cartan involution*, which splits the algebra into a compact part  $\mathfrak{k}$  (rotations) and a non-compact part  $\mathfrak{p}$  (boosts). The difference  $\dim \mathfrak{p} - \dim \mathfrak{k}$  is the *signature*.



**Figure 19.** The three real forms of  $E_8$ , drawn as the split  $248 = \dim \mathfrak{k} + \dim \mathfrak{p}$ . The committed “split” form of the repository is  $E_{8(8)}$ ; the Hesse clocks select  $E_{8(-24)}$ .

## 7.2 A sign law, and the height twist

On the committed Chevalley basis the Killing form obeys an exact law,

$$K(e_\alpha, e_{-\alpha}) = -60 (-1)^{\text{ht } \alpha},$$

where  $\text{ht } \alpha$  is the height of the root (the sum of its simple-root coefficients). From it one reads off the *compact* conjugation  $\sigma_c(e_\alpha) = (-1)^{\text{ht } \alpha} \overline{e_{-\alpha}}$ , verified to be an automorphism on all 30 628 basis pairs. The repository’s certified *split* involution  $\Theta$  turns out to be  $\sigma_c$  composed with the *height parity*  $(-1)^{\text{ht}} = \exp(i\pi\rho^\vee)$ , where  $\rho^\vee$  is the Weyl co-vector. **COMPUTATION** (`w33_pass10949_freudenthal_quasiconformal_clock_cone.py`; prior certificate `w33_e8_split_real_form_involution.py`)

## 7.3 The three tick lines are a Klein four-group

The three roots  $\theta$ ,  $-(\theta - \alpha_8)$  and  $-\alpha_8$  that define the tick line and its companions sum to zero, have pairwise products  $-1$ , and are orthogonal to  $E_6$ : they form the *external*  $A_2$  of  $E_6 \times A_2$ . Each defines a parity:  $(-1)^{a_8}$ ,  $(-1)^{a_7}$ ,  $(-1)^{a_7+a_8}$ .

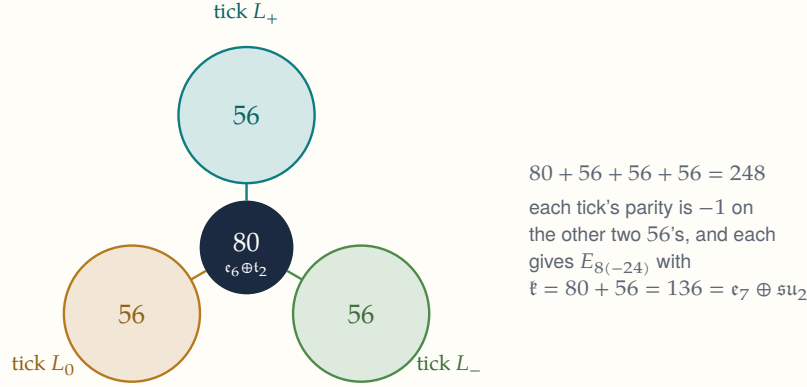
### Theorem 7.1 (tick lines give $E_{8(-24)}$ )

**COMPUTATION**

Composed with  $\sigma_c$ , each of the three tick-line parities defines the real form  $E_{8(-24)}$ . The three commute and form a Klein four-group whose simultaneous eigenspaces decompose

$$E_8 = \underset{\mathfrak{e}_6 \oplus \mathfrak{t}_2}{80} + 56 + 56 + 56,$$

one Freudenthal 56 per tick line. `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`



**Figure 20.** The three tick lines of a Hesse clock, and the decomposition of  $E_8$  under the Klein four-group of their parities.

#### 7.4 What each twist does to $E_7$ and $E_6$

| twist of $\sigma_c$                         | $E_8$        | contact Levi $E_7$     | $E_6$      |
|---------------------------------------------|--------------|------------------------|------------|
| tick $L_+$ : $(-1)^{a_8}$                   | $E_{8(-24)}$ | compact                | compact    |
| tick $L_0$ : $(-1)^{a_7}$                   | $E_{8(-24)}$ | $E_{7(-25)}$           | compact    |
| tick $L_-$ : $(-1)^{a_7+a_8}$               | $E_{8(-24)}$ | $E_{7(-25)}$           | compact    |
| height $(-1)^{\text{ht}}$ (committed split) | $E_{8(8)}$   | $E_{7(7)}$             | $E_{6(2)}$ |
| height $\cdot L_+$ / height $\cdot L_0$     | $E_{8(8)}$   | $E_{7(7)} / E_{7(-5)}$ | $E_{6(2)}$ |
| height $\cdot L_-$                          | $E_{8(-24)}$ | $E_{7(-5)}$            | $E_{6(2)}$ |

##### Two supergravities in one table

$E_{7(7)}$  is the hidden symmetry (“U-duality”) of maximal  $\mathcal{N} = 8$  supergravity in four dimensions, and  $E_{8(8)}$  that of maximal supergravity in three.  $E_{7(-25)}$  is the duality group of the “magic”  $\mathcal{N} = 2$  supergravity built on the octonions. The committed split form of the repository lands on the first; the Hesse clock lands on the second. The black-hole quartic of Section 6 is the entropy formula in both.

More precisely, the clock’s chain  $E_{6(-26)} \subset E_{7(-25)} \subset E_{8(-24)}$  (all three computed here, the first in Section 8) is exactly the chain of duality groups of the octonionic magic  $\mathcal{N} = 2$  supergravity of Günaydin, Sierra and Townsend in five, four and three space-time dimensions [21]:  $E_{6(-26)}$  acts on the 27 charges in five dimensions,  $E_{7(-25)}$  on the 56 in four, and  $E_{8(-24)}$  is the symmetry of the three-dimensional theory obtained by one more reduction. The split chain plays the same role for maximal supergravity.

BRIDGE

#### 7.5 The Lorentzian slice lives only in the clock real form

The contact Levi  $E_{7(-25)}$  has a Hermitian symmetric structure: on the 27-dimensional space  $\mathfrak{p}^+ = \{a_8 = 0, a_7 = +1\}$  the triple product

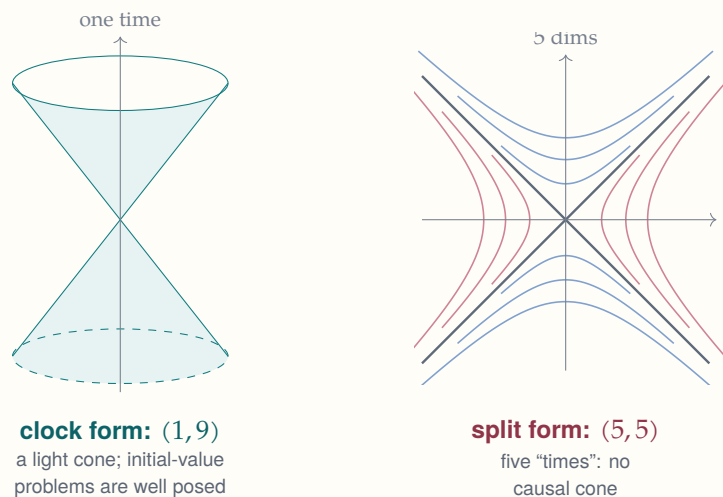
$$\{x, y, z\} = -[[x, \sigma_c y], z]$$

makes  $\mathfrak{p}^+$  a Hermitian Jordan triple system. A *tripotent* is an element with  $\{e, e, e\} = 2e$  (the normalisation used here).

**Theorem 7.2 (signature is a real-form datum)**
**COMPUTATION**

- (i) Under the clock conjugation all 27 root vectors of  $\mathfrak{p}^+$  are tripotents; under the committed split conjugation only 12 of 27 are.
- (ii) For the tripotent  $e$  of a triad frame,  $A = \{x : \{e, x, e\} = 2x\}$  with  $x \circ y = \frac{1}{2}\{x, e, y\}$  is a 27-dimensional commutative Jordan algebra with *positive-definite* trace form, inertia  $(27, 0, 0)$ : a Euclidean Jordan algebra of rank three — the Albert algebra  $H_3(\mathbb{O})$  [13].
- (iii) For a primitive idempotent of the frame, the Peirce spaces have dimensions  $1 + 16 + 10$ , and the rank-two determinant on the 10-dimensional Peirce-0 space has inertia  $(1, 9)$  — Lorentzian.
- (iv) In the committed split form the same construction gives inertia  $(5, 5)$ . Over  $\mathbb{F}_3$  the two quadrics are both of type  $O^+(10, 3)$  and cannot be told apart.

w33\_pass10949\_freudenthal\_quasiconformal\_clock\_cone.py



over  $\mathbb{F}_3$ : both are  $O^+(10, 3)$  — indistinguishable

**Figure 21.** The same Peirce-0 quadric in two real forms of  $E_8$ . Only the real form selected by the Hesse clock is Lorentzian.

**What it means physically: one time, not five**

Tegmark observed that only space-times with exactly one time direction support predictable physics: with several, the equations of motion are ultrahyperbolic and initial data do not determine the future [43]. Theorem 7.2 says the finite geometry does not decide between one time and five — over  $\mathbb{F}_3$  they are the same — but that the real form picked out by its own clock is the one-time, Lorentzian choice. The committed split form, chosen earlier for computational convenience, is the other one. **BRIDGE**

*Signature is not visible over a finite field. It is a choice of real form, and the Hesse clock's tick lines make that choice: they select  $E_{8(-24)} \supset E_{7(-25)}$ , and with it a Euclidean Albert algebra whose Peirce slice is a  $1 + 9$  light cone.*

## 8 The Albert algebra: ten dimensions, a spinor, and matter parity

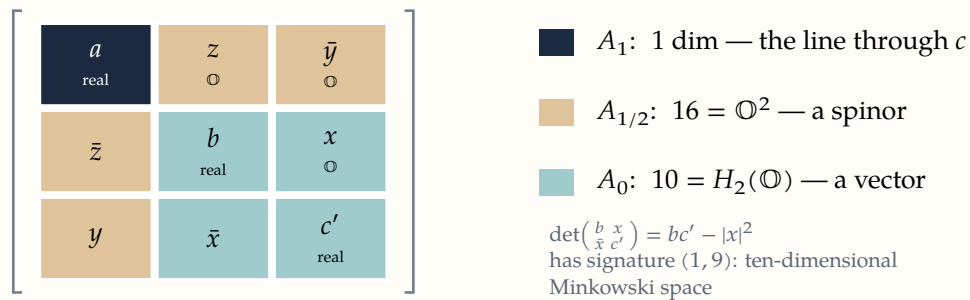
Section 7 produced, from the clock, a 27-dimensional Euclidean Jordan algebra. By the Jordan–von Neumann–Wigner classification there is exactly one such algebra that is “exceptional”: the *Albert algebra*  $H_3(\mathbb{O})$  of  $3 \times 3$  Hermitian matrices over the octonions. Since the 1930s it has been suspected of having something to do with quantum mechanics and particle physics; this section computes its symmetries on the committed basis and finds the Lorentz group of ten-dimensional space-time, a chiral spinor, and the repository’s own matter parity.

### In plain words

The *octonions* are the largest of the four number systems in which you can divide: the real numbers, the complex numbers, the quaternions, and the octonions, of dimensions 1, 2, 4, 8. Octonions do not even obey  $(ab)c = a(bc)$ , yet  $3 \times 3$  “Hermitian” matrices of them — real numbers on the diagonal, octonions above it, their conjugates below — form a consistent algebra of dimension  $3 + 3 \times 8 = 27$ . Its symmetries are the exceptional groups  $F_4$  and  $E_6$  [3]. Manogue, Dray and Wilson have recently described the Standard Model through octonionic  $E_8$  structures of this kind [35].

### 8.1 Anatomy of a $3 \times 3$ octonionic matrix

Choose the idempotent  $c = E_{11}$  (a 1 in the top-left corner). Every element of the algebra splits by how it behaves under multiplication by  $c$  — its *Peirce decomposition*, with eigenvalues  $1, \frac{1}{2}, 0$ :



**Figure 22.** The Peirce decomposition  $27 = 1 + 16 + 10$  of the Albert algebra with respect to a primitive idempotent. The lower-right  $2 \times 2$  block is a copy of ten-dimensional Minkowski space; the two octonions in the first row and column form a sixteen-component spinor.

That the  $2 \times 2$  Hermitian octonionic matrices form ten-dimensional Minkowski space, with the determinant as the Minkowski interval, is a classical observation (Sudbery; Kugo and

Townsend; Baez and Huerta), and it is the algebraic reason the superstring lives in ten dimensions [4, 30]. The new content is that the forty-point clock reaches it, on the committed basis.

## 8.2 The symmetries, computed

### Theorem 8.1 (the clock Albert algebra)

COMPUTATION

For the 27-dimensional clock algebra  $A$  of Theorem 7.2:

- (i) the derivations  $\text{Der}(A) = \text{span}[L_x, L_y]$  form a 52-dimensional Lie algebra with negative-definite trace form  $(0, 52)$ : the compact  $F_{4(-52)}$ ;
- (ii) the reduced structure algebra  $\text{str}_0(A) = \text{Der}(A) \oplus L(A_{\text{tr}=0})$  is 78-dimensional, obeys  $[D, L_x] = L_{Dx}$  for all  $52 \times 27$  pairs, and has inertia  $(26, 52)$ :  $E_{6(-26)}$ ;
- (iii) the derivations fixing  $c$  form a compact 36-dimensional  $\mathfrak{spin}(9)$ ; adding the nine “boosts”  $L_y$  ( $y \in A_0$ , orthogonal to  $e - c$ ) gives a 45-dimensional algebra that preserves  $A_0$  and its determinant of inertia  $(1, 9)$ , and acts faithfully: it is  $\mathfrak{so}(1, 9)$ ;
- (iv) on  $A_{1/2}$  the commutant of  $\mathfrak{so}(1, 9)$  is one-dimensional: the 16 is absolutely irreducible — the Majorana–Weyl spinor of  $\mathfrak{so}(1, 9)$ .

w33\_pass10950\_clock\_albert\_lorentz\_spinor.py

|                           | ordinary space-time<br>$H_2(\mathbb{C})$ : 4 dims | superstring space-time<br>$H_2(\mathbb{O})$ : 10 dims | exceptional space-time<br>$H_3(\mathbb{O})$ : 27 dims |
|---------------------------|---------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|
| rotations (automorphisms) | $SO(3)$                                           | $\text{Spin}(9)$ ✓                                    | $F_{4(-52)}$ ✓                                        |
| Lorentz (norm-preserving) | $SO(1, 3)$                                        | $\text{Spin}(1, 9)$ ✓                                 | $E_{6(-26)}$ ✓                                        |
| conformal                 | $SO(2, 4)$                                        | $SO(2, 10)$                                           | $E_{7(-25)}$ ✓                                        |
| quasiconformal            | —                                                 | —                                                     | $E_{8(-24)}$ ✓                                        |

✓ computed in this paper on the committed  $E_8$  basis, in the real form selected by the Hesse clock (Passes 10949–10950); unmarked cells are classical.

**Figure 23.** The ladder of space-times. A Jordan algebra of Hermitian matrices is a space-time; its determinant is the interval. For  $2 \times 2$  complex matrices it is ordinary Minkowski space, for  $2 \times 2$  octonionic matrices the ten-dimensional space of the superstring, and for  $3 \times 3$  octonionic matrices the 27-dimensional “exceptional space-time” of Günaydin [22]. Each has rotations, a Lorentz group, a conformal group and (for the last) a quasiconformal group. The Hesse clock’s real form of  $E_8$  produces the whole right-hand column, and the middle column inside it.

#### What it means physically: the clock picks the division algebra

The last row of Freudenthal’s “magic square” of Lie algebras can be built from the octonions in two ways: from the division algebra  $\mathbb{O}$ , giving the chain  $F_{4(-52)} \subset E_{6(-26)} \subset E_{7(-25)} \subset E_{8(-24)}$  of automorphism, Lorentz, conformal and quasiconformal groups of  $H_3(\mathbb{O})$ , or from the split octonions, giving the split forms. The Hesse clock selects

the first: the *division-algebra* octonions, the ones with a positive norm, the ones that make  $H_2(\mathbb{O})$  a genuine Minkowski space. The repository's earlier "split" choice lands on the other side. **COMPUTATION** (the four real forms) **BRIDGE** (the reading)

### 8.3 Down to four dimensions

Choose a further splitting of the 10 into 4 + 6: time  $u = e - c$ , one frame direction, and a complex line inside the octonions supplied by the complex structure of  $\mathfrak{p}^+$  itself. The subalgebra of  $\mathfrak{so}(1, 9)$  preserving the 4 and the 6 is 21-dimensional,  $\mathfrak{so}(1, 3) \oplus \mathfrak{so}(6) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{su}(4)$ , and its commutant on the 16 is two-dimensional and contains a complex structure. Hence

$$16 = (2, 4) \oplus \overline{(2, 4)}.$$

**COMPUTATION** (`w33_pass10950_clock_albert_lorentz_spinor.py`)

#### Two readings of the same branching

A ten-dimensional chiral spinor, seen by a four-dimensional observer, becomes four left-handed Weyl spinors transforming as the 4 of  $SU(4)$ . If  $SU(4)$  is read as the rotation group of six hidden dimensions, this is the fermion content of maximally supersymmetric Yang–Mills theory. If it is read as Pati–Salam colour, with lepton number as the fourth colour, it is one generation of quarks and leptons. The branching is exact; which reading nature uses is not decided by it. **BRIDGE**

### 8.4 Matter parity is a rotation by $2\pi$

The repository carries an independent certificate for a "matter charge"  $Q_\psi$  on the 27: it splits  $27 = 16_1 + 10_{-2} + 1_4$ , and on the 45 triads the charges fall into the pattern  $40 \times (-2, 1, 1) + 5 \times (-2, -2, 4)$ . Matter parity is  $(-1)^{Q_\psi}$ : odd on the 16, even on the 10 and 1. In the Minimal Supersymmetric Standard Model a parity of exactly this kind is what forbids the proton from decaying through dimension-four operators.

#### Theorem 8.2 (matter parity is the Peirce symmetry)

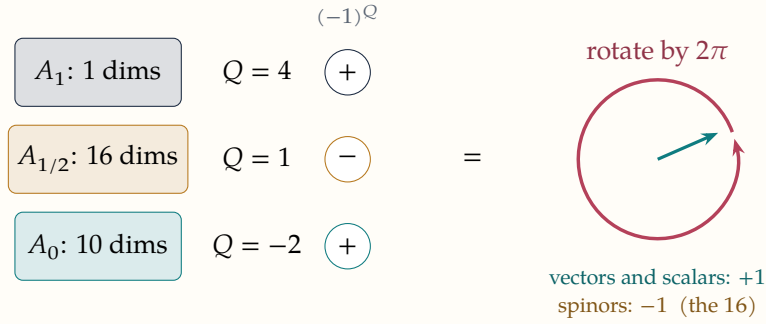
**COMPUTATION**

For every one of the 27 coordinate idempotents  $c$ , the Peirce charge

$$Q = 6L_c - 2 \quad (\text{eigenvalue } 4 \text{ on } A_1, 1 \text{ on } A_{1/2}, -2 \text{ on } A_0)$$

reproduces the certified spectrum of  $Q_\psi$ , is conserved on all 45 triads, and gives exactly the pattern  $40 \times (-2, 1, 1) + 5 \times (-2, -2, 4)$ . Its parity  $(-1)^Q$  is the Peirce symmetry  $U_{2c-e}$ , an automorphism of  $A$  that is +1 on  $A_1 \oplus A_0$  and -1 on the spinor  $A_{1/2}$ . For a single-plane rotation generator  $D \in \mathfrak{spin}(9)$ ,  $\exp(2\pi D) = U_{2c-e}$  (numerical agreement  $2.7 \times 10^{-15}$ ). `w33_pass10950_clock_albert_lorentz_spinor.py`

#### What it means physically



**Figure 24.** Matter parity in the clock Albert algebra: the sign that a full turn gives a spinor.

A rotation by  $2\pi$  returns every vector to itself, but multiplies every spinor by  $-1$ : the mathematical root of the difference between fermions and bosons. In  $SO(10)$  grand unification it is known that matter parity can be realised by exactly such a central element of  $\text{Spin}(10)$ , acting as  $-1$  on the matter **16** and  $+1$  on the Higgs **10** (cf. [36]). Theorem 8.2 is the objectwise version of that fact inside the forty-point clock: the repository's matter parity *is* the  $2\pi$  rotation of the  $\text{Spin}(9)$  fixing a primitive idempotent. The same  $27 = 1 + 10 + 16$  that a GUT reads as singlet, Higgs and matter is, in the clock real form, scalar, Lorentz vector and Weyl spinor. **BRIDGE** (reading) **COMPUTATION** (identity)

### 8.5 The clock's fourth tick is the same sign

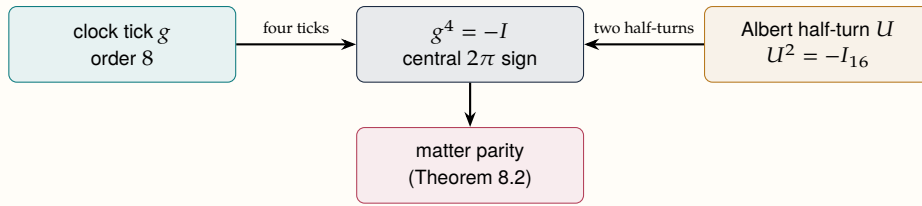
Section 4 met a clock whose fourth tick is a sign flip. The other research track has since shown that this is not an analogy but one spin structure, seen at both ends of the ladder. We cite their results (certificates named) rather than re-derive them.

#### Theorem 8.3 (clock, spin and matter parity; Passes 10951–10959)

**COMPUTATION**

- (i) The negacyclic clock tick lifts uniquely to  $g = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in GL(2, 3)$  of order eight, with  $g^4 = -I$ ; under an explicit isomorphism  $SL(2, 3) \cong 2T \subset \text{Spin}(3)$  the element  $-I$  is the quaternion  $-1$ , the central  $2\pi$  spin sign. `w33_pass10951_clock_pin_spin_central_sign_bridge.py`
- (ii) As a signed permutation of the four clock coordinates, determinant-odd ticks exchange the two  $D_4$  half-spin sheets and determinant-even ticks preserve them. `w33_pass10955_d4_halfspin_clock_bridge.py`
- (iii) On the Albert Peirce 16, the  $\text{Spin}(8)$  fixing a spatial axis splits it as  $8_s + 8_c$ ; an exact rational  $\text{Spin}(9)$  half-turn  $U$  exchanges them, with  $U^2 = -I_{16}$  — the matter-parity sign of Theorem 8.2 — and  $U^4 = I_{16}$ . `w33_pass10956_albert_spin8_halfspin_normalizer.py`, `w33_pass10958_exact_albert_mu4_missing_mode_intertwiner.py`
- (iv) One Albert 16 cannot carry the whole order-eight clock as a  $GL(2, 3)$  representation; the minimal carrier is  $16 \oplus 16$ . `w33_pass10959_albert_clock_extension`

obstruction.py

**Where this stops**

The  $\mathfrak{so}(1, 9)$  here is the stabiliser of an idempotent inside  $E_{6(-26)}$ , an *internal* symmetry of the algebra; it is not derived to be the Lorentz group of physical space-time, and the  $2\pi$  rotation is a rotation in that internal  $\text{Spin}(9)$ . The identification of  $Q_\psi$  with  $6L_c - 2$  holds up to the  $W(E_6)$ -transitive choice of coordinate idempotent. Neither chirality selection, masses nor gauge dynamics follow.

*The exceptional ladder closes: from one qutrit, through  $W(3, 3)$ ,  $E_8$ , a Hesse clock and its real form, to the Albert algebra  $H_3(\mathbb{O})$  — whose automorphisms are  $F_4$ , whose structure group is  $E_{6(-26)}$ , and inside which sit ten-dimensional Minkowski space, its Lorentz algebra  $\mathfrak{so}(1, 9)$ , a Majorana–Weyl spinor, and, as a rotation by  $2\pi$ , the matter parity that protects the proton.*

## 9 The Standard Model on forty points

Sections 2–8 established exact structure. This section asks the question the programme exists to answer: how much of the actual physics of our universe does that structure carry? We give the answer at three levels — what the finite geometry itself contains, what a string compactification built on it produces, and what is still missing — and we end with a score-card.

***The string route, completed.** Nine families of heterotic orbifolds carry the forty-point twist ( $\mathbb{Z}_3$ ,  $\mathbb{Z}_3 \times \mathbb{Z}_3$ ,  $\mathbb{Z}_6$ -I,  $\mathbb{Z}_6$ -II,  $\mathbb{Z}_{12}$ -I,  $\mathbb{Z}_2 \times \mathbb{Z}_6$ -I,  $\mathbb{Z}_2 \times \mathbb{Z}_6$ -II,  $\mathbb{Z}_3 \times \mathbb{Z}_6$ ,  $\mathbb{Z}_6 \times \mathbb{Z}_6$ ). They produce 560 inequivalent three-generation Standard Models. On the selection rules the string's own worldsheet establishes, not one has a supersymmetric vacuum with a stable proton and a clean spectrum, and every escape we could find — other discrete symmetries, rotations of the hidden dimensions, heavy superpartners, flavour suppression, hidden-sector condensates — fails for a stated, computed reason. In eight families the mechanism is the same: the one field that can cancel the anomalous  $U(1)$ 's Fayet–Iliopoulos term has the quantum numbers of a right-handed sneutrino, and its vacuum value switches on the couplings that make the proton decay. In the ninth,  $\mathbb{Z}_3 \times \mathbb{Z}_3$ , the parity survives and the vacua are supersymmetric — but every one of them carries six massless particles of electric charge  $1/3$ , forced by a counting argument no hidden dynamics removes. The forty points host the Standard Model; this route does not make it our universe.*

|          |                                                                             |
|----------|-----------------------------------------------------------------------------|
| EXACT    | Quantum measurement structure (qutrit Paulis, contexts, Clifford group)     |
| EXACT    | Exceptional algebras $E_6, E_7, E_8, F_4$ and the Albert algebra            |
| EXACT    | A finite light cone; clocks; one Lorentzian real form $\mathfrak{so}(1, 9)$ |
| HOSTED   | Gauge group $SU(3) \times SU(2) \times U(1)$                                |
| HOSTED   | Three generations                                                           |
| EXACT    | One chiral generation as a spinor (16 of $\text{Spin}(1, 9)$ )              |
| NO-GO    | Which chirality: exchanged only by antiunitary (CPT-type) symmetries        |
| EXCLUDED | Matter parity in a supersymmetric vacuum (seven orbifold families)          |
| CLOSED   | A realistic vacuum of the heterotic orbifold route                          |
| OPEN     | Masses, mixing angles, coupling constants                                   |
| OPEN     | Gravity, a dynamical space-time, the cosmological constant                  |

**Figure 25.** Scorecard. EXACT: a theorem or exact computation in this paper. HOSTED: realised in an explicit construction built on the geometry, not forced by it. EXCLUDED: exhaustively ruled out in a stated scope (Section 9.4). NO-GO: proved impossible from inside. OPEN: not addressed by any exact result.

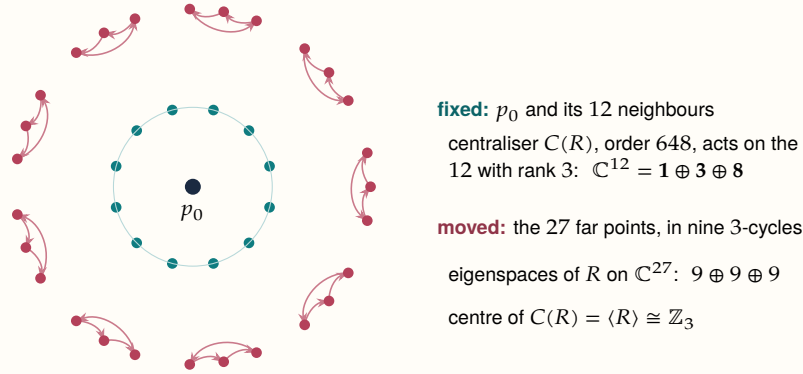
## 9.1 What the finite geometry itself contains

**The anatomy of one symmetry.** Pick a point  $p_0$  of  $W(3, 3)$  and the *transvection*  $R$  centred on it: the symplectic map  $x \mapsto x + \langle x, v \rangle v$ , of order three. It fixes the 13 points of the plane  $p_0^\perp$  —  $p_0$  and its 12 neighbours — and permutes the other 27 points in nine orbits of three.

### Proposition 9.1 (the transvection spine)

COMPUTATION

The centraliser  $C(R)$  of a transvection has order 648 (it is the point stabiliser  $3^{1+2}:2A_4$ ) and centre  $\langle R \rangle \cong \mathbb{Z}_3$ . Its permutation module on the 12 neighbours of  $p_0$  decomposes as  $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8}$ . On the 27 far points,  $R$  has eigenspaces  $9 \oplus 9 \oplus 9$ . `bt886_standard_model_`



**Figure 26.** One transvection of  $W(3,3)$ . Its centraliser acts on the twelve fixed neighbours with the dimensions of the Standard Model’s gauge algebra,  $1 + 3 + 8$ , and it splits the twenty-seven moved points into three equal eigenspaces.

spine.py

### How far this goes

The dimensions 1, 3, 8 are those of the Lie algebras  $\mathfrak{u}(1)$ ,  $\mathfrak{su}(2)$ ,  $\mathfrak{su}(3)$ , and the corpus has sometimes read Proposition 9.1 as “the Standard Model gauge group follows with zero free parameters”. That is an over-read. The  $1 \oplus 3 \oplus 8$  here are representations of a *finite* group of order 648; they are not Lie algebras, they do not come with gauge fields, and nothing fixes their couplings. The honest statement is that the local symmetry of one point of the geometry has the *shape* of the Standard Model’s gauge algebra.

BRIDGE

**Three from the Steinberg module.** The first homology of  $W(3,3)$  (its 40 points, 240 edges and 160 triangles in lines) is 81-dimensional and irreducible under  $\mathrm{PSp}(4,3)$ : it is the *Steinberg module*, the most rigid representation a group of Lie type has, and the logical space of the repository’s  $[[240, 81, 3]]_3$  quantum code. **COMPUTATION** (bt861\_code\_register\_is\_steinberg.py)

**Theorem 9.2 (every order-three symmetry splits matter  $27 + 27 + 27$ )**

**THEOREM**

Let  $g \in \mathrm{PSp}(4,3)$  have order three. Then  $g$  acts on the 81-dimensional Steinberg module with eigenvalues  $1, \omega, \omega^2$ , each with multiplicity exactly 27.

*Proof.* The Steinberg character vanishes on every element whose order is divisible by the defining characteristic  $p = 3$  (a theorem of Steinberg; see [24]). So  $\chi(g) = \chi(g^2) = 0$ , and the multiplicity of each eigenvalue  $\omega^j$  is  $\frac{1}{3}(\chi(1) + \omega^{-j}\chi(g) + \omega^{-2j}\chi(g^2)) = 81/3 = 27$ .  $\square$

### Three is forced, but not selected

Theorem 9.2 holds for *every* element of order three, so it produces a threefold splitting without choosing one. And the string-theory track (below) found that in the explicit

compactifications the number of families is the number of complex planes of the internal torus, not a property of  $W(3,3)$ . We therefore record the Steinberg splitting as a structural host for three generations, not a derivation of them. **BRIDGE**

**The qutrit register is  $SU(9)$ , not  $E_6$ .** One might hope that the  $E_6$  of grand unification acts directly on qutrit registers.

**Theorem 9.3 (no Pauli  $E_6$ )**

**THEOREM** **COMPUTATION**

Conjugation by the three-qutrit Pauli group splits  $\mathfrak{gl}(27)$  into 729 distinct one-dimensional character spaces, so every conjugation-stable subalgebra is spanned by Pauli operators. A simple Pauli-spanned Lie algebra has dimension  $3^a - 3^b$ , i.e. one of 8, 24, 72, 80, 216, 240, 648, 720, 728. Since  $\dim \mathfrak{e}_6 = 78$  is not among them,  $E_6$  cannot act on a qutrit register through Pauli-stable generators. `Holotrade 1cdelf1`

The natural gauge algebra of the qutrit register is instead  $\mathfrak{sl}_9$  — the 80 of Section 5 — which is exactly what appears in the string construction.

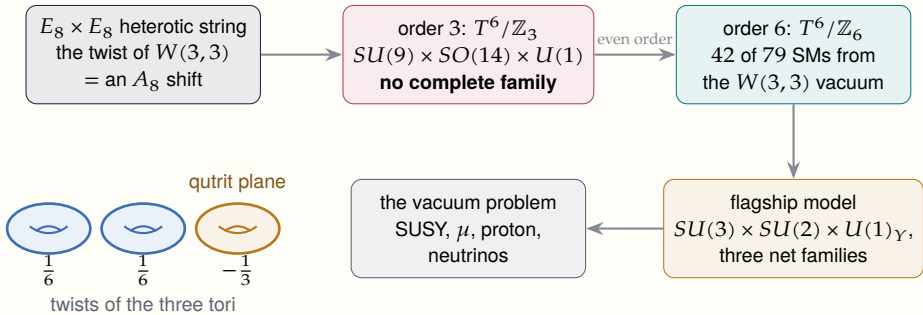
9.2 The string that carries the geometry

The most developed attempt to turn this structure into physics uses the  $E_8 \times E_8$  heterotic string compactified on an orbifold [10]: six extra dimensions curled up as three two-dimensional tori, divided by a finite rotation group. Its advantage is that everything is computable exactly, with a public tool, the ORBIFOLDER [37].

**Theorem 9.4 ( $W(3,3)$  lives in the twisted sector of  $E_8$ )**

**COMPUTATION**

For the  $\mathbb{Z}_3$  twist of the  $E_8$  lattice vertex algebra, the commutator form of the twisted-sector Pauli group equals twice the lattice form on every rung, and the resulting commutation geometry is  $W(3,3)$ . The twist is anomalous on a single  $E_8$  (ground-state energy  $\rho = 4/9$ ), consistent exactly when  $3 \mid k$  on  $E_8^k$ ; the  $E_8^3$  orbifold is the Niemeier lattice vertex algebra  $V_{A_8^3}$  with algebra  $\mathfrak{sl}_9^3$ . `Holotrade 68df33f`



**Figure 27.** The heterotic route. At order three no Wilson-line breaking completes a Standard-Model family (a theorem); at even order the  $W(3,3)$  vacuum is the most productive source of Standard Models; the open problem is a realistic vacuum. The third torus, the unique order-three plane, is where the Higgs lives.

**Order three fails, for a reason.** In the  $T^6/\mathbb{Z}_3$  orbifold built on this twist, Wilson lines produce three-family  $SU(5)$  and  $SO(10)$  models; but an order-three Wilson line keeps at most one of  $Q, u^c, e^c$  in each **10**, so no breaking completes a Standard-Model family. The general law: for a  $\mathbb{Z}_N$  orbifold,  $u^c$  and  $e^c$  survive together only when  $N$  is even. This explains, from first principles, why heterotic model builders moved to even-order orbifolds. An exhaustive census of 127 980 classes confirms it, and 64 000 fresh Wilson-line draws on the unique modular-invariant  $W(3,3)$  twist of  $T^6/\mathbb{Z}_3$  find no Standard Model (P10979). The law is a statement about a *cyclic* group: two independent order-three twists,  $T^6/\mathbb{Z}_3 \times \mathbb{Z}_3$ , evade it, and there the  $W(3,3)$  twist gives 13 Standard Models — consistent with the known  $\mathbb{Z}_3 \times \mathbb{Z}_3$  landscape of model builders [38]. **THEOREM** **COMPUTATION** (Holotrade ed1eaca, a6e1c69, a779053)

**Order six succeeds.** Among the 58  $\mathbb{Z}_6$ -I and 61  $\mathbb{Z}_6$ -II shifts, the  $W(3,3)$  vacuum  $SU(9) \times SO(14) \times U(1)$  is the most productive class: of 79 inequivalent Standard Models found in it, 42 come from the  $\mathbb{Z}_6$ -I  $W(3,3)$  vacuum. The flagship has gauge group  $SU(3)_C \times SU(2)_L \times U(1)_Y$  with three net families and no anomalies, and at one fixed point the local gauge group is  $SU(9)$  with all four simple roots of the Standard Model's  $SU(5)$  inside it: the Standard Model's unification group descends from the twist's own  $\mathfrak{sl}_9$ . **COMPUTATION** (Holotrade 5b3f3ad)

#### Measured regularities of the $W(3,3)$ Standard Models

**COMPUTATION**

- **The breaking is an entangling gate.** The order-three Wilson line's  $\mathfrak{sl}_9$  holonomy is of two-qutrit controlled-Z type (5,2,2) in 45 of 45  $\mathbb{Z}_6$ -I models, against a 33% background (chance  $\approx 2.6 \times 10^{-22}$ ). (d772153)
- **Two commuting holonomies cut out  $SU(3) \times SU(2)$ :** in all 25 models with parity-type  $\theta^3$ , against a 30% background ( $\approx 8.6 \times 10^{-14}$ ). (94df376)
- **A heavy top is allowed:** 57 of 60 models admit a renormalisable top Yukawa coupling. (93b34e1)
- **The Higgs sits alone in the qutrit plane:** in all 87  $\mathbb{Z}_6$ -I models the weak doublets come only from the third torus (the unique order-three plane) and colour triplets only from the other two; this one fact gives both the top coupling and **doublet-triplet splitting**, solved in 55 of 87 models by a missing-partner mechanism. (3caf15e, a6f1cae)

#### What it means physically

The Standard Model's most puzzling structural feature — that the Higgs doublet is light while its would-be colour-triplet partner, which would make protons decay, is absent — is here a consequence of geometry: the Higgs lives on the one internal plane whose twist has order three, the “qutrit plane”, and that plane carries no colour triplet. And the symmetry breaking that produces the Standard Model is, in qutrit language, an entangling gate. These are measured regularities of an explicit string construction, with controls; they are not theorems about all constructions.

### 9.3 Where the string route stands

A model is not a universe until it has a stable vacuum with the right low-energy properties. Here the honest record is mixed, and it has been corrected more than once.

| Requirement           | Current exact status                                                                                                                                                                                                                          | Badge                            |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Supersymmetric vacuum | In the doublet–triplet-solving vacuum, $F$ - and $D$ -flatness conflict: one $U(1)$ direction gives every mass singlet a positive charge. Gordan’s alternative (“ $D$ -flat or empty superpotential, never both”) verified on all 215 models. | <b>REFUTED</b><br>(that vacuum)  |
| $D$ -flat directions  | Exist in 23 of 87 ( $\mathbb{Z}_6$ -I) and 105 of 128 ( $\mathbb{Z}_6$ -II); 83 of the 211 anomalous models have none.                                                                                                                        | <b>COMPUTATION</b>               |
| Proton stability      | No supersymmetric singlet vacuum preserves a matter parity built from the gauge torus and the space group: 0 of 215 (Section 9.4). Baryon triality is impossible in all 215. Heavy superpartners would need $m_t \leq 171$ GeV.               | <b>COMPUTATION</b><br>(excluded) |
| Parity from rotations | Appeared in one model under the old per-plane rule; with the established rules (Bizet et al.) it closes: 0 of 128.                                                                                                                            | <b>COMPUTATION</b><br>(excluded) |
| The $\mu$ term        | Arises at order 3 or 4 in every model; none is $\mu$ -free through order five. Must come from a vacuum symmetry.                                                                                                                              | <b>OPEN</b>                      |
| Neutrino masses       | In the splitting-solving models, one to two orders of magnitude too light.                                                                                                                                                                    | <b>OPEN</b>                      |

Figure 28 gathers the whole search in one picture. Each band is an exact filter; the number is how many models or vacua pass it, and the tag names the computation that decided it.

### 9.4 The Fayet–Iliopoulos term forces a right-handed sneutrino

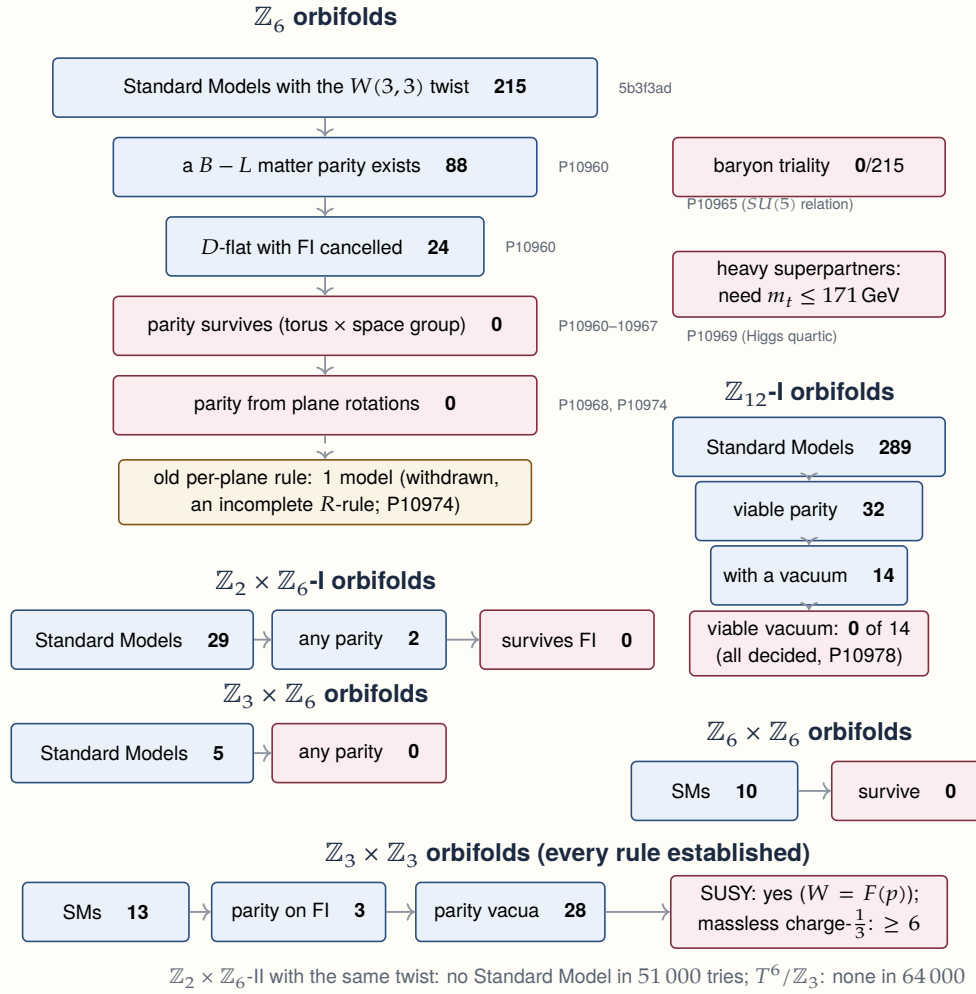
Every model of the class has an anomalous  $U(1)$ , whose Fayet–Iliopoulos term must be cancelled by giving vacuum values to some singlet fields ( $D$ -flatness), with  $\langle n \rangle / M_s \approx 0.2$ – $0.4$ . Proton stability needs a *matter parity* — a  $\mathbb{Z}_2$  under which quarks and leptons are odd and Higgs fields even — to survive that condensation, so the fields that condense must all be even. The earlier record found every tested  $D$ -flat direction contained an odd singlet, but it had *sampled* 60 choices of  $B - L$  per model and said explicitly that this was not a proof; the raw spectra needed to finish it were in no repository. They were recovered for this paper from the original scan and frozen as an exact-rational ledger of all 215 models.

#### Theorem 9.6 (no matter parity survives the FI term)

**COMPUTATION**

Over the *entire* rational solution set  $B - L = x_0 + Nt$  of each model:

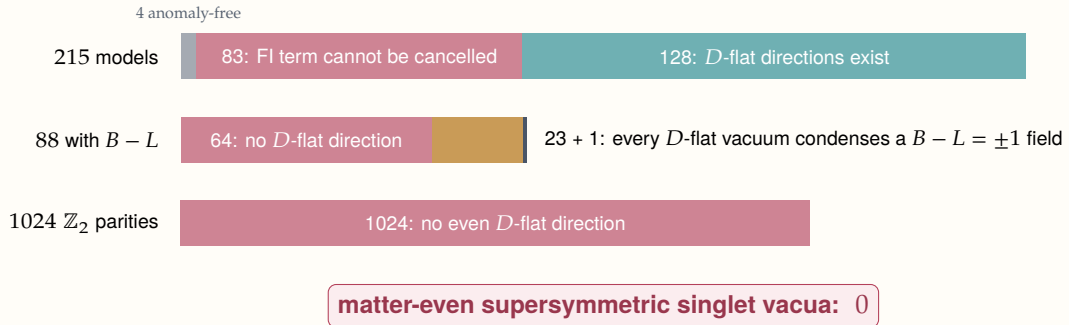
- (i) 64 of the 88 models with a  $B - L$  direction have no  $D$ -flat singlet direction at all (an exact Farkas vector certifies that the FI term cannot be cancelled);
- (ii) in the other 23  $\mathbb{Z}_6$ -I models, the singlets that are even for *some* choice of  $B - L$  are never  $D$ -flat (one exact Farkas vector each); the only singlets that are never even carry  $3(B - L) = 3$ , i.e.  $B - L = +1$ , fixed by the Standard Model itself, and every  $D$ -flat direction uses one;



**Figure 28.** The vacuum gauntlet. Blue: filters passed; red: dead ends; gold: a door that looked open and was withdrawn. The last row is the one family where the parity survives: its vacua are supersymmetric, and a hidden-representation index leaves six massless charge- $\frac{1}{3}$  states in each (P10979, P11024). Every count is exact (Farkas certificates, Smith normal forms, lattice tests and integer programs), and each is regenerated by the script named in the ledger.

- (iii) in the one remaining ( $\mathbb{Z}_6$ -II) model, all 304 FI-cancelling extreme rays of the  $D$ -flat cone (3334 rays, computed exactly) are obstructed by an integer (Smith normal form) parity certificate; its never-even singlets carry  $B - L = \pm 1$ .

Independently, all 1024  $\mathbb{Z}_2$  characters of the charge lattice that are odd on every  $q, u^c, d^c, e^c$  give no even  $D$ -flat direction. Controls: the same exact machinery finds the  $23 + 105 = 128$   $D$ -flat models and the 88  $B - L$  models of the earlier record. `w33_pass10960_matter_even_dflat_closure.py`



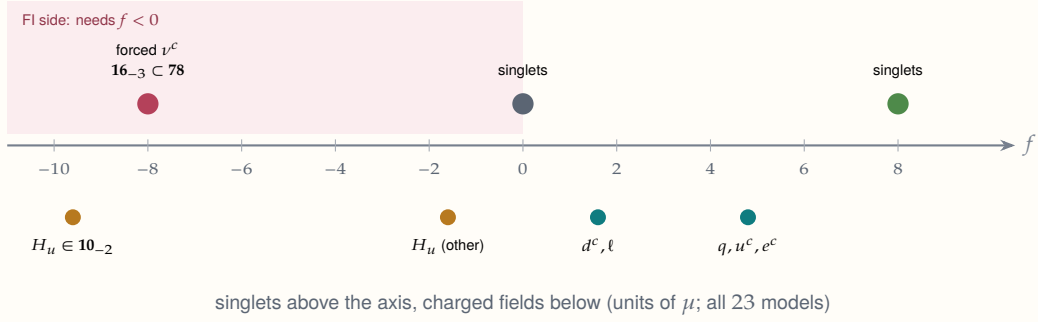
**Figure 29.** The exhaustive closure of Theorem 9.6. Every one of the 215 models is accounted for; no branch leaves a matter-parity-preserving supersymmetric vacuum in singlet directions.

#### What it means physically: the sneutrino that breaks R-parity

A singlet with  $B - L = +1$  has the quantum numbers of a right-handed (s)neutrino. Theorem 9.6 says the anomalous  $U(1)$  forces such a field to condense. Martin's criterion [36] is exactly relevant here: matter parity survives the breaking of  $B - L$  only if the fields that break it have even  $3(B - L)$  — which is why realistic  $SO(10)$  models break  $B - L$  with **126**-type fields ( $B - L = \pm 2$ ) rather than **16**-type ones ( $B - L = \pm 1$ ). In this class the FI term leaves no choice:  $B - L$  is broken by one unit, and the baryon- and lepton-number-violating couplings that matter parity was protecting are no longer forbidden by any symmetry, at the scale of the singlet vacuum values.

**Why: one  $E_6$ -shaped charge.** The obstruction has a single mechanism. In each of the 23 models there is an exact certificate  $f$  — a combination of the anomalous and ordinary  $U(1)$  charges — with a prescribed physical shape:  $f = \frac{24}{5}$  on every  $q, u^c, e^c$  and  $f = \frac{8}{5}$  on every  $d^c, \ell$  (in units of a positive scale), and  $f \geq 0$  on every singlet that could be even. On the singlets  $f$  takes only the values  $-8, 0$  and  $8$ , and the negative ones are exactly the forced  $B - L = +1$  fields. Because  $D$ -flatness makes the  $f$ -charge-weighted sum of condensates equal the (negative) FI contribution, one of them must condense. On the families  $f = 4Q_\psi - \frac{4}{5}Q_\chi$ , the  $U(1)$  combination of  $E_6 \supset SO(10) \times U(1)_\psi \supset SU(5) \times U(1)_\chi \times U(1)_\psi$  with  $Q_\chi = 4Y - 5(B - L)$ ; the value  $-8$  at  $(B - L, Y) = (1, 0)$  then identifies the forced field as the right-handed-sneutrino direction of the matter-odd **16**<sub>-3</sub> in the adjoint **78** of  $E_6$ . **COMPUTATION** (`w33_pass10964_e6_shaped_fi_mechanism.py`) **BRIDGE** (the  $E_6$  reading)

**The hidden sector does not rescue it.** Theorem 9.6 covers singlets of every nonabelian group. Fields that are Standard-Model singlets but charged under the *hidden* gauge group



**Figure 30.** The canonical certificate  $f = 4Q_\psi - \frac{4}{5}Q_\chi$  on the families. Among singlets, only the right-handed-sneutrino direction of the  $E_6$  adjoint's  $16_{-3}$  sits on the side that can balance the Fayet–Iliopoulos term.

can also condense, and their parity can be compensated by a hidden gauge element. Pass 10962 attacks this loophole in tiers:

- **exact, for every hidden direction:** a parity-preserving vacuum must be  $D$ -flat for the hidden maximal torus with component-wise parity (up to a hidden torus element); for 32 of the 88 models even this relaxation has an exact Farkas certificate;
- **exact over explicit invariant generators:** by the Luty–Taylor and Kempf–Ness correspondence a parity-preserving  $D$ -flat vacuum exists iff an FI-cancelling product of matter-even hidden-group invariants does; over singlets, all quadratic invariants,  $SU(N)$  baryons and  $SU(5)$  cubics, the other 55 models close by exact Farkas vectors, and the last by a parity obstruction;
- the four anomaly-free models (no FI term) admit no matter parity at all.

**COMPUTATION** (`w33_pass10962_hidden_sector_dflat_tiers.py`)

#### Scope

For the 56 models not closed at the first tier, completeness of the invariant generators is not proved (they contain bi-fundamental,  $SU(4)$  **6** or  $SU(5)$  **10** hidden fields), so for them the hidden-sector statement is “no rescue among the standard invariants” rather than a theorem. Whether higher-order selection rules suppress the regenerated couplings in a particular model is not addressed. [OPEN](#)

**The other discrete guardian is excluded too.** Matter parity is not the only discrete symmetry that can keep the proton alive. *Baryon triality*  $B_3 = \exp(2\pi i(B - 2Y)/3)$  (Ibáñez–Ross) forbids the baryon-number violating couplings  $u^c d^c d^c$  and  $qqql$  while allowing the lepton-number violating ones. Could the class carry it instead? As a gauge symmetry it would have to be a  $\mathbb{Z}_3$  character of the  $U(1)$  charge lattice, equal to  $B_3$  on every quark and lepton up to a hypercharge rotation. Pass 10965 shows that this linear system over  $\text{GF}(3)$  is inconsistent in all 215 models, and names the reason: in 214 of them single copies obey the exact identity

$$Q(u^c) + Q(e^c) = 2Q(q)$$

of  $U(1)$  charge vectors — the extra  $U(1)$ s act on the  $SU(5)$  **10** through the **10** itself. On that combination  $B_3$  is  $1 + 1 - 0 = 2$  while every hypercharge shift is  $2 + 0 - 2 = 0 \bmod 3$ , so nothing repairs it; the remaining model has an explicit integer relation with the same property. The zero-target control is solvable in all 215. This is the structural reason behind Holotrade’s observation that the three dimension-four operators are locked together (b81ef8c).

**COMPUTATION** (w33\_pass10965\_baryon\_triality\_impossible.py)

#### In plain words

Of the two textbook discrete symmetries that protect the proton, the first dies at the Fayet–Iliopoulos term and the second is ruled out by the grand-unified shape of the charges. What remains is an  $R$ -symmetry — the anomaly-free  $\mathbb{Z}_4^R$  of Lee, Raby, Ratz, Ross, Schieren, Schmidt-Hoberg and Vaudrevange — which in a string orbifold lives in the internal rotation sector rather than in the gauge lattice. That is the one open route.

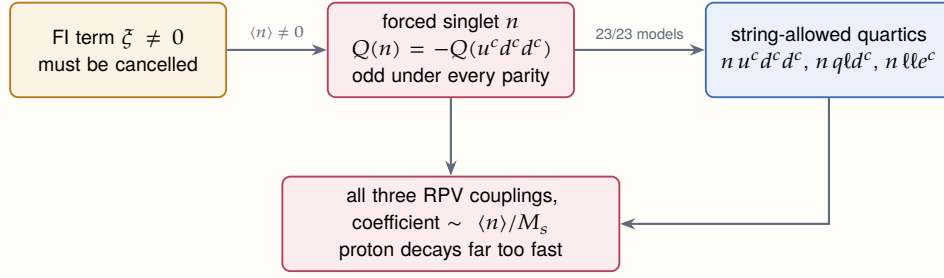
**The vacuum switches the forbidden couplings back on.** Two gaps remained. First, the searches above used the  $B - L$  family and  $\mathbb{Z}_2$  characters that are  $\pm 1$  on *every* field. A matter parity only has to be  $-1$  on quarks and leptons and  $+1$  on the condensate; exotic fields may carry any phase. Second, the string has discrete symmetries of its own, the *space-group selection rules*, and these are not part of the gauge lattice. Pass 10967 closes both. It takes the most general candidate: any element of the gauge torus times the space-group group (the point-group  $\mathbb{Z}_6$ , a  $\mathbb{Z}_3$ , and in  $\mathbb{Z}_6$ -II a further  $\mathbb{Z}_2 \times \mathbb{Z}_2$  from fixed points). It then decides exactly, with a Smith normal form, which singlets such an element can leave even. The answer is unchanged: such parities exist in the same 88 models and survive the FI term in none.

The reason is a single identity. In 290 of the 302 forced singlets, single copies of the fields satisfy

$$Q(n) = -Q(u^c d^c d^c) = -Q(q \ell d^c) = -Q(\ell \ell e^c),$$

which is the  $SO(10)$  quartic invariant **16**<sup>4</sup> evaluated at  $\nu^c$ . Each cubic is odd under every matter parity, so  $n$  is too. The string’s own coupling rules then finish the argument (Figure 31). In each of the 23  $D$ -flat  $\mathbb{Z}_6$ -I models, none of the three  $R$ -parity-violating cubics is allowed at the renormalisable level. Yet all nine forced singlets of every model appear in allowed quartics  $n u^c d^c d^c$ ,  $n q \ell d^c$  and  $n \ell \ell e^c$ . The vacuum that supersymmetry requires turns all three back on, with strength  $\langle n \rangle / M_s \approx 0.2$ – $0.4$ . Holotrade had noted that its class closure (b81ef8c) held only if the vacuum breaks matter parity; this is that vacuum, named field by field. **COMPUTATION** (w33\_pass10967\_fi\_vacuum\_regenerates\_rpv.py)

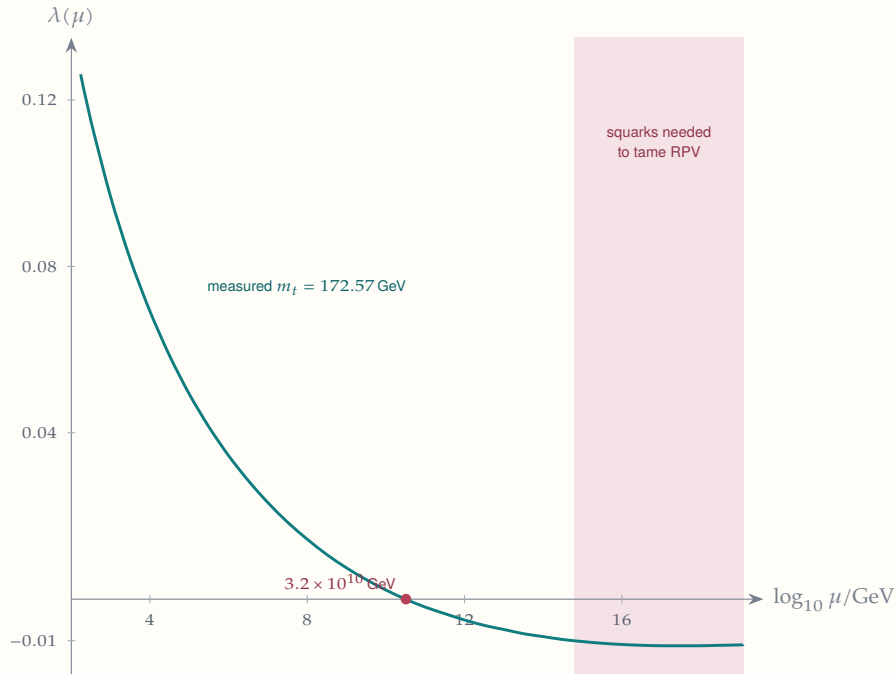
**Could heavy superpartners make it harmless?** Another part of this programme argues that the geometry’s supersymmetry is *structural*, with no light superpartners. Heavy squarks do tame  $R$ -parity violation, because proton decay through  $\lambda' \lambda''$  falls as  $1/\tilde{m}^4$ . With  $\lambda' \sim \lambda'' \sim \langle n \rangle / M_s \approx 0.2$ – $0.4$ , the bound  $|\lambda' \lambda''| \lesssim 10^{-27} (\tilde{m}/100 \text{ GeV})^2$  needs  $\tilde{m} \gtrsim 6 \times 10^{14}$ – $1.2 \times 10^{15}$  GeV. The Higgs sets a ceiling. Running the Standard-Model couplings up at two loops (checked against the published three-loop value at the Planck scale), the Higgs self-coupling turns negative at  $3.2 \times 10^{10}$  GeV (Figure 32). Supersymmetric matching requires it to be non-negative wherever the superpartners appear. Pushing that scale to  $10^{15}$  GeV would need a



**Figure 31.** Why supersymmetric vacua of the class violate  $R$ -parity. The only fields that can cancel the Fayet–Iliopoulos term carry exactly the conjugate charge of the three dimension-four operators. Every string selection rule allows the resulting quartic couplings, so the vacuum turns the cubic operators back on.

top quark of at most 171.0 GeV, 5.4 standard deviations below the measured  $172.57 \pm 0.29$  GeV.

**COMPUTATION** (w33\_pass10969\_rpv\_heavy\_superpartners\_vs\_higgs\_quartic.py)



**Figure 32.** The Higgs self-coupling run to high energy (two loops, NNLO matching). It turns negative near  $3 \times 10^{10}$  GeV, far below the  $\gtrsim 6 \times 10^{14}$  GeV that heavy squarks would need to make the regenerated  $R$ -parity violation harmless (shaded).

**The last door: a parity made of rotations — and why it closes.** The discrete groups tested so far live in the gauge lattice and the space-group rules. The orbifold also has *rotational* symmetries, turning each internal plane by its own angle, and these act on the supersymmetry charges ( $R$ -symmetries). Combinations of them that leave the superpotential alone can act as a matter parity. With the rules built into the standard orbifold code, this reopens exactly one model,  $\mathbb{Z}_6\text{II}_{23}$ , and the model then shows a striking two-sided trade: vacua kept flat by an unbroken rotation keep the exotic colour triplets massless at every order (202 found, 202 with massless exotics), while larger vacua that break the rotation make every exotic heavy but either lose the light Higgs pair or keep it only at the price of a superpotential that is not

zero:

| vacuum (old rotation rules)           | flat by symmetry | exotics  | a light Higgs pair |
|---------------------------------------|------------------|----------|--------------------|
| rotation unbroken (202 found)         | yes              | massless | protected          |
| rotation broken, 68 directions        | no               | heavy    | not protected      |
| rotation partly broken, 23 directions | no               | heavy    | protected          |

Even the best of these, the certified 23-direction vacuum, would not have described our world: the Higgs pair it protects is protected by an exact symmetry that also forbids any mass for its superpartners (massless Higgsinos, long excluded), and although its top-quark coupling is of order one, its down-quark and lepton couplings show no hierarchy at all.

Then the rules themselves were checked. Bizet, Kobayashi, Mayorga Peña, Parameswaran, Schmitz and Zavala derived the  $R$ -charge selection rule directly from the symmetries of the worldsheet instantons that mediate string couplings, and found that the textbook rule — conservation of the rotation charge plane by plane — holds only for planes whose rotation has prime order. On a plane of order six the rule acquires an extra contribution from the phases with which twisted strings at different fixed points combine. The door of  $\mathbb{Z}_6\text{II}_{23}$  had used the uncorrected rule on its order-six plane. Keeping only the rotation symmetries that are established (the order-3 and order-2 planes), and separately adding the corrected order-six charge, the model closes: all 304 Fayet–Iliopoulos cancelling directions break the parity. The whole  $\mathbb{Z}_6\text{-II}$  class is then excluded, 0 of 128, and three further orbifold families built on the same twist,  $\mathbb{Z}_2 \times \mathbb{Z}_6$  (29 Standard Models),  $\mathbb{Z}_3 \times \mathbb{Z}_6$  (5) and  $\mathbb{Z}_6 \times \mathbb{Z}_6$  (10), have no surviving parity either, and  $\mathbb{Z}_2 \times \mathbb{Z}_6\text{-II}$  produces no Standard Model at all. For  $\mathbb{Z}_{12}\text{-I}$ , whose lattice is not a product of planes, the admissible rule follows from the symmetries of the  $E_6$  lattice itself: the only rotation commuting with the twist beyond the twist’s own powers is a half-turn of one plane, and with it every one of the 14 candidate vacua is decided — eleven keep exotic triplets massless, and the most promising one, which does protect a single Higgs pair, has a singlet entering its superpotential linearly at fourth order whose absorption destroys that protection. A final escape — that flavour structure might make the regenerated couplings tiny for the first generation, where the proton is sensitive — fails as well: the first-generation couplings stay at  $10^{-3}$ – $10^{-1}$ , far above the  $10^{-10}$  the proton would need.

The one family in which every rule is beyond doubt is the last to be told. In  $T^6/\mathbb{Z}_3 \times \mathbb{Z}_3$  every plane has prime order, so every rotation symmetry the orbifold reports is established. Here, for the first time, the parity *survives* the Fayet–Iliopoulos term: three of the 13 Standard Models keep it on cancelling directions, 36 ways. Each minimal vacuum repeats the familiar pattern — the superpotential vanishes on the vacuum and an exotic triplet stays massless, not one because of the other but both because the vacuum leaves a discrete  $R$ -symmetry unbroken (the pattern is common, not a law: P11086 finds  $\mathbb{Z}_{12}\text{-I}$  vacua where it fails in each direction) — but singlets outside it enter linearly at third and fourth order, and absorbing them yields 28 vacua with the parity intact, the exotics massive, the Fayet–Iliopoulos term cancelled with every field switched on, and in 22 of them a Higgs pair massless to all orders.

Are they supersymmetric? On each of these vacua the superpotential turns out to be a function of just one or two composite fields  $p_i$  — products of three or four of the vacuum fields — and, to all orders, only the powers with  $a_1 + a_2 \equiv 1 \pmod{3}$  appear:  $W = F(p)$  with  $F(\omega p) = \omega F(p)$ , a  $\mathbb{Z}_3$  rotation symmetry acting on the composites. That is why no two

terms can balance below twelfth order ( $12 = 4 \times 3$ ), and it is also why they *can* balance there: the  $F$ -terms vanish exactly where  $\nabla_p F = 0$ , a system of one or two equations with the Bezout number of solutions. Every solution, for random couplings, lifts to an exact supersymmetric vacuum with every field switched on — in all 28 vacua — at field values near the string scale ( $0.8\text{--}1.1 M_s$  for couplings of order one). The door was real. What closes it is not supersymmetry but charge. A mass term needs a particle and an antiparticle with conjugate charges under *every* gauge group, the hidden ones included; counting both sides class by class, these models carry 48 more states of electric charge  $\pm 1/3$  on one side than on the other, six of them neutral under every hidden group. In any vacuum that leaves the hidden sector unbroken — all 28 do — those six are exactly massless, and both hidden groups are infrared free, so no confinement binds them. Breaking the hidden  $SU(4)$  is the only way to give them partners; adding any one of its 70 possible meson condensates to any of the 28 vacua never pairs them all. Massless particles of charge  $1/3$  are excluded outright. Letting hidden-sector condensates into the vacuum itself is the last thing to try, and the counting says exactly which ones can help: the charge- $\frac{1}{3}$  excess disappears if and only if one particular hidden factor is broken. Such vacua exist — 54 of them keep the parity with the hidden  $SU(4)$  condensed, all supersymmetric, all with exactly the Standard Model's quarks and one Higgs pair light — and in every one the excess is gone. Yet the couplings that would pair the balanced states do not exist: at least 120 fractionally charged particles stay massless in each, and most of them are protected by an exact discrete symmetry the vacuum leaves unbroken (always a 3-group, of order  $3^4$  or  $3^6$ ), so no correction of any kind can give them mass. A sibling model offers a near miss in which the needed condensate pairs every class in number, but its parity cannot survive two condensates of the same hidden group. COMPUTATION  
(w33\_pass10979\_z3xz3\_parity\_vacua\_f\_obstruction.py, w33\_pass11024\_z3xz3\_susy\_vacua\_fractional\_charge\_index.py)

Four questions remain about that verdict, and each has an exact answer.

*What protects those massless states?* Not one symmetry. The unbroken group is assembled from three kinds of remnant: of the broken  $U(1)$ s, of the translations of the orbifold's space group, and of its discrete  $R$ -rotations. Almost half of the forbidden mass terms are forbidden twice over, so there is no single symmetry whose breaking would release them.

*Would a different condensate help?* No. The census was rerun over every hidden invariant, including the baryonic ones and mixtures of two kinds. It finds 63 further parity-preserving vacua in which the curing factor is condensed, some with both  $SU(4)$ s broken. Every one keeps at least 142 fractionally charged multiplets massless.

*Could smoothing out the orbifold help?* No. The fractional part of electric charge turns out to be a character of the orbifold's space group. On every one of the 946 fixed points of all 13 models, the class  $3Q + t \pmod{3}$  is constant ( $t$  is the colour triality), and it is a single linear function of the fixed point's labels. The 516 fixed points where it is nonzero carry only fractionally charged states. So no neutral field lives there, and they cannot be blown up without breaking colour or electromagnetism. Even discarding every discrete selection rule and keeping only gauge invariance leaves at least fifteen fractional multiplets massless in every vacuum.

*Would giving up supersymmetry help?* Not here.

- The obstructions concern fermions and gauge charges, which supersymmetry breaking does not touch.

- The  $\mathbb{Z}_3$  orbifolds admit no non-supersymmetric twin with the same order-three shifts. For every order-three shift of  $E_8$ ,  $9V^2$  is even, while attaching  $(-1)^F$  to an order-three twist makes  $9v^2$  odd, and modular invariance needs the two to agree. A non-supersymmetric model on the  $\mathbb{Z}_3$  lattice needs a shift of order six, which makes it a different model.
- The one  $W(3,3)$  family whose Standard Models do have such twins is  $\mathbb{Z}_6$ -I: all 87 of them, with the twist  $(\frac{1}{6}, \frac{1}{6}, \frac{2}{3})$  of Font and Hernández [14]. Without superpartners they have no dimension-four or dimension-five proton-decay operators.

We computed the 87 twins after repairing five faults that stop the standard spectrum program at  $N = 0$ . Every twin is anomaly-free up to a single Green–Schwarz  $U(1)$ , keeps its parent’s gauge group, and has exactly three net generations. And every one has tachyons at its  $\theta$  fixed points. The reason is one level-matching fact: the tachyonic right-moving level lies exactly  $\frac{1}{6}$  below the massless one, and  $\frac{1}{6}$  is the smallest twisted oscillator mode. So a fixed point of the twin is tachyonic exactly when its supersymmetric parent has a massless oscillator-excited state there, and all 87 parents do. Orbifolds of the separate non-supersymmetric  $SO(16) \times SO(16)$  string, which has its own software [12], are a different construction and are not covered by this statement.

**COMPUTATION** (w33\_pass11087\_unbroken\_three\_group\_protection.py, w33\_pass11088\_baryonic\_hidden\_condensates.py, w33\_pass11089\_w33\_twist\_without\_supersymmetry.py, w33\_pass11090\_resolution\_cannot\_free\_fractional\_charges.py, w33\_pass11092\_nonsusy\_z6i\_twins\_are\_tachyonic.py)

The same level-matching fact decides the whole  $\mathbb{Z}_6$ -I scan, not only its Standard Models.

- For 47 of its 58 base shifts, the fixed point without a Wilson line is already tachyonic, so every model built on those shifts is tachyonic whatever its Wilson lines.
- On the other 11 shifts we drew 66 000 random Wilson lines. 9 576 twins are tachyon-free and 705 parents are Standard Models, and no model is both. If the two properties were independent, about 79 would be.

The spectrum shows why. The Standard-Model Wilson lines always leave half-charged exotic states one oscillator step above the tachyonic level, and those states become the tachyons of the twin. Every tachyon carries electric charge  $\pm \frac{1}{2}$  or colour, so any condensate of them breaks electromagnetism. The twins of the other  $W(3,3)$  families behave the same way (333 Standard Models of  $\mathbb{Z}_{12}$ -I,  $\mathbb{Z}_2 \times \mathbb{Z}_6$ -I,  $\mathbb{Z}_3 \times \mathbb{Z}_6$  and  $\mathbb{Z}_6 \times \mathbb{Z}_6$ ): none is both tachyon-free and a three-generation Standard Model.

**COMPUTATION** (w33\_pass11093\_tachyon\_free\_filter\_z6i\_scan.py, w33\_pass11094\_tachyons\_break\_electromagnetism.py, w33\_pass11096\_twins\_of\_the\_other\_w33\_families.py)

The separate non-supersymmetric string does better. In the  $SO(16) \times SO(16)$  heterotic string [12, 39] the  $(-1)^F$  twist is a freely acting  $\mathbb{Z}_2$  of its own, with its own gauge shift  $V_0$ .

- The  $W(3,3)$  shift of  $T^6/\mathbb{Z}_3$ , the  $A_8$  Kac pair, admits 793 508 such  $V_0$ , in six gauge-group classes. The Witten shift always breaks the  $SU(9)$ .
- On the four classes the code accepts we drew 80 000 random Wilson lines. Every model is tachyon-free, and 153 of them (104 inequivalent) pass the Standard-Model test of the non-supersymmetric orbifold.
- We re-derived every one of them with an independent engine. Each is tachyon-free and anomaly-free, and has exactly three chiral generations, no net fractionally charged

fermion, and at least 6 Higgs-doublet scalar fields.

These are non-supersymmetric, tachyon-free, three-generation string models built on the  $W(3,3)$  twist. Three things remain to be shown: masses for their vector-like exotics, some of them fractionally charged; a stabilised vacuum, which requires the one-loop cosmological constant and dilaton tadpole of a non-supersymmetric string; and realistic Yukawa couplings. Standard-Model-like models of this string are known in large numbers [39]; what is new here is that the  $W(3,3)$  shift is among them. Two independent codes agree on the massless spectra of every sample model the non-supersymmetric orbifolder ships. **COMPUTATION**

(w33\_pass11095\_sol16xsol16\_a8\_standard\_models.py)

How good are these models? We asked the three questions that decide a string vacuum.

*Can the fractional charges be removed?* Unlike in  $\mathbb{Z}_3 \times \mathbb{Z}_3$ , they are not protected by a symmetry. Let the Standard-Model-neutral, hidden-singlet scalars take expectation values, and impose the full set of string selection rules (gauge, the Witten  $\mathbb{Z}_2$ , point group, space group and  $H$ -momentum).

- Masses for *all* fractionally charged fermions are allowed in 33 of the 104 models.
- In 6 of those 33, a single scalar suffices, through cubic couplings. In every one of the 33 all such masses appear by order 7.

*Is the top heavy, and alone?* In 73 models, no. Their up-type Yukawas are untwisted and carry the  $\epsilon$ -structure of the ten-dimensional gauge coupling. The singular values are  $(1, 1, 0)$  for every Higgs direction: top and charm degenerate, the up quark massless. In the other 31 the quarks and Higgs doublets sit at fixed points. A light Higgs localized at one fixed point then gives a single heavy top, with charm and up suppressed by  $e^{-\text{area}}$ . The two are degenerate at tree level, because the three fixed points of the  $\mathbb{Z}_3$  torus form an equilateral triangle; this is the known two-valued structure of  $\mathbb{Z}_3$  Yukawas. In 27 models the bottom behaves the same way. Twelve models pass every test so far: they are tachyon-free, have three generations, can give all their fractional charges masses (at third order), and have a heavy top and bottom from a localized Higgs.

*Is the vacuum stable?* We recomputed the ten-dimensional  $SO(16) \times SO(16)$  one-loop integral:  $\Lambda_{10} = +363 M^{10}$ , positive, as Alvarez-Gaumé, Ginsparg, Moore and Vafa found [2]. In every model the large-volume vacuum energy is one third of it times the volume, and the massless Bose–Fermi counts are never degenerate. The potential runs away, as it does for every symmetric non-supersymmetric orbifold [20].

The  $W(3,3)$  models of the  $SO(16) \times SO(16)$  string thus remove the proton-decay and fractional-charge obstructions of the supersymmetric routes. Twelve of them even single out the top, but charm and up are degenerate at tree level. All of them meet the universal vacuum-energy problem of non-supersymmetric strings. The other  $W(3,3)$  geometries give no tachyon-free Standard Model at all: 77 600 draws on  $\mathbb{Z}_3 \times \mathbb{Z}_3$ ,  $\mathbb{Z}_6$ -I and  $\mathbb{Z}_{12}$ -I. **COMPUTATION** (w33\_pass11097\_fractional\_fermion\_masses.py, w33\_pass11098\_yukawa\_textures.py, w33\_pass11099\_vacuum\_energy\_sol16\_a8.py, w33\_pass11100\_sol16\_other\_w33\_geometries.py, w33\_pass11101\_localized\_higgs\_heavy\_top.py)

Can the scalar vacuum split charm from up? Extending every Yukawa entry to all orders with the same scalars that give the fractional charges their masses, the answer is no. In all 12 survivors, and for each of the 72 Higgs doublets with a tree-level coupling, the charm–

up (and strange–down) block is symmetric at every order under the point reflection of the generation line of fixed points about the top’s. The two lighter quarks of each type stay parametrically degenerate; a hierarchy would need a fine-tuned cancellation. **COMPUTATION**

(w33\_pass11102\_higher\_order\_quark\_hierarchy.py)

The reflection has a name. The symmetry group of the  $\mathbb{Z}_3$  space-group rule on one torus, found by brute force over every permutation and phase of the three fixed points, has order 54: it is  $\Delta(54)$ , the flavour group of a Wilson-line-free  $\mathbb{Z}_3$  torus [27]. Its Heisenberg subgroup  $H_{27} = \langle X, Z \rangle$  (translation of the fixed points and the phase of the selection rule) has as its centre the orbifold twist itself,  $ZX = \omega^I XZ$ , and the reflection is the  $-I$  of  $SL(2, 3)$  acting on  $H_{27}/Z = \mathbb{F}_3^2$ . This is the qutrit Heisenberg group of the  $E_8$  chapters, now realised on the three generations of an explicit string vacuum—with only  $\{\pm I\}$  of its Clifford normaliser (index 12), because the space-group rule forbids quadratic phases. In all 104 models exactly one torus is free of Wilson lines, so the geometry realises one qutrit and never two: the two-qutrit Pauli group, whose commutation graph is  $W(3, 3)$ , is not the flavour group of these vacua. A Higgs localised at one fixed point leaves an  $S_3$  acting irreducibly on the other two generations, so by Schur’s lemma  $m_c = m_u$  exactly while it is unbroken.

Breaking it spontaneously—giving the twisted scalars at the three fixed points different expectation values—does not help the up sector: for every alignment we tried (six patterns, small and large torus area, hidden group unbroken or broken), charm and up stay at the same order in all 36 Higgs cases. The strange–down pair splits only through mixing with vector-like  $\bar{d}$  states, in exactly the 8 models that have them. The heavy generation fares better. Top and bottom share a doublet only if  $H_u$  and  $H_d$  sit at the same fixed point; then  $V_{tb} \approx 1$ ,  $|V_{cb}| \sim |V_{ub}| \sim \epsilon^2$  and the  $\tau$  is singled out—but  $V_{cb} \sim m_c/m_t$ , a factor of  $\sim 10$  below the data, and  $\mu$  and  $e$  are degenerate like charm and up. **COMPUTATION**

(w33\_pass11103\_reflection\_breaking\_vacua.py, w33\_pass11104\_ckm\_and\_leptons.py, w33\_pass11105\_delta54\_flavour\_join.py)

Finally, the vacuum energy at the orbifold point. We built the full one-loop integrand: all 36 sectors of  $\mathbb{Z}_2^W \times \mathbb{Z}_3$  with Wilson lines, the twisted sectors as 24 modular images of one  $\Gamma_1(6)$ -invariant block per fixed point, the Witten sectors as sums over the Narain lattice. It is modular invariant to  $10^{-12}$ , reproduces the ten-dimensional integral and the large-volume limit, and its massless coefficient matches the spectrum sector by sector (which exposed that the earlier Bose–Fermi counts had omitted the  $U(1)$  gauge bosons; corrected, they are never zero). The Kähler modulus of the family torus is stabilised at its self-dual point  $T = \rho$ : the potential in that direction is exactly  $SL(2, \mathbb{Z})$ -invariant and is lowest there. The Wilson-line moduli are not: the potential falls monotonically as their tori shrink, stays positive, and below a critical radius a winding state of the Witten-twisted sector becomes tachyonic—in every one of the 12 survivors, at a radius between 2 and the  $SU(3)$  point. The instability of the  $O(16) \times O(16)$  string on small tori [18] is where these vacua roll. **COMPUTATION**

(w33\_pass11106\_one\_loop\_orbifold\_point.py)

Built state by state, the tachyon has the right-moving ground state of the Witten-twisted sector and a winding in the Wilson-line tori; its level at  $\text{Im } T = 1$  is  $\Delta = -(3 - \sqrt{3})/6$ , which the one-loop integrand confirms. The critical radius is  $\sqrt{3}$  or 1.51 (one model is safe on the whole imaginary axis and fails only at  $\rho$ ). Every tachyonic state is a colour singlet, and in 10 of the 12 models every one carries fractional electric charge  $(\pm \frac{1}{3}, \pm \frac{2}{3})$ : a condensate would break electromagnetism, the disease of the non-supersymmetric twins again. Over

the moduli the potential can reach, it has no minimum in the tachyon-free region:  $\Lambda > 0$  at all 20 sampled points of model 2, lowest right at the tachyon boundary, and in the Einstein frame the dilaton runs to weak coupling. Among all 104 models only one stays tachyon-free down to the  $SU(3)$  point, and it fails the earlier tests (massless fractional charges, top = charm, no down-type Yukawa); a fresh scan of 387 new models finds 4 more—all five with untwisted quarks, hence top = charm and no down-type Yukawa. Classifying all 491 quark sectors gives 0 of 152 twisted-quark models safe against 5 of 339 untwisted: suggestive, not significant (Fisher  $p = 0.16$ ). **COMPUTATION** (w33\_pass11107\_winding\_tachyons.py, w33\_pass11108\_tachyon\_free\_moduli\_rescan.py, w33\_pass11110\_moduli\_map.py)

The stabilised family modulus also undoes the heavy top. Worldsheet instantons make the  $\mathbb{Z}_3$  Yukawa couplings theta functions of the  $A_2$  lattice in  $T$  [31]; the one-loop minimum  $T = \rho$  is a fixed point of the duality group, where  $|Y_{\text{distinct}}/Y_{\text{same}}| = \frac{1}{2}$  exactly. At tree level the vacuum therefore gives  $m_c = m_u = m_t/2$ ; the observed  $m_c/m_t$  needs  $\text{Im } T \approx 3.2$ , where the potential is 2.4 times higher. **BRIDGE** (w33\_pass11109\_instanton\_yukawas\_at\_rho.py)

Nor can the generations carry more than one qutrit. The  $k$ -qutrit Pauli group  $3^{1+2k}$  acts non-trivially through its centre only in dimensions divisible by  $3^k$ ; the centre here is the orbifold twist, so three twisted generations force  $k = 1$ —the single free torus of all 104 models is a theorem, not an accident. The two-qutrit group whose commutation graph is  $W(3,3)$  cannot be the family group, and it cannot be assembled from family and colour either: the twist is not a gauge-centre element on the quarks of any survivor. **BRIDGE** (w33\_pass11111\_no\_243\_family\_group.py)

Three further attempts, and one opening. *The Higgs direction.* With a general light Higgs in the  $\Delta(54)$  triplet the up-type mass determinant is a member of the Hesse pencil  $x^3 + y^3 + z^3 - 3\lambda xyz$  in the Higgs direction (forced by  $\Delta(27)$ , whose invariant cubics are exactly that pencil—the other track's Hesse configuration is the geometry of the quark masses), and the modulus picks the member: at the stabilised point  $T = \rho$  it is exactly the singular one,  $\lambda = \omega^2$ . Yet no alignment beats a localised Higgs: the second singular value never falls below  $|Y_{\text{dist}}/Y_{\text{same}}| = \frac{1}{2}$  of the first. *Duality.* Along the zero- $B$  axis of the Wilson-line tori the potential must turn around at small radius, but the turnaround never lands in a tachyon-free window: one model's duality extremum is its tachyonic self-dual radius, and the survivor safe furthest down rolls into a charged tachyon. *The other track's groups.* The centre of the family  $H_{27}$ —the orbifold twist—acts on the massless spectrum as a gauge element in all 104 models (as the anomalous- $U(1)$   $\mathbb{Z}_3$  itself in three), cf. the gauge origin of  $\Delta(54)$  [6]; no  $SU(3)$  factor, visible or hidden, can be glued to it, so the two-qutrit group is realised on no matter at all.

*The opening.* In the two survivors with neutral winding tachyons, the neutral ones wind in one Wilson-line torus and the charged ones in the other, and the one-loop potential pulls the neutral torus down faster. The first instability these vacua meet is therefore Standard-Model-neutral: its condensation breaks extra  $U(1)$ s only, and electromagnetism survives. Where it ends is the open question. **COMPUTATION** (w33\_pass11112\_neutral\_tachyon\_corridor.py, w33\_pass11113\_rho\_tachyon\_correlation.py, w33\_pass11114\_hesse\_mass\_determinant.py, w33\_pass11115\_axis\_duality\_and\_window.py, w33\_pass11116\_one\_qutrit\_vs\_other\_track.py)

Following the opening further. The condensate breaks exactly one  $U(1)$  (a combination with a component along the anomalous one), leaves the Standard Model and every hidden

non-abelian factor intact, and is a singlet of the family  $\Delta(54)$ —so it cannot help the flavour problem; no cubic term exists, so the instability is Higgs-like. The full gradient flow over both Wilson-line radii of model 57 reaches the neutral boundary from equal radii, but the margin is narrow: a 1.5% head start for the charged torus reverses it. Across all 491 models, 60 place their neutral tachyons on a Wilson-line torus of their own (38 with no charged tachyon on either torus alone), and every one of them has twisted quarks, the heavy-top class; and 19 of the new ones pass every earlier test—fractional charges removable, a single heavy top and bottom from a localised Higgs—giving 21 survivors with a neutral exit. In 8 of them only neutral tachyons appear even when both Wilson-line tori shrink together, and in 6 the whole zero- $B$  direction stays neutral down to  $\text{Im } T = 0.6$ : charged tachyons need a  $B$ -field at small radius or the  $SU(3)$  point, far below the neutral onset at  $\text{Im } T \approx 1.5$ – $1.7$ . For these six the first instability the potential leads to is Standard-Model-neutral with a wide margin, not a knife-edge; their open problems are the class-wide ones (family degeneracy, positive vacuum energy). No closed-form lattice criterion decides the  $SU(3)$ -point tachyon: the gauge cells give a sharp necessary condition, but the torus offsets are needed too. **COMPUTATION** (w33\_pass11117\_neutral\_condensate.py, w33\_pass11118\_wilson\_line\_flow.py, w33\_pass11119\_neutral\_tori\_across\_491.py, w33\_pass11120\_rho\_cell\_criterion.py, w33\_pass11121\_hesse\_gauge\_bridge.py)

Following the six to the end of the opening. Along the zero- $B$  diagonal their one-loop potential falls monotonically to the neutral onset with no barrier; it is a volume law,  $\Lambda_\beta \approx 160 y^2 - (5-13)$  in units of  $M^4$ , identical across the six, so the drive is the large-volume Bose-Fermi term and the approach is steep ( $d\Lambda/dy \approx 600$ ), not slow roll. The critical radii are exact:  $y_c = \sqrt{3}$  in two models and  $\varphi^2/\sqrt{3}$  ( $\varphi$  the golden ratio) in four, where the neutral states have  $P_L^2 = 2$ ,  $p_R^2 = 1$ ; no pair of them makes a massless vector, so the onset is not an enhanced-symmetry point. Their tree-level four-point amplitude is a Kawai-Lewellen-Tye beta function whose low-energy limit is pure exchange,  $4C\pi(1/t + 1/u) + 3C\pi s^2/(tu)$ , with no contact quartic, and the tachyon mass falls as the radius shrinks ( $dm^2/dy > 0$ ): condensate and radius run together, into the winding-tachyon regime where the cycle pinches off (Adams-Liu-McGreevy-Saltman-Silverstein), outside perturbative control. A  $B$ -field does not change the order: charged tachyons need both  $B \neq 0$  and  $\text{Im } T \leq 1$ , always at least 0.4 below the neutral onset, and in the four golden models  $B \geq 0.4$  removes the neutral tachyon at  $\text{Im } T = 1.4$ . The condensate carries no quantum numbers on the family torus, so it cannot move  $T^*$  from  $\rho$  or touch  $m_c = m_u$ . What it does change is the selection rules: its charge lies outside the span of the singlet vacuum, and every Yukawa target in the new direction needs exactly one condensate insertion. With it, charged-lepton, down-quark and neutrino-Dirac entries forbidden at every order become allowed (orders 2–6, 1863 entries, 1854 with a twisted field), while up-type Yukawas and the  $\mu$ -term stay forbidden. These are individual entries of matrices that already have allowed ones: the light doublets have tree-level lepton and down Yukawas (see below), so this is not a new source of the missing masses, as first read. A first encoding that treated the winding as an exactly conserved  $\mathbb{Z}_3$  found nothing; that zero was an artefact, since winding modulo  $(1 - \theta)\Lambda$  is the fixed-point class. **COMPUTATION** (w33\_pass11122\_six\_potentials.py, w33\_pass11123\_critical\_point\_amplitude.py, w33\_pass11124\_condensate\_unlocked\_couplings.py, w33\_pass11125\_bfield\_charged\_onset.py, w33\_pass11126\_family\_modulus\_under\_condensation.py)

Where the unlocked entries matter, and where the opening ends. For the three light Higgs

doublets (those with a tree-level down coupling) the lepton and down textures are  $[0, 1, 1]$  with and without the condensate, and the cubic lepton entries are symmetric under the point reflection that forces  $m_c = m_u$ , so  $m_\mu = m_e$  survives. But in a non-supersymmetric model with one light Higgs  $h$ —the doublet that gives the top—the bottom and tau come from  $h^*$ , and  $h^*$  is not a tree-level down doublet: in all six the conjugates of the three top doublets are exactly the heavier doublets. There the condensate lowers  $m_b/m_t$  from  $\epsilon^3$  to  $\epsilon^2$  in four models and brings the tau to the same order, so  $m_b/m_\tau = O(1)$  with  $\epsilon \approx 0.12\text{--}0.15$ , i.e.  $\langle T \rangle \langle S \rangle / M^2 \approx 0.01\text{--}0.02$ ; the light generations stay degenerate. Across all 21 neutral-exit survivors the condensate never unlocks an up-type Yukawa or a  $\mu$ -term (down entries in 14, lepton in 17). The endpoint itself is not a nearby vacuum: shrinking the neutral torus deepens the tachyon along an exact winding law,  $p_R^2 = y_1/\sqrt{3}$  (plus  $1/(3\sqrt{3}y_2)$  in the golden models), whose diagonal root  $y^2 - \sqrt{3}y + \frac{1}{3} = 0$  is where the golden radius  $\phi^2/\sqrt{3} = (3 + \sqrt{5})/(2\sqrt{3})$  comes from; as  $y \rightarrow 0$  it approaches the  $\Delta = -\frac{1}{2}$  ground state of the T-dual tachyonic string (cf. Fraiman–Graña–Parra De Freitas–Sethi), a single complex species that exists only through winding, with no tachyon-free island. The  $B$ -field is a near-flat direction ( $|\partial_B \Lambda| \lesssim 5 \times 10^{-5} |\partial_y \Lambda|$ ), with  $B = 0$  a shallow minimum just above the onset. **COMPUTATION** (w33\_pass11127\_lepton\_texture\_under\_condensation.py, w33\_pass11128\_pinched\_cycle.py, w33\_pass11129\_offaxis\_potential.py, w33\_pass11130\_single\_higgs\_bottom\_tau.py, w33\_pass11131\_unlocked\_across\_21.py)

The geometry of the exit is exact. On the diagonal of the two Wilson-line moduli  $T = B + iy$  every tachyon domain is a hyperbolic disk centred at a special point of  $\Gamma_0(3)$ : the charged one at the elliptic point  $(3+i\sqrt{3})/6$  with radius  $\ln(2+\sqrt{3})$ , the neutral one either the horodisk  $\text{Im}(-1/T) > 1/\sqrt{3}$  at the cusp 0 or the disk at the Fricke point  $i/\sqrt{3}$  with radius  $2 \ln \phi$ —which is where the golden ratio comes from. The neutral top lies at least 0.197 above the charged one for every  $B$ , so the exit is SM-neutral on the whole  $B$  circle. Its consequence for flavour is now class-wide: with one light Higgs the bottom falls to  $\epsilon^2$  in exactly the 14 of 21 models with unlocked down entries, with the tau at the same order for at least two of the three candidate Higgs, so  $m_b/m_\tau = O(1)$  with  $\epsilon_S \epsilon_T \approx m_b/m_t$ . The light generations do not follow: every entry of the conjugate sector has the same options, so no ratio of the two VEV scales splits them (anarchy). The size of the condensate is not fixed: the supersymmetric FI formula would give  $\langle T \rangle / M_P = 0.15\text{--}0.20$ , close to what is needed, but without supersymmetry there is no FI potential for a complex scalar, so this is a coincidence of scales, not a prediction. **COMPUTATION** (w33\_pass11132\_bottom\_tau\_across\_21.py, w33\_pass11134\_dual\_tachyon\_species.py, w33\_pass11135\_fi\_scale\_no\_mechanism.py, w33\_pass11136\_tachyon\_domains.py, w33\_pass11137\_neutral\_exit\_scorecard.py)

The end of the exit and the flavour question, closed from four sides. The duality group of the Wilson-line modulus is  $\Gamma_0(3)$ :  $T \mapsto T/(3T + 1)$  and the order-three elliptic element are exact symmetries of the partition function,  $S$  and  $\Gamma^0(3)$  are broken, and the Fricke map is only approximate. Its one elliptic point is exactly the centre of the charged tachyon disk and its cusp at 0 is tachyonic in 15 of 21 models (in the other six it is the approximate Fricke image of the large-volume region, below the golden disk), so every point the symmetry singles out is tachyonic or a runaway: the Wilson-line modulus cannot be stabilised at a self-dual point. At zero radius of the neutral torus the massless vectors form  $\mathfrak{su}(16) \oplus \mathfrak{u}(1)$  in all 21 models — the non-supersymmetric  $U(16)$  heterotic string, whose  $2^1 = 2$  tachyons are exactly the two norm-one states found there (Fraiman, Graña, Parra De Freitas and Sethi [51]) — so the endpoint of the neutral condensation, perturbatively, is that string.

Read thermally, the exit is a Hagedorn transition that exists only because of the Wilson-line holonomy: with the Wilson lines switched off no Witten-sector tachyon appears at any radius. On flavour: no pattern of singlet expectation values (108 random hierarchies) and no two-doublet mixture of Higgs fields (576 tried) splits the light generations; the degeneracy is structural. A new hypothesis — that the condensate makes the neutrino Dirac matrix anarchic, the known route to large lepton mixing — fails: the light-Higgs Dirac texture is the up-quark one at tree level ( $m_D(\nu) \sim m_u$  in texture, as in  $SO(10)$ ) and is untouched by the condensate, while the anarchic conjugate sector that would give large lepton mixing gives large quark mixing too. **COMPUTATION** (`w33_pass11138_singlet_vev_hierarchies.py`, `w33_pass11139_wilson_line_duality_group.py`, `w33_pass11140_zero_radius_ul6.py`, `w33_pass11141_no_symmetric_stabilization.py`, `w33_pass11142_higgs_mixing_up_sector.py`, `w33_pass11145_neutrino_anarchy.py`, `w33_pass11146_thermal_reading.py`)

Two refinements. The zero-radius vector lattice is identified, not only counted: it is even, of rank 16 and determinant 4 with the hyperbolic discriminant form on  $\mathbb{Z}_2^2$  — the genus of the vector lattices of the non-supersymmetric heterotic strings, which that genus exhausts [51] — and its root system is  $A_{15}$ , so it is the  $U(16)$  lattice, in all 21 models. And the tachyon-free region near the cusp at 0 in the six golden-type models is, to 0.3–0.5%, the Fricke mirror of the large-volume region: the potential falls steeply into the golden disk from below, so it holds no minimum either. **COMPUTATION** (`w33_pass11150_cusp0_potential.py`, `w33_pass11151_ul6_lattice.py`)

The apparent opening was an artefact of an incomplete selection rule; it was caught by reading the literature on the rules themselves, not by the computation. **COMPUTATION** (`w33_pass10974_valid_r_rules_close_the_rotation_door.py`; the conditional analysis is `w33_pass10968_r_symmetry_parity_and_massless_exotics.py`)

## 9.5 Handedness: host, not select

The weak force acts only on left-handed particles. Can  $W(3,3)$  explain why?

### Theorem 9.5 (the chirality no-go)

**THEOREM**
**COMPUTATION**

The two half-spin representations  $S^\pm$  of the  $D_5$  structure built on the geometry are exchanged by the substrate's own outer controller  $T$ , which generates  $\text{PGSp}(4,3) = \text{PSp}(4,3):2 = W(E_6)$  together with  $\text{PSp}(4,3)$  and has  $\det T = -1$ . Consequently no invariant of the full automorphism group of  $W(3,3)$  can distinguish  $S^+$  from  $S^-$ . `w33_pass346_the_chirality_no_go.py`

### In plain words

The forty-point geometry is perfectly capable of *carrying* a left-handed world: the chiral spinors are there (Section 8). What it cannot do is *choose* left over right, because one of its own symmetries swaps them. Any explanation of the handedness of the weak force must come from outside the geometry — from a choice of vacuum, of orientation, or of dynamics. This is the same distinction as in Section 2: the two groups of order 51 840 differ precisely by whether that swapping symmetry is present.

**What it means physically: the swap is antiunitary**

Which symmetries do the swapping? For the qutrit, the determinant of a symplectic label map decides: determinant  $+1$  is a unitary Clifford operation, determinant  $-1$  can only be implemented antiunitarily (Pass 5730, `w33_pass5727_5730_torsion_fam-ily_heisenberg.py`). The decisive statement was proved on the 32 orientations of a complete two-qutrit factorisation frame (Holotrade `e92a047`, `frame_chirality_is_reversed_only_by_antiunitaries.py`): they split  $16+16$  exactly as the two half-spin weight sets of  $D_5$ , unitary Cliffords preserve each half, and antiunitary ones — and nothing else — exchange them. The other track has since found the same law on the clock: determinant-odd ticks exchange the two  $D_4$  half-spin sheets (Pass 10955) and are not completely positive (Pass 10952). So the geometry's chirality is invariant under every *unitary* symmetry and exchanged only by *antiunitary* ones. That is the structure of a chiral quantum theory obeying CPT: the mirror of a chiral world is related to it by an antiunitary map, and which of the two is called “left” is a convention that no internal measurement can fix. The no-go is the finite form of this, not a defect peculiar to  $W(3,3)$ . **BRIDGE** (`w33_pass10952_clock_complete_positivity_firewall.py`, `w33_pass10955_d4_halfspin_clock_bridge.py`)

**Corollary 9.7 (one coset reverses time, acts antiunitarily, and swaps chirality)****THEOREM**

Fix a “now” (the Bell line  $B$ ). Its stabiliser in  $\mathrm{PGSp}(4,3)$  has order 1296, and its elements outside  $\mathrm{PSp}(4,3)$  form a single coset of size 648. That coset is exactly

- (i) the set of symmetries that *reverse* the unique global orientation of the history graph (Theorem 4.4);
- (ii) the set that can only be implemented *antiunitarily* on qutrits (Pass 5730);
- (iii) the set that *exchange* the two chiral half-spin modules (Theorem 9.5; Holotrade `e92a047`).

*Proof.* Each item identifies its set with the complement of  $\mathrm{PSp}(4,3)$  in the relevant group: (i) is the statement of Theorem 4.4(ii) (the 648 elements of the Bell stabiliser in  $\mathrm{PSp}(4,3)$  fix the orientation cycle, the other 648 negate it); (ii) symplectic similitudes of multiplier  $+1$  lift to unitary Clifford operations and those of multiplier  $-1$  only to antiunitary ones; (iii)  $\mathrm{PSp}(4,3)$  preserves  $S^+$  and  $S^-$  (they are inequivalent) while the outer element  $T$  exchanges them. Intersecting with the stabiliser of  $B$  gives the same coset in all three cases.  $\square$

**What it means physically: a finite CPT theorem**

In quantum field theory, the CPT theorem says that the only way to relate a chiral world to its mirror image by a symmetry is antiunitary, and that the same operation reverses the direction of time. Corollary 9.7 is that structure, exactly, in the forty-point geometry: the operations that reverse time's orientation are the antiunitary ones, and they are precisely the ones that exchange left and right. Choosing an orientation of time does not choose a chirality — but it makes chirality a conserved label, and the world of opposite chirality is the time-reversed one. This answers, in the finite model,

the question of whether the datum that fixes the arrow and the datum that fixes handedness are related: they live in the same coset. **BRIDGE**

The coincidence in Corollary 9.7 is forced rather than accidental. The similitude multiplier  $\mu$  (defined by  $M^T J M = \mu J$ ) maps  $\mathrm{PGSp}(4, 3)$  onto  $\mathbb{Z}_2$  with kernel  $\mathrm{PSp}(4, 3)$ . Because  $\mathrm{Sp}(4, 3)$  is perfect — its commutators generate all 51 840 of its elements —  $\mu$  is the *only* nontrivial sign character on the symmetries of the forty points. Holotrade had already made this identification (5419c27, 98cafff): frame orientation parity is the index-two character of  $W(E_6)$ , and the Pass 346 half-spin swap is the same character. It also separated this relative chirality from two other  $\mathbb{Z}_2$  labels that are *not* characters: the point/line type, which is invariant and can be selected, and the local charge conjugations of the 45 factorisations. The other track later found two further labels. Its cubic “tick” orientation is again  $\mu$ . Its chamber-sheet sign is flipped by elements with  $\mu = +1$ , so it is defined only relative to a chosen flag. So however the geometry is probed, it has exactly one global sign to fix: which chirality goes with which direction of time. **THEOREM** **COMPUTATION** (w33\_pass11091\_one\_global\_orientation\_bit.py)

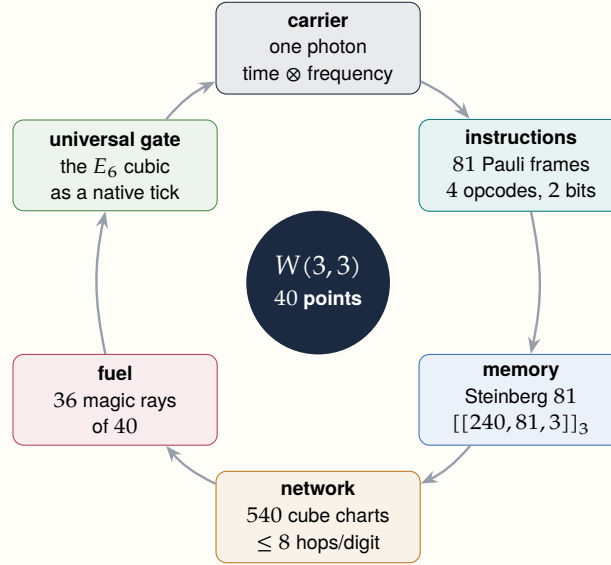
## 9.6 What we do not count as evidence

The older manuscripts contain many numerical coincidences:  $1/\alpha \approx 137$  written as a combination of cyclotomic values of  $q = 3$ , spectral indices of the cosmic microwave background, spanning-tree counts read as gravitational entropies. Some are striking. None is derived from a dynamical principle with its uncertainties, and a finite geometry with many natural integers can match many targets. We therefore do not use any of them as evidence in this paper. (The value  $\sin^2 \theta_W = 3/8$  at unification, often quoted alongside them, is the standard prediction of any  $SU(5)$  or  $SO(10)$  grand unification and is not specific to  $W(3, 3)$ .) **BRIDGE** at most.

*The geometry carries the Standard Model’s shapes — its gauge algebra’s dimensions, three equal matter sectors, a chiral spinor, a matter parity — and a string compactification built on its twist produces genuine three-family Standard Models whose Higgs sits in the “qutrit plane”. It cannot choose the handedness of the weak force — only an antiunitary, CPT-type symmetry exchanges the two — and, exhaustively, none of its supersymmetric singlet vacua preserves a matter parity: the anomalous  $U(1)$  forces a  $B - L = \pm 1$  condensate.*

## 10 The machine: a computer read off the geometry

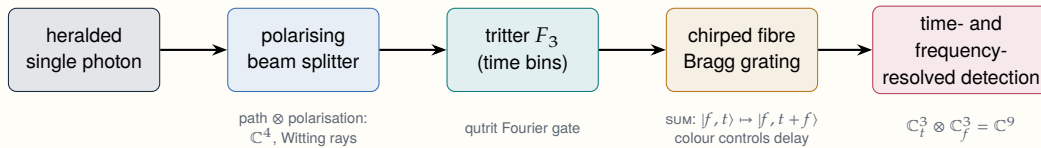
The same forty points that carry  $E_8$  and a light cone also describe, with no design choices, a quantum computer: an instruction set, a memory, a network and a supply of the non-classical resource that makes quantum computation powerful. The Holonet papers develop this at length; here we give the parts that are exact, and mark carefully what has and has not been built.



**Figure 33.** The Holonet stack. Every layer is a structure of  $W(3,3)$  read as engineering: the Pauli frames are the 81 points of  $\mathbb{F}_3^4$ , the memory is the Steinberg module, the network is the geometry’s atlas of cube charts, the fuel is its set of magic states, and the universal gate is the  $E_6$  cubic of Section 5.

### 10.1 The carrier: two qutrits in one photon

A single photon has several independent degrees of freedom. Two of them — which of three *time bins* it arrives in, and which of three *frequency bins* it occupies — give two qutrits, and together the  $\mathbb{C}^9$  on which the two-qutrit Paulis of Section 2 act. Its path and polarisation give a second register,  $\mathbb{C}^4$ , whose 40 Witting rays realise  $W(3,3)$  as *states* (Theorem 5.3 and the box after it).



**Figure 34.** One photon, two qutrits. The controlled-add (SUM) gate between the frequency and time qutrits has been demonstrated with a single passive element, a chirped fibre Bragg grating, at basis fidelity  $0.92 \pm 0.01$  [25].

### 10.2 The instruction set

The machine’s classical control state is a *Pauli frame*: a label  $x \in \mathbb{F}_3^4$  saying which Pauli correction is pending. There are 81 frames. Instructions are symmetries of the geometry: the two Fourier gates, the two phase gates, the two controlled-add gates, and the Pauli translations.

**Theorem 10.1 (the complete instruction group)**

COMPUTATION

The six Clifford opcodes generate exactly  $\text{Sp}(4, 3)$  (order 51 840); adjoining one translation generates the affine group

$$\text{ASp}(4, 3) = \mathbb{F}_3^4 \rtimes \text{Sp}(4, 3), \quad |\text{ASp}(4, 3)| = 81 \times 51\,840 = 4\,199\,040.$$

No one or two linear opcodes suffice; exactly six triples do, so *four* instructions (three Clifford, one translation) in a *two-bit* opcode field generate everything, with worst-case program length 19 for the best triple. The minimal arithmetic core synthesises to 43 logic cells at 208.86 MHz on an iCE40 HX8K FPGA. `holonet_machine_blueprint_body.tex`, Passes 2778, 2789, 2882

**In plain words**

Four instructions are enough to perform every one of more than four million possible frame transformations, and no program needs more than nineteen steps. The instruction set was not designed: it is the list of symmetries of the forty-point shape, and the theorem says which symmetries you can leave out.

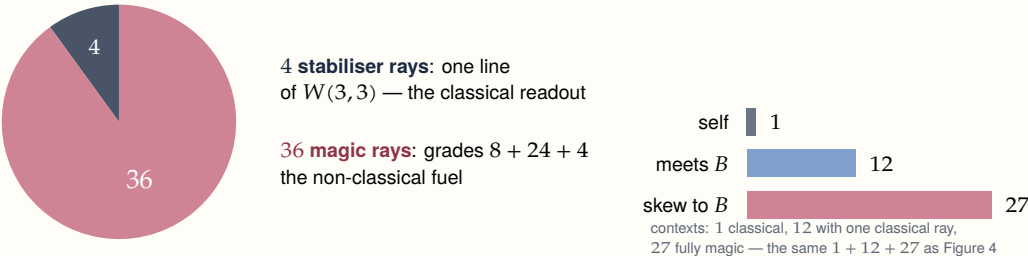
**10.3 The fuel: matter is magic**

Clifford operations alone can be simulated efficiently on an ordinary computer; quantum advantage needs a non-Clifford resource, usually supplied as *magic states*; contextuality is what supplies them [23].

**Theorem 10.2 (magic is everything except one line)**

COMPUTATION

Of the 40 Witting rays, exactly the 4 computational-basis rays are stabiliser states; the other 36 are magic. The 36 split by stabiliser fidelity into grades 8 + 24 + 4. A classical  $\{0, 1\}$  assignment can satisfy the “exactly one” rule on at most 36 of the 40 contexts, so the contextual fraction is exactly  $1/10$ ; the independence number of the collinearity graph is 7, attained by heptads of seven rays with all pairwise overlaps  $1/3$ . `bt822_magic_census_matter_is_magic.py`, `bt818_ovoid_nogo_theta_gap.py`



**Figure 35.** The machine’s quantum resource is the geometry itself: remove one line and every remaining point is a magic state.

**The cubic gate.** Clifford gates are “degree two”: they act on frame labels by linear symplectic maps and on phases by quadratic forms. The classical continuous-variable theorem of Lloyd and Braunstein [34] says that quadratic operations plus a single cubic one are universal. In Section 5 the cubic appeared as the  $E_6$  form  $N$ , the rule by which three first-order moves reach the top of the ladder. Pass 10947 compiles that signed cubic into a native gate on a seven-qutrit virtual machine: ten operations in seven parallel layers, both ancillas returned to zero. **COMPUTATION** (`w33_pass10947_five_front_execution.py`) The same object is, in the language of Section 5, the Yukawa coupling of  $E_6$  unification, and in the language of this section, the machine’s universal non-Clifford gate. **BRIDGE**

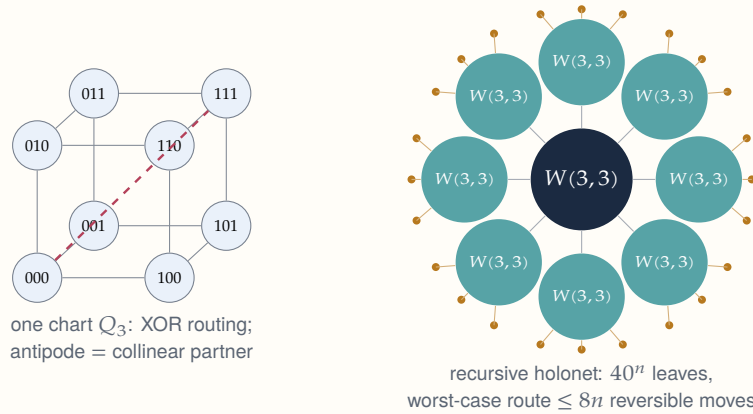
## 10.4 Memory and error correction

### Theorem 10.3 (the memory is the Steinberg module)

**COMPUTATION**

The CSS code  $[[240, 81, 3]]_3$  on the edges of  $W(3, 3)$ , with distances  $(d_X, d_Z) = (3, 4)$ , stores its 81 logical qutrits in  $H_1(W(3, 3))$ , which is the irreducible Steinberg module of  $\mathrm{PSp}(4, 3)$ . By Schur’s lemma any symmetry-covariant logical error acts as a scalar. Separately, a ternary Golay  $[[11, 1, 5]]_3$  lane with two verified syndrome rounds has 552 fault locations in a declared model, 2 576 205 malignant fault triples, and a conservative threshold crossing at  $p \approx 5.32 \times 10^{-4}$ . `bt861_code_register_is_steinberg.py`, `w33_pass10947_five_front_execution.py`

## 10.5 The network



**Figure 36.** Left: a local routing chart. Each of the 540 pairs of disjoint lines of  $W(3, 3)$  carries a cube of eight points with XOR addressing. Right: substituting a copy of  $W(3, 3)$  at every point gives a network with  $40^n$  leaves and logarithmic routing.

$W(3, 3)$  contains exactly 540 pairs of disjoint lines; each pair’s eight points carry the cube graph  $Q_3$ , in which the antipode of a vertex is its unique collinear partner. The charts are glued along the 1620 apartments of the Tits building of  $\mathrm{Sp}(4, 3)$ . A route needs at most 3 in-chart moves plus 5 chart hops, so each digit of a recursive address costs at most 8 reversible moves. **COMPUTATION** (`bt773_involution_cube_theorem.py`, `bt777_hypercube_chart_web.py`, `bt827_holonet_fractal_architecture.py`)

**What is measured, what is proved, what is not built**

**Measured in a laboratory (prior art):** the single-photon time–frequency sum gate, fidelity  $0.92 \pm 0.01$  [25]. **Proved exactly:** the group theory, the instruction semantics, reachability, code parameters, routing bounds, and FPGA synthesis figures. **Not built:** an integrated device; a magic-state injection circuit with a threshold; a physical demonstration of the network. The machine is a specification whose every number can be checked, not a device.

*The forty points are also a computer: 81 frames, four instructions generating 4 199 040 operations, a memory that is the Steinberg module, a network of 540 cube charts, and a quantum fuel that is every point off one line. The  $E_6$  cubic that is the Yukawa coupling in physics is the universal gate in the machine.*

## 11 Epilogue: what a theory of everything still needs

We began with a qutrit acting on itself and followed one chain of exact identifications: forty points; a finite present with a light cone; four clocks;  $E_8$  from nine histories; the Freudenthal quartic and a 57-dimensional light cone; a real form in which time has one dimension; the Albert algebra, the Lorentz algebra  $\mathfrak{so}(1, 9)$ , a chiral spinor, and matter parity as a  $2\pi$  rotation. On top of that chain a heterotic string construction produces three-family Standard Models, and the same geometry specifies a quantum computer.

| exact                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | bridge / hosted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | open                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• <math>W(3, 3)</math> as qutrit commutation geometry</li> <li>• Bell line = unitary channels</li> <li>• 27-event light cone, <math>t^2 - x^2 - y^2</math></li> <li>• tetracode = four clocks</li> <li>• <math>E_8 = 80 + 84 + 84</math>; <math>240 = 40 \times 6</math></li> <li>• clock gradings; cubic ladder</li> <li>• Freudenthal quartic; 57-cone</li> <li>• clock real form <math>\Rightarrow (1, 9)</math></li> <li>• <math>F_4</math>, <math>E_{6(-26)}</math>, <math>\mathfrak{so}(1, 9)</math>, 16</li> <li>• matter parity = <math>2\pi</math> rotation</li> <li>• chirality no-go (antiunitary swap)</li> <li>• string route closed: no parity-keeping vacuum in 7 families</li> <li>• ISA, memory, magic census</li> </ul> | <ul style="list-style-type: none"> <li>• “null” = shared measurement</li> <li>• lifted history = proper time</li> <li>• forgetting 2 trits = arrow</li> <li>• cubic ladder = cubic time</li> <li>• quartic = black-hole entropy</li> <li>• <math>1 + 3 + 8</math> = gauge shape</li> <li>• <math>27 + 27 + 27</math> = generations</li> <li>• <math>W(3, 3)</math> twist <math>\Rightarrow</math> heterotic SMs</li> <li>• Higgs in the qutrit plane</li> <li>• SM breaking = CZ gate</li> </ul> | <ul style="list-style-type: none"> <li>• a dynamical principle</li> <li>• a continuum limit; gravity</li> <li>• which chirality, and why</li> <li>• a stable universe beyond heterotic orbifolds</li> <li>• masses, mixings, couplings</li> <li>• the 54 Fourier intertwiner</li> <li>• magic-state injection</li> <li>• a built device</li> </ul> |

**Figure 37.** The state of the programme, in three columns.

### 11.1 What stands, and what closed

Two different kinds of result are in this paper, and they deserve to be read differently.

*The mathematics stands.* Everything in Sections 2–8 is exact and independent of any physical

reading: the forty points as the commutation geometry of a qutrit acting on itself, the finite light cone, the four clocks and the tetracode,  $E_8$  from nine histories and its 6:1 fibration over the forty points, the Freudenthal quartic and its 57-dimensional cone, the real form in which time has one dimension, and the Albert algebra with its  $\mathfrak{so}(1,9)$  and chiral **16**. None of it depends on a string construction, and none of it was touched by the corrections below.

*The string route closed.* Section 9 asked whether heterotic orbifolds carrying the forty-point twist give our universe. The answer, reached by exact computation over nine orbifold families and checked against the selection rules the worldsheet establishes, is no in all nine: in eight, their supersymmetric vacua cannot keep the proton stable; in the ninth,  $\mathbb{Z}_3 \times \mathbb{Z}_3$ , the parity survives and the vacua are supersymmetric, but a counting argument over hidden representations leaves six massless particles of charge  $1/3$  in every one. That is a finished negative result — it tells the programme where the forty points do *not* lead, and it isolates what a successful route must do differently: provide a vacuum in which the Fayet–Iliopoulos term is cancelled by something other than a sneutrino-like field, with a hidden sector whose representations pair up.

## 11.2 Seven open problems, stated sharply

**1. A realistic vacuum.** Every parity built from the gauge torus, the space group and the established rotation symmetries is excluded in  $\mathbb{Z}_6$ -I,  $\mathbb{Z}_6$ -II,  $\mathbb{Z}_{12}$ -I (where the  $E_6$  lattice itself fixes the admissible rule),  $\mathbb{Z}_2 \times \mathbb{Z}_6$ ,  $\mathbb{Z}_3 \times \mathbb{Z}_6$  and  $\mathbb{Z}_6 \times \mathbb{Z}_6$ . The one family where it survives is  $\mathbb{Z}_3 \times \mathbb{Z}_3$ : its 28 parity vacua are supersymmetric ( $W = F(p)$ ,  $\mathbb{Z}_3$ -graded) but carry six massless charge- $\frac{1}{3}$  states; its 54 parity vacua with the hidden  $SU(4)$  condensed cure that count but leave at least 120 fractional charges massless, protected by an unbroken discrete 3-group. Baryonic and mixed condensates do not free them, and neither does any resolution of the orbifold or any breaking of supersymmetry. The 87 non-supersymmetric twins of the  $\mathbb{Z}_6$ -I Standard Models are consistent three-generation models, but every one is tachyonic. What remains:

- for the  $SO(16) \times SO(16)$  models with the  $W(3,3)$  shift, a source of the light-generation hierarchies (the  $\Delta(54)$  flavour symmetry keeps  $m_c = m_u$  and  $m_\mu = m_e$  at the same order under every alignment of the scalar vacuum, and the stabilised family modulus  $T = \rho$  gives  $m_c = m_u = m_t/2$ ) and a vacuum at all (no minimum in the tachyon-free region: the Wilson-line moduli roll into a winding tachyon and the dilaton runs to weak coupling—but 21 survivors have a neutral exit and in six of them every nearby instability is Standard-Model-neutral; there the potential drives them steeply to it, the condensate never touches the up sector but, for a single light Higgs, lowers  $m_b/m_t$  to  $\epsilon^2$  with the tau at the same order in 14 of 21 models; the exit is SM-neutral on the whole  $B$  circle (exact hyperbolic disks at the  $\Gamma_0(3)$  points, the duality group itself), its endpoint is the non-supersymmetric  $U(16)$  string, not a nearby vacuum, every symmetric point of the modulus is tachyonic or a runaway, and the light generations stay degenerate under every vacuum and Higgs choice tried);
- twists and lattices not yet scanned;
- a mechanism other than a parity for proton stability.

**2. A dynamical principle.** Every result in this paper is kinematics. What action, Hamil-

tonian or spectral principle on  $W(3,3)$  has the structures found here as its symmetries, and what does it predict? In particular: does such a principle select the clock real form  $E_{8(-24)}$  over the split form — does the geometry *choose* one time?

3. **A continuum limit.** The finite light cones (in  $2 + 1$ ,  $3 + 1$ ,  $1 + 9$  and the quartic 57) are exact algebraic objects. Is there a limit, a tower over  $W(3, q)$  or a lattice construction, in which one of them becomes a smooth Lorentzian manifold with dynamical curvature?
4. **Chirality from outside.** Corollary 9.7 shows that reversing time's orientation, acting antiunitarily and exchanging chirality are one coset, so fixing an arrow makes chirality conserved but does not choose it. What physical datum correlates the two — which chirality goes with which direction of time — and can it come from a vacuum? (There is exactly one such global sign to fix: the symplectic multiplier is the only nontrivial character of  $\mathrm{PGSp}(4, 3)$ ; Holotrade 98cafff, P11091.)
5. **The 54 intertwiner.** The diagonal clock/phase weld projects onto 54 Fourier directions, while the clock grading supplies a canonical  $27 \otimes 2$  at degree one. Is there an explicit equivariant isomorphism?
6. **Numbers.** The up-type Yukawa couplings of the  $\mathbb{Z}_6$ -I class have a  $2 + 1$  block structure qualitatively like the observed quark mixing. Can any mass ratio or mixing angle be computed, with an error, from a vacuum of the class?
7. **The machine.** Specify a magic-state injection circuit for the 36 Witting rays with a decoder and a threshold, and build the two-qutrit photonic core.

### 11.3 Corrections we kept on the page

A research record is trustworthy only if it keeps its errors. These were believed, written down, and then shown false; each correction made the structure *more* natural, not less.

| Was believed                                                            | Is true                                                                                                                |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| there is a natural bijection 240 edges $\leftrightarrow$ 240 roots      | no equivariant bijection; a 6:1 fibration of roots over <i>points</i> (§5.4)                                           |
| the Freudenthal 56 is the clock's 54 plus 2                             | the 56 contains only one 27 of the 54 (§6.1)                                                                           |
| the Albert algebra has "27 minimal idempotents" as frames               | the frames are the 45 triads (§6.4)                                                                                    |
| three time bins hold a past and a future qutrit                         | $\mathbb{C}^3$ cannot hold two qutrits; use time $\otimes$ frequency (§3.4)                                            |
| the 9 fixed modes of the spectral clock are the qutrit's 9 histories    | same count, different module (§4.4)                                                                                    |
| the independence number of $W(3,3)$ is 10 (an ovoid)                    | it is 7; $W(3,3)$ has no ovoid (§10.3)                                                                                 |
| at most 34 of 40 contexts are classically satisfiable                   | exactly 36 (§10.3)                                                                                                     |
| the $\mathbb{Z}_6$ -I class is excluded unconditionally                 | excluded only if matter parity is broken (§9.3)                                                                        |
| every $D$ -flat support contains a matter-odd singlet (from 60 samples) | exhaustively true, and the odd singlet always has $B - L = \pm 1$ (§9.4)                                               |
| the three families come from $W(3,3)$                                   | in the orbifolds they come from the three tori (§9.1)                                                                  |
| directions each compatible with a parity form a parity-even vacuum      | they must be even under <i>one</i> element; the 84-direction vacuum was withdrawn, a 68-direction one certified (§9.4) |
| the Higgs quartic is positive at unification (+1/40)                    | measured running gives $-0.011$ at $10^{16}$ GeV (§9.4)                                                                |
| a parity from plane rotations reopens one model                         | it used an $R$ -rule valid only on prime planes; with the established rules it closes (§9.4)                           |

*The forty points do not yet give a theory of everything. They give something rarer: a single finite object in which quantum measurement, a light cone, clocks,  $E_8$ , the octonionic Albert algebra, ten-dimensional Lorentz symmetry, a chiral spinor, matter parity and a quantum computer are exact aspects of one structure — with every claim checkable, and the remaining distance to physics mapped rather than hidden.*

## A The ledger: every claim, its status, its witness

Every script named below lives in `analysis/` of the repository `wilcompute/W33-Theory` and writes its certificate to `data/`; most have a regression test in `tests/`. Commit hashes marked "Holotrade" refer to the second research track and are recorded row by row in the ledger of the  $W(3,3)$  paper.

| Claim                                                                                                                                                                                                                | Badge               | Witness                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------|
| <i>The object and one qutrit (§§2–3)</i>                                                                                                                                                                             |                     |                                                         |
| $W(3, 3)$ : 40 points, 40 lines,<br>SRG(40, 12, 2, 4), spectrum<br>$12, 2^{24}, (-4)^{15}$                                                                                                                           | THEOREM             | Props. 2.1–2.2 (proofs in text)                         |
| two-sided Weyl maps span $\mathfrak{sl}_9$ ;<br>commutation geometry is $W(3, 3)$                                                                                                                                    | THEOREM COMPUTATION | w33_20260924_temporal_<br>su9_e8_compiler.py            |
| Bell line = unitary-channel diagonal;<br>swap is anti-symplectic                                                                                                                                                     | THEOREM COMPUTATION | w33_20260924_liouville_<br>swap_cp_outer_bridge.py      |
| transpose induces the outer class (up<br>to labelling gauge)                                                                                                                                                         | COMPUTATION         | w33_20260924_intrinsic_<br>transpose_outer.py           |
| memory witness: Bell fidelity<br>$1 - 8p/9$ , threshold $p = 3/4$                                                                                                                                                    | COMPUTATION         | w33_20260924_temporal_<br>quantum_memory_<br>witness.py |
| modular flow of a qutrit state has $A_2$<br>frequency pattern                                                                                                                                                        | COMPUTATION         | w33_20260924_modular_a2_<br>event_clock.py              |
| space and time are two $GQ(3, 3)$ on<br>the same 40 points sharing the 16<br>product lines; partial transpose =<br>bijection 216 entangled states $\leftrightarrow$ 216<br>Clifford gates; common symmetries<br>2304 | THEOREM COMPUTATION | w33_pass11143_space_<br>time_two_quadrangles.py         |
| CGLMP: $W(3, 3)$ resources give<br>exactly 2 in space and time; onset<br>$2 + \frac{\pi^2}{6}t^2$ ; one qutrit in time<br>$3.1628 > 2.9149$                                                                          | COMPUTATION         | w33_pass11144_<br>spookiness_budget.py                  |
| thermofield double = Choi state of<br>Euclidean half-evolution; negativity<br>$((\sum \sqrt{p_i})^2 - 1)/2$                                                                                                          | COMPUTATION         | w33_pass11146_thermal_<br>reading.py                    |
| polygamy in time: pairwise temporal<br>negativities sum to 3; space $\leq 1.139$ ,<br>$N_{AB} + N_{AC} \leq 4/\sqrt{15}$                                                                                             | COMPUTATION         | w33_pass11148_polygamy_<br>in_time.py                   |
| temporal CGLMP beats spatial for<br>every $d \geq 3$ (3.163, 3.283, 3.377 vs<br>$1 + \sqrt{11/3}$ , 2.973, 3.016)                                                                                                    | COMPUTATION         | w33_pass11147_temporal_<br>cglmp_by_dimension.py        |
| Pauli-only CGLMP optimum<br>$2\sqrt{2} \cos(\frac{1}{3} \arccos \frac{1}{2\sqrt{2}})$ (root of<br>$x^3 - 6x - 2$ ): magic in the state suffices                                                                      | COMPUTATION         | w33_pass11149_magic_for_<br>spookiness.py               |
| perfect space–time gates:<br>$\det S_{AA} + \det S_{BA} = 1$ ; $13824 = 24^3$<br>perfect (AME(4,3)), one local orbit;<br>tetra-code gate is the time-reversing<br>representative                                     | THEOREM COMPUTATION | w33_pass11156_perfect_<br>spacetime_gates.py            |
| Pauli-only Bell operator:<br>characteristic polynomial<br>$(x^3 - 6x - 2)(x^3 - 3x + 1)^2$ exactly                                                                                                                   | COMPUTATION         | w33_pass11152_pauli_<br>bell_charpoly.py                |

| Claim                                                                                                                                                                      | Badge               | Witness                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------|
| in time, Pauli measurements give exactly 2 for every start: magic must sit in the measurements                                                                             | COMPUTATION         | w33_pass11153_temporal_magic_budget.py                |
| $4/\sqrt{15}$ optimum: spectra $(7, 4, 4)/15$ , $(11, 2, 2)/15$ ; $N_{AB} = N_{AC} = 2/\sqrt{15}$ , $N_{BC} = 1/15$                                                        | COMPUTATION         | w33_pass11154_polygamy_optimum_structure.py           |
| temporal CGLMP $d = 6, 7$ : 3.453, 3.517 (spatial 3.050, 3.078); deficit from 4 $\sim d^{-0.65}$                                                                           | COMPUTATION         | w33_pass11155_temporal_cglmp_trend.py                 |
| ninth roots explained:<br>$B^3 - 3B + 1 = (B + 1)(I + U + U^2)$ ,<br>$U = (XZ^2) \otimes (XZ^2)$                                                                           | COMPUTATION         | w33_pass11157_ninth_roots_explained.py                |
| general law $\sum_i \det S_{ij} = 1$ ; perfect three-qutrit gates exist ( $\approx 3.1\%$ of $\text{Sp}(6, 3)$ ), columns $(1, 1, -1)$                                     | THEOREM COMPUTATION | w33_pass11158_perfect_three_qutrit_gates.py           |
| $4/\sqrt{15}$ derived:<br>$N(x) = \sqrt{(1 + 15x)(1 - x)}/4$ , max at $x = 7/15$ ; explicit optimal state                                                                  | COMPUTATION         | w33_pass11159_explicit_polygamy_state.py              |
| temporal CGLMP $d = 8, 9, 10$ : 3.576, 3.621, 3.658 (spatial 3.101, 3.122, 3.140); deficit $\sim d^{-0.75}$                                                                | COMPUTATION         | w33_pass11160_temporal_cglmp_high_d.py                |
| $I_3 = 2 - \text{rank } S_{AA} - \text{rank } S_{BA} \leq 0$ for every two-qutrit Clifford gate; perfect = maximal scramblers; tick + forgetting the partner = Theorem 4.7 | COMPUTATION         | w33_pass11161_scrambling_and_arrow.py                 |
| $4/\sqrt{15}$ a certified local maximum: partial-transpose gap $2/15$ , Hessian 32 negative, 20 flat = local-unitary orbit; 96/96 restarts reach it                        | COMPUTATION         | w33_pass11162_polygamy_local_maximum.py               |
| exactly $286\,654\,464 = 2^{17}3^7$ perfect three-qutrit gates, 128/4095 of $\text{Sp}(6, 3)$ (control: 13824 for two)                                                     | COMPUTATION         | w33_pass11163_perfect_three_qutrit_count.py           |
| no $\mathbb{F}_3$ tetracode analogue for three qutrits; over $\mathbb{F}_9$ , 12288 perfect unitaries in $U(3, \mathbb{F}_9)$ , e.g. $K = P^2 + i1$                        | COMPUTATION         | w33_pass11164_three_qutrit_tetracode.py               |
| perfect three-qutrit tick secret-shares each past (0 to every proper subset of futures, 2 to all three); cf. [54]                                                          | COMPUTATION         | w33_pass11165_three_qutrit_arrow.py                   |
| spectral clock tick = Clifford propagator $e^{-iq(p)}$ ; translation-invariant clocks never perfect (off-diagonal rank $\leq 1$ )                                          | COMPUTATION         | w33_pass11166_clock_tick_clifford.py                  |
| point/line firewall (27/108/36 coincidences, non-isomorphic)                                                                                                               | COMPUTATION         | w33_20260924_history_pointline_cospectral_firewall.py |
| <i>Time (§4)</i>                                                                                                                                                           |                     |                                                       |

| Claim                                                                                                     | Badge               | Witness                                                       |
|-----------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------|
| $40 = 1 + 12 + 27$ ; big cell of $Q(4, 3)$                                                                | COMPUTATION         | w33_20260924_history_bigcell_q43_compactification.py          |
| finite light cone $t^2 - x^2 - y^2$ ;<br>$1 + 8 + 6 + 12$                                                 | THEOREM COMPUTATION | same                                                          |
| $\text{Sym}_2(\mathbb{F}_3) = \mathbb{F}_3^4/\langle 1111 \rangle$ ;<br>$D_0 + D_\infty + D_+ + D_- = 3I$ | THEOREM COMPUTATION | w33_20260924_positive_null_counter_cover.py                   |
| null graph spectra; $U^3 = I$ at<br>$t^* = 2\pi/9$                                                        | COMPUTATION         | w33_20260924_null_history_spectral_clock.py                   |
| fixed 9 is not the qutrit module                                                                          | REFUTED             | w33_20260924_spectral_clock_fixed9_group_probe.py             |
| 36 triangles; unique orientation cycle;<br>$82 = 1 + 81, 81 = 35 + 46$                                    | COMPUTATION         | w33_20260924_history_invariant_cycle_orientation.py           |
| filled temporal $H_1$ injects into<br>$H_1(W(3, 3))$                                                      | COMPUTATION         | w33_20260924_history_to_global_h1_intertwiner.py              |
| universal cover: 8-regular tree,<br>$\pi_1 = F_{82}$                                                      | THEOREM COMPUTATION | w33_20260924_temporal_universal_cover.py                      |
| Weil phase census $1^{13}i^4(-1)^6(-i)^4$ ;<br>$(FG)^3 = iF^2$                                            | COMPUTATION         | w33_20260924_temporal_maslov_mu12.py                          |
| common $S_4$ of null directions, Hesse<br>clocks, $A_2$ factors                                           | COMPUTATION         | w33_20260924_null_hesse_4a2_s4_intertwiner.py                 |
| tetracode = clock evaluation code                                                                         | COMPUTATION         | w33_pass10946_clock_code_cone_objectwise.py                   |
| negacyclic hyperbolic pair, $N^4 = -I$<br>(other track)                                                   | COMPUTATION         | w33_pass10948_clock_tetracode_negacyclic_hyperbolic_bridge.py |
| Page–Wootters diagonal weld; 24 vs<br>248                                                                 | COMPUTATION         | w33_20260924_page_wootters_diagonal_weld.py                   |
| Hermitian 3 + 1: $21 + 30 + 30$ ; null<br>directions = spread                                             | COMPUTATION         | w33_20260924_hermitian_3pl_spread_bridge.py                   |
| no compatible complex structure $J$                                                                       | REFUTED             | same                                                          |
| forgetting two trits gives the<br>depolarising channel                                                    | THEOREM             | Thm. 4.7 (proof in text);<br>bt823_the_closure.py             |
| $E_8$ (§5)                                                                                                |                     |                                                               |
| $E_8 = \mathfrak{sl}_9 + \Lambda^3 + \Lambda^6$ as packet<br>operations                                   | COMPUTATION         | w33_20260924_temporal_a8_e8_process_bracket.py                |
| Hesse $4A_2$ ; $240 = 72 + 24 + 144$                                                                      | THEOREM COMPUTATION | w33_20260924_temporal_hesse_4a2_a8_bridge.py                  |
| no equivariant edge–root bijection                                                                        | COMPUTATION         | w33_pass1020_e8_transitive_51840.g                            |
| $240 = 40 \times 6$ fibration onto points                                                                 | COMPUTATION         | w33_pass1021_e8_fibration_over_forty.g                        |

| Claim                                                                                                               | Badge               | Witness                                                                                                |
|---------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------|
| Hesse clock gradings<br>$1+56+134+56+1$ and $ 3 $                                                                   | COMPUTATION         | w33_20260925_hesse_clock_e8_gradings.py                                                                |
| $\mathbb{Z}_3$ grading is the mod-3 shadow                                                                          | COMPUTATION         | w33_20260925_e8_parabolic_cubic_clock_lift.py                                                          |
| cubic ladder $54 \rightarrow 81 \rightarrow 83$ ; 45 triads $\times 12$                                             | COMPUTATION         | w33_20260925_cubic_clock_bracket_ladder.py                                                             |
| <i>Freudenthal, real forms, Albert</i> (§§6–8)                                                                      |                     |                                                                                                        |
| 56 shares one 27 with the 54;<br>$54 + 2 = 56$ refuted                                                              | COMPUTATION REFUTED | w33_pass10949_freudenthal_quasiconformal_clock_cone.py                                                 |
| Freudenthal normal form of the quartic, no sign repair                                                              | COMPUTATION         | same                                                                                                   |
| 57-cone, Heisenberg law, GKN inversion                                                                              | COMPUTATION         | same                                                                                                   |
| $K(e_\alpha, e_{-\alpha}) = -60(-1)^{\text{ht } \alpha}$ ; split = height twist                                     | COMPUTATION         | same                                                                                                   |
| tick lines: $E_{8(-24)}$ , Klein four-group, $80 + 3 \times 56$                                                     | COMPUTATION         | same                                                                                                   |
| Euclidean Albert algebra, Peirce $1 + 16 + 10$ , $(1, 9)$ vs $(5, 5)$                                               | COMPUTATION         | same                                                                                                   |
| $\text{Der} = F_{4(-52)}$ , $\text{str}_0 = E_{6(-26)}$                                                             | COMPUTATION         | w33_pass10950_clock_albert_lorentz_spinor.py                                                           |
| $\mathfrak{so}(1, 9)$ , Majorana–Weyl 16;<br>$16 = (2, 4) + \overline{(2, 4)}$                                      | COMPUTATION         | same                                                                                                   |
| $Q_\psi = 6L_c - 2$ ; matter parity<br>$= \exp(2\pi D)$                                                             | COMPUTATION         | same                                                                                                   |
| clock $g^4 = -I$ is the central $2\pi$ spin sign (other track)                                                      | COMPUTATION         | w33_pass10951_clock_pin_spin_central_sign_bridge.py                                                    |
| Albert $16 = 8_s + 8_c$ ; half-turn<br>$U^2 = -I_{16}$ ; minimal full-clock carrier<br>$16 \oplus 16$ (other track) | COMPUTATION         | w33_pass10956_albert_spin8_halfspin_normalizer.py, w33_pass10959_albert_clock_extension_obstruction.py |
| <i>Physics</i> (§9)                                                                                                 |                     |                                                                                                        |
| transvection spine: $648, 1 \oplus 3 \oplus 8, 9 \oplus 9 \oplus 9$                                                 | COMPUTATION BRIDGE  | bt886_standard_model_spine.py                                                                          |
| Steinberg: every order-3 element splits $27 + 27 + 27$                                                              | THEOREM             | Thm. 9.2 (proof in text)                                                                               |
| no Pauli-spanned $E_6$ on qutrit registers                                                                          | THEOREM COMPUTATION | Holotrade 1cde1f1                                                                                      |
| $W(3, 3) = \text{Pauli geometry of } \mathbb{Z}_3\text{-twisted } V_{E_8}$                                          | COMPUTATION         | Holotrade 68df33f                                                                                      |
| order-3 orbifold completes no SM family                                                                             | THEOREM COMPUTATION | Holotrade ed1eaca, a6e1c69, a779053                                                                    |

| Claim                                                                                                                                                   | Badge               | Witness                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------|
| $\mathbb{Z}_6$ : 42 of 79 SMs from the $W(3,3)$ vacuum                                                                                                  | COMPUTATION         | Holotrade 5b3f3ad                                                                                                 |
| CZ-type breaking 45/45;<br>$SU(3) \times SU(2)$ cut 25/25                                                                                               | COMPUTATION         | Holotrade d772153, 94df376                                                                                        |
| Higgs alone in the qutrit plane; DTS 55/87                                                                                                              | COMPUTATION         | Holotrade 3caf15e, a6f1cae                                                                                        |
| $F/D$ conflict; Gordan alternative 215/215                                                                                                              | COMPUTATION         | Holotrade babfd48                                                                                                 |
| matter parity in all 87; class closure conditional                                                                                                      | COMPUTATION         | ledger commit 78ec0a0fc                                                                                           |
| every sampled $D$ -flat support has a matter-odd singlet                                                                                                | COMPUTATION OPEN    | ledger commit a3dbdf2d7                                                                                           |
| chirality no-go                                                                                                                                         | THEOREM COMPUTATION | w33_pass346_the_chirality_no_go.py                                                                                |
| chirality swap is exactly the antiunitary (determinant-odd) coset                                                                                       | COMPUTATION BRIDGE  | Holotrade e92a047; w33_pass10952_clock_complete_positivity_firewall.py, w33_pass10955_d4_halfspin_clock_bridge.py |
| no matter-even SUSY singlet vacuum: 0/88 over full $B-L$ , 0/1024 $\mathbb{Z}_2$ parities; forced $B-L = \pm 1$ condensate                              | COMPUTATION         | w33_pass10960_matter_even_dflat_closure.py                                                                        |
| hidden sector: 32 exact (all directions), 55 over invariant generators, 1 by 62,309 rays; anomaly-free models have no parity                            | COMPUTATION         | w33_pass10962_hidden_sector_dflat_tiers.py, w33_pass10962_tierc_rays.py                                           |
| time reversal = antiunitary = chirality swap (one coset)                                                                                                | THEOREM             | Corollary 9.7 (proof in text)                                                                                     |
| FI acts through $f = 4Q_\psi - \frac{4}{5}Q_\chi$ ; singlet values $\{-8, 0, 8\}$ ; forced field = $\nu^c$ of $\mathbf{16}_{-3} \subset \mathbf{78}$    | COMPUTATION BRIDGE  | w33_pass10964_e6_shaped_fi_mechanism.py                                                                           |
| Baryon triality impossible in 215/215: $Q(u^c) + Q(e^c) = 2Q(q)$ ( $SU(5)$ )                                                                            | COMPUTATION         | w33_pass10965_baryon_triparity_impossible.py                                                                      |
| Most general matter parity (torus $\times$ space group) survives FI in 0/215; forced $Q(n) = -Q(u^c d^c d^c)$ in 290/302; quartic RPV regenerated 23/23 | COMPUTATION         | w33_pass10967_fi_vacuum_regenerates_rpv.py                                                                        |
| (Withdrawn by P10974)                                                                                                                                   | COMPUTATION         |                                                                                                                   |
| Plane-rotation $R$ -parity under the old per-plane rule: 3/128; $\mathbb{Z}_6 \Pi_{23}$ 38 vacua, exotics massless                                      | COMPUTATION         | w33_pass10968_r_symmetry_parity_and_massless_exotics.py                                                           |
| 68-direction vacuum: one parity element, exotics full rank by order 5, Higgs rank 6                                                                     | COMPUTATION         | w33_pass10968_leaf1_vacuum_certificate.json                                                                       |
| 23-direction vacuum: exotics full rank, Higgs rank 5/6 at all orders, $W _S \neq 0$                                                                     | COMPUTATION OPEN    | w33_pass10968_approxR_vacuum_certificate.json                                                                     |

| Claim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Badge       | Witness                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------|
| $\mathbb{Z}_{12}$ -I: 289 SMs, 14 parity vacua, exotics massive in none; $\mathbb{Z}_2 \times \mathbb{Z}_6$ -I: 29 SMs, 0 survive                                                                                                                                                                                                                                                                                                                                                                                                                                         | COMPUTATION | w33_pass10968_z12_left_chiral_ledger.json.gz               |
| Harmless RPV needs<br>$\tilde{m} \gtrsim 6 \times 10^{14}$ GeV; $\lambda < 0$ above<br>$3.2 \times 10^{10}$ GeV, so $m_t \leq 171.0$ GeV<br>( $5.4\sigma$ low)                                                                                                                                                                                                                                                                                                                                                                                                            | COMPUTATION | w33_pass10969_rpv_heavy_superpartners_vs_higgs_quartic.py  |
| Established $R$ -rules (prime planes + $\gamma$ -corrected G2 plane): $\mathbb{Z}_6 \Pi_{23}$ closes, $\mathbb{Z}_6$ -II 0/128; $\mathbb{Z}_3 \times \mathbb{Z}_6$ 0/5                                                                                                                                                                                                                                                                                                                                                                                                    | COMPUTATION | w33_pass10974_valid_rules_close_the_rotation_door.py       |
| $C_{\text{Aut}(E_6)}(c) = \langle c \rangle \times \langle -1 \rangle$ : single admissible $\mathbb{Z}_2^R$ ; all 14 $\mathbb{Z}_{12}$ -I parity vacua decided, 0 viable                                                                                                                                                                                                                                                                                                                                                                                                  | COMPUTATION | w33_pass10978_z12_e6_admissible_r_rule.py                  |
| $T^6/\mathbb{Z}_3$ : 0 SMs in 64 000; $\mathbb{Z}_3 \times \mathbb{Z}_3$ (all rules established): 13 SMs, parity on FI rays in 3, 28 parity vacua, $W$ exponent vectors independent through degree 11 in all 28                                                                                                                                                                                                                                                                                                                                                           | COMPUTATION | w33_pass10979_z3xz3_parity_vacua_f_obstruction.py          |
| $\mathbb{Z}_3 \times \mathbb{Z}_3$ parity vacua: $W _S = F(p)$ , $F(\omega p) = \omega F(p)$ to all orders (28/28); SUSY roots exist (28/28, $\langle \phi \rangle \approx M_s$ ); hidden-refined index: 48 chiral charge- $\frac{1}{3}$ states, 6 hidden singlets, massless in every hidden-unbroken vacuum; hidden groups IR-free; 54 $SU(4)$ -condensed parity vacua cure the index but keep $\geq 120$ fractional states massless (unbroken discrete 3-group of order $3^4$ or $3^6$ ); exact-lattice certificate of the 10978–10980 integer programs (0 differences) | COMPUTATION | w33_pass11024_z3xz3_susy_vacua_fractional_charge_index.py  |
| “Flat-or-massive” is an unbroken $R$ -symmetry, not a law: where $W _S \equiv 0$ and the exotic triplet is massless, both are forbidden by the same unbroken symmetry (46/48); counterexamples both ways in $\mathbb{Z}_{12}$ -I (2 and 11)                                                                                                                                                                                                                                                                                                                               | COMPUTATION | w33_pass11086_flat_or_massive_is_an_unbroken_r_symmetry.py |
| The unbroken 3-group protecting the $\mathbb{Z}_3 \times \mathbb{Z}_3$ fractional charges is assembled from broken- $U(1)$ remnants, space-group translations and $\mathbb{Z}_3^R$ (point group and hidden centres contribute nothing); 47% of forbidden mass terms redundantly protected; the 28 singlet vacua also leave a massless $U(1)'$ coupled to SM matter                                                                                                                                                                                                        | COMPUTATION | w33_pass11087_unbroken_three_group_protection.py           |

| Claim                                                                                                                                                                                                                                                                                                                        | Badge               | Witness                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------|
| Baryonic and mixed hidden condensates (all 775/802/277 single-factor invariants, exact subset-sum parity engine): 63 more parity vacua with the curing factor condensed, every one keeping $\geq 142$ fractional multiplets massless                                                                                         | COMPUTATION         | w33_pass11088_baryonic_hidden_condensates.py                   |
| No non-SUSY twin of $T^6/\mathbb{Z}_3$ or $\mathbb{Z}_3 \times \mathbb{Z}_3$ with an order-3 shift ( $9V^2$ even, $9v'^2$ odd); $\mathbb{Z}_6$ -II has no non-SUSY twin; all 87 $\mathbb{Z}_6$ -I $W(3,3)$ SMs have twins with the same shifts ( $N = 0$ )                                                                   | THEOREM COMPUTATION | w33_pass11089_w33_twist_without_supersymmetry.py               |
| $3Q + t \bmod 3$ is constant on all 946 fixed points of the 13 $\mathbb{Z}_3 \times \mathbb{Z}_3$ SMs and is a linear character of the space group; 516 fractional fixed points host no neutral field; gauge-only rules still leave $\geq 15$ fractional multiplets massless                                                 | COMPUTATION         | w33_pass11090_resolution_cannot_free_fractional_charges.py     |
| $\mathrm{Sp}(4,3)$ is perfect, so the multiplier is the unique nontrivial character of $\mathrm{PGSp}(4,3)$ (identification with half-spins: Holotrade); the other track's cubic tick is this character, its chamber-sheet sign is flag-relative                                                                             | THEOREM COMPUTATION | Holotrade 98cafff; w33_pass11091_one_global_orientation_bit.py |
| The 87 non-SUSY $\mathbb{Z}_6$ -I twins (five orbifold $N = 0$ faults repaired): anomaly-free up to one GS $U(1)$ , parent gauge group, 3 net generations, 87/87; all tachyonic (2–20 states in the $\theta$ sector); tachyonic fixed point $\Leftrightarrow$ parent has a massless oscillator-excited state there (261/261) | COMPUTATION         | w33_pass11092_nonsusy_z6i_twins_are_tachyonic.py               |
| Tachyon-freedom and the SM exclude each other in the $\mathbb{Z}_6$ -I $W(3,3)$ scan: 47/58 base shifts tachyonic for any Wilson lines; 66 000 draws on the rest give 9 576 tachyon-free twins and 705 SM parents, overlap 0 (78.7 expected); SM exotics $v, w, x$ become the tachyons                                       | COMPUTATION         | w33_pass11093_tachyon_free_filter_z6i_scan.py                  |
| Every $\theta$ -sector tachyon of the 87 $\mathbb{Z}_6$ -I twins has charge $\pm \frac{1}{2}$ or colour, none neutral (two engines; mass-level orbifold validated 87/87 and against the non-SUSY orbifold)                                                                                                                   | COMPUTATION         | w33_pass11094_tachyons_break_electromagnetism.py               |

| Claim                                                                                                                                                                                                                                                         | Badge       | Witness                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------|
| Non-SUSY orbifold cross-check<br>19/19 identical; $SO(16) \times SO(16)$ with the $A_8$ shift: 793 508 Witten shifts in 6 classes; 153 tachyon-free three-generation models (104 inequivalent), all re-verified                                               | COMPUTATION | w33_pass11095_sol6xsol6_a8_standard_models.py    |
| Twins of $\mathbb{Z}_{12}$ -I (289/289), $\mathbb{Z}_3 \times \mathbb{Z}_6$ (5/5), $\mathbb{Z}_6 \times \mathbb{Z}_6$ (10/10) all tachyonic; $\mathbb{Z}_2 \times \mathbb{Z}_6$ -I: 8 tachyon-free, none with SM chirality; corrects P11089's mixed condition | COMPUTATION | w33_pass11096_twins_of_the_other_w33_families.py |
| $SO(16) \times SO(16)$ $A_8$ models: fractional fermions not symmetry-protected – all massive in 33/104 (gauge, Witten $\mathbb{Z}_2$ , PG, SG, $H$ -momentum), 6/33 at cubic order, all by order 7                                                           | COMPUTATION | w33_pass11097_fractional_fermion_masses.py       |
| Tree-level Yukawas, generic Higgs: up-type singular values (1, 1, 0) ( $\epsilon$ -structure) in 73/104, full rank in 31; down-type zero in 77                                                                                                                | COMPUTATION | w33_pass11098_yukawa_textures.py                 |
| $\Lambda_{10}(SO(16)^2) = +363 M^{10}$ (massless term $-2112 = n_B - n_F$ ); $A_8$ models: $\Lambda_4 \rightarrow (V/3)\Lambda_{10} > 0$ , $n_B - n_F \in [-480, -48]$ (corrected in Pass 11106)                                                              | COMPUTATION | w33_pass11099_vacuum_energy_sol6_a8.py           |
| $SO(16) \times SO(16)$ with $W(3,3)$ shifts on $\mathbb{Z}_3 \times \mathbb{Z}_3$ , $\mathbb{Z}_6$ -I, $\mathbb{Z}_{12}$ -I: 77 600 draws, no tachyon-free SM (only $T^6/\mathbb{Z}_3$ works)                                                                 | COMPUTATION | w33_pass11100_sol6_other_w33_geometries.py       |
| Localized light Higgs: single heavy top $(0, w, w)$ in all 31 twisted-up models, single heavy bottom in 27; $m_c = m_u$ at tree level (equilateral fixed points); 12 models pass every test (fractional masses at $T^* = 3$ )                                 | COMPUTATION | w33_pass11101_localized_higgs_heavy_top.py       |
| Higher orders do not split charm from up: in all 12 survivors the non-top $2 \times 2$ block is symmetric under the point reflection of the generation line at every order (72/72 Higgs cases); $m_c/m_u = O(1)$ parametrically                               | COMPUTATION | w33_pass11102_higher_order_quark_hierarchy.py    |
| No alignment of the scalar VEVs (fixed-point dependent, 6 patterns, 2 area regimes, hidden group unbroken/broken) splits charm from up (0/36 cases); strange–down split only via vector-like $\bar{d}$ (the 8 models with them)                               | COMPUTATION | w33_pass11103_reflection_breaking_vacua.py       |

| Claim                                                                                                                                                                                                                                                                                     | Badge       | Witness                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------------------------------|
| Top and bottom in one doublet iff $H_u, H_d$ at the same fixed point (36/108); then $V_{tb} \approx 1$ , $V_{cb} \sim V_{ub} \sim \epsilon^2 \sim m_c/m_t$ ; single heavy $\tau$ , $m_\mu = m_e$ at the same order                                                                        | COMPUTATION | w33_pass11104_ckm_and_leptons.py            |
| Symmetry group of the $\mathbb{Z}_3$ space-group rule<br>$= \Delta(54) = H_{27} \rtimes \langle -I \rangle$ (order 54; centre of $H_{27}$ = point-group twist; index 12 in Clifford-648); one Wilson-line-free torus in 104/104: two-qutrit $3^{1+4}$ ( $W(3,3)$ graph) realised in 0/104 | BRIDGE      | w33_pass11105_delta54_flavour_join.py       |
| Full one-loop integrand (36 sectors, Wilson lines) modular invariant, massless term = -252 sector by sector; family modulus minimal at $T = \rho$ ; Wilson-line moduli roll to small radius, winding tachyon in 12/12; $\Lambda > 0$ throughout                                           | COMPUTATION | w33_pass11106_one_loop_orbifold_point.py    |
| Winding tachyons built explicitly: $\Delta(\text{Im } T = 1) = -(3 - \sqrt{3})/6$ ; critical radius $\sqrt{3}$ (4 models) or 1.51 (7), model 77 only at $\rho$ ; colour singlets, fractionally charged in 10/12 (neutral directions in 53, 57); corrects Pass 11106's onset               | COMPUTATION | w33_pass11107_winding_tachyons.py           |
| Tachyon-free over the Wilson-line moduli: 104/104 at $2i$ , 55 at $i$ , 1 at $\rho$ (model 18, which fails fractional charges, top = charm, no down Yukawa); rescan: 4/387 new models free at $\rho$ , all with untwisted quarks (correlation not significant, Pass 11113)                | COMPUTATION | w33_pass11108_tachyon_free_moduli_rescan.py |
| $\mathbb{Z}_3$ Yukawas = $A_2$ theta functions of $T$ (duality closes only for the double-triangle action); $ Y_{\text{dist}}/Y_{\text{same}}  = \frac{1}{2}$ at $\rho$ , $(\sqrt{3} - 1)/2$ at $i$ ; $m_c/m_t$ needs $\text{Im } T \approx 3.2$                                          | BRIDGE      | w33_pass11109_instanton_yukawas_at_rho.py   |
| Model 2 potential at 20 tachyon-free moduli points: $\Lambda > 0$ everywhere, family minimum at $\rho$ (2.4× lower than at $3.2i$ ), Wilson-line minimum on the tachyon boundary; dilaton runaway                                                                                         | COMPUTATION | w33_pass11110_moduli_map.py                 |

| Claim                                                                                                                                                                                                                                                   | Badge       | Witness                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------|
| $3^{1+2k}$ has faithful degree $3^k$ : three twisted generations force $k = 1$ ; family $\otimes$ colour gluing impossible (twist $\notin$ SM gauge centre on quarks, 12/12)                                                                            | BRIDGE      | w33_pass11111_no_243_family_group.py       |
| Models 53, 57: neutral tachyons wind in one Wilson-line torus, charged in the other (degenerate for any common $B$ ); the one-loop potential shrinks the neutral torus first ( $\Delta\Lambda = -1.3, -2.3; -0.9, -1.7$ ): SM-neutral first instability | COMPUTATION | w33_pass11112_neutral_tachyon_corridor.py  |
| $\rho$ -safety vs quark sector over 491 models: 0/152 twisted vs 5/339 untwisted, Fisher $p = 0.16$ (not significant; corrects Pass 11108's reading)                                                                                                    | COMPUTATION | w33_pass11113_rho_tachyon_correlation.py   |
| $\det M(h)$ is a Hesse-pencil cubic in the Higgs direction; at $T = \rho, \lambda = \omega^2$ (singular member); $\min_h \sigma_2/\sigma_1 =  Y_{\text{dist}}/Y_{\text{same}}  = \frac{1}{2}$ (localised Higgs optimal)                                 | BRIDGE      | w33_pass11114_hesse_mass_determinant.py    |
| $B = 0$ axis: approximate Fricke duality in model 10 (extremum at tachyonic $1/\sqrt{3}$ ); model 77 tachyon-free to 0.87, $\Lambda$ monotone to its onset                                                                                              | COMPUTATION | w33_pass11115_axis_duality_and_window.py   |
| Family- $H_{27}$ centre (orbifold twist) is a gauge element in 104/104 ( $U(1)$ s alone 97; FI $\mathbb{Z}_3$ exactly in 3); no $SU(3)$ gluing in any model: two-qutrit group on no matter                                                              | COMPUTATION | w33_pass11116_one_qutrit_vs_other_track.py |
| Neutral condensate (53, 57): breaks exactly one $U(1)$ (component along the anomalous one), SM and hidden non-abelian intact, $\Delta(54)$ singlet, no cubic term                                                                                       | COMPUTATION | w33_pass11117_neutral_condensate.py        |
| Model 57 gradient flow on a $6 \times 6$ grid: neutral boundary from equal radii; separatrix at a 1–2% head start for the charged torus; no interior minimum                                                                                            | COMPUTATION | w33_pass11118_wilson_line_flow.py          |

| Claim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Badge       | Witness                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------|
| Neutral tachyons on their own<br>Wilson-line torus in 60/491 models<br>(38 neutral-only), all with twisted<br>quarks; gauntlet passes:<br>A8SM_20260928_2233_TF,<br>A8SM_20260928_533_TF,<br>A8SM_20260941_7061_TF,<br>A8SM_20260941_7243_TF,<br>A8SM_20260942_36621_TF,<br>A8SM_20260942_46043_TF,<br>A8SM_20260951_24165_TF,<br>A8SM_20260971_14027_TF,<br>A8SM_20260971_18172_TF,<br>A8SM_20260971_19247_TF,<br>A8SM_20260971_3818_TF,<br>A8SM_20260971_40521_TF,<br>A8SM_20260971_5285_TF,<br>A8SM_20260971_5904_TF,<br>A8SM_20260972_17224_TF,<br>A8SM_20260972_19688_TF,<br>A8SM_20260972_2080_TF,<br>A8SM_20260972_31386_TF,<br>A8SM_20260972_5273_TF,<br>A8SM_20260982_1645_TF,<br>A8SM_20260982_47048_TF | COMPUTATION | w33_pass11119_neutral_<br>tori_across_491.py           |
| $\rho$ -safety: necessary sparse-cell<br>condition (2 norm-1, 10 norm- $\frac{5}{3}$ cells),<br>not sufficient (15 tachyonic share it);<br>no norm invariant decides                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | COMPUTATION | w33_pass11120_rho_cell_<br>criterion.py                |
| Bridge to the other track: Hesse<br>parameter $\lambda(i) = 1 + \sqrt{3}$ ( $j = 1728$ ),<br>$\lambda(\rho) = \omega^2$ (singular); object table<br>(H27 realised, Clifford-648 partial,<br>243 not)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | BRIDGE      | w33_pass11121_hesse_<br>gauge_bridge.py                |
| Six neutral-exit survivors on the $B = 0$<br>diagonal: monotone to the onset,<br>volume law $\Lambda_\beta \approx 160y^2 - (5-13)$ ,<br>$d\Lambda/dy \approx 600$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | COMPUTATION | w33_pass11122_six_<br>potentials.py                    |
| Critical radii $\sqrt{3}$ (2) and $\varphi^2/\sqrt{3}$ (4);<br>no massless vector; KLT amplitude<br>with no contact quartic; radion slope<br>$1/(2\sqrt{3})$ , 0.247                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | BRIDGE      | w33_pass11123_critical_<br>point_amplitude.py          |
| Condensate unlocks forbidden lepton<br>(6/6), down (4/6), $\nu$ Dirac (6/6)<br>entries at orders 2–6, one insertion<br>each; up and $\mu$ untouched;<br>separate-winding zero was an<br>artefact; “missing Yukawas” reading<br>corrected in 11127                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | COMPUTATION | w33_pass11124_<br>condensate_unlocked_<br>couplings.py |

| Claim                                                                                                                                                                                                                                              | Badge       | Witness                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------|
| <i>B</i> -field map: charged tachyons need $B \neq 0$ and $\text{Im } T \leq 1$ ; $B \geq 0.4$ lifts the neutral tachyon at 1.4 in exactly the four golden models                                                                                  | COMPUTATION | w33_pass11125_bfield_charged_onset.py              |
| Condensate has zero family-torus quantum numbers: $T^* = \rho$ and $m_c = m_u$ unchanged                                                                                                                                                           | BRIDGE      | w33_pass11126_family_modulus_under_condensation.py |
| Light-doublet lepton/down textures $[0, 1, 1]$ unchanged by the condensate; reflection symmetry keeps $m_\mu = m_e$ (18/18); corrects the 11124 reading                                                                                            | COMPUTATION | w33_pass11127_lepton_texture_under_condensation.py |
| Pinched cycle: exact winding law $p_R^2 = y_1/\sqrt{3} (+1/(3\sqrt{3}y_2))$ ; golden radius from $y^2 - \sqrt{3}y + \frac{1}{3} = 0$ ; $\Delta \rightarrow -\frac{1}{2}$ , no tachyon-free island                                                  | BRIDGE      | w33_pass11128_pinched_cycle.py                     |
| Off-axis potential:<br>$\Lambda(B) - \Lambda(0) = A(1 - \cos 2\pi B)/2$ ,<br>$ A  \leq 5 \times 10^{-5}$ of the radial force;<br>$B = 0$ a minimum near onset                                                                                      | COMPUTATION | w33_pass11129_offaxis_potential.py                 |
| Top Higgs conjugates are the heavier doublets; single-Higgs $m_b/m_t$ :<br>$\epsilon^3 \rightarrow \epsilon^2$ (4/6), tau same order; light generations degenerate                                                                                 | COMPUTATION | w33_pass11130_single_higgs_bottom_tau.py           |
| Across 21 neutral-exit survivors: up and $\mu$ never unlocked; down 14, lepton 17, $\nu$ 20; reproduces 11124                                                                                                                                      | COMPUTATION | w33_pass11131_unlocked_across_21.py                |
| Single light Higgs across 21: bottom $\epsilon^3 \rightarrow \epsilon^2$ in exactly the 14 down-unlocked models, tau at the same order for $\geq 2$ of 3 Higgs                                                                                     | COMPUTATION | w33_pass11132_bottom_tau_across_21.py              |
| Two VEV scales ( $r = \log \epsilon_T / \log \epsilon_S$ ): 0 splittings; conjugate sector anarchic                                                                                                                                                | COMPUTATION | w33_pass11132_bottom_tau_across_21.py              |
| Zero-radius tachyon: one complex species ( $N = \pm 1 \bmod 3$ , none at $N = 0$ ), $l_1^2 \in \{\frac{2}{9}, \frac{4}{9}, \frac{5}{9}, \frac{7}{9}\}$ ; not a vector multiplet                                                                    | BRIDGE      | w33_pass11134_dual_tachyon_species.py              |
| FI scale $\langle T \rangle / M_P = 0.15\text{--}0.20$ (SUSY formula) near the needed $\epsilon$ , but no FI mechanism without SUSY: coincidence, not prediction                                                                                   | COMPUTATION | w33_pass11135_fi_scale_no_mechanism.py             |
| Tachyon domains = hyperbolic disks at $\Gamma_0(3)$ points: charged at $(3 + i\sqrt{3})/6$ , radius $\ln(2 + \sqrt{3})$ ; neutral horodisk at cusp 0 or Fricke point $i/\sqrt{3}$ , radius $2 \ln \varphi$ ; neutral first on the whole $B$ circle | BRIDGE      | w33_pass11136_tachyon_domains.py                   |
| Scorecard of the neutral-exit vacua (works / fails / open)                                                                                                                                                                                         | COMPUTATION | w33_pass11137_neutral_exit_scorecard.py            |

| Claim                                                                                                                                                           | Badge       | Witness                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------------|
| per-singlet VEV hierarchies (108 draws): light generations stay degenerate                                                                                      | COMPUTATION | w33_pass11138_singlet_vuv_hierarchies.py                                                                    |
| Wilson-line duality group $\Gamma_0(3)$ ; $S$ , $\Gamma^0(3)$ broken; Fricke approximate                                                                        | COMPUTATION | w33_pass11139_wilson_line_duality_group.py                                                                  |
| zero-radius gauge algebra $\mathfrak{su}(16) \oplus \mathfrak{u}(1)$ in 21/21: the $U(16)$ string, 2 tachyons                                                   | COMPUTATION | w33_pass11140_zero_radius_u16.py                                                                            |
| elliptic point = charged-disk centre ( $\Delta = -\frac{1}{6}$ ); cusp 0 tachyonic in 15/21, Fricke image of large volume in 6: no symmetric stabilisation      | COMPUTATION | w33_pass11141_no_symmetric_stabilization.py                                                                 |
| no two-doublet light Higgs splits charm from up (0/576)                                                                                                         | COMPUTATION | w33_pass11142_higgs_mixing_up_sector.py                                                                     |
| neutrino anarchy refuted: Dirac texture up-like $[0, 1, 1]$ at tree level, unchanged by the condensate; conjugate-sector anarchy mixes quarks too               | COMPUTATION | w33_pass11145_neutrino_anarchy.py                                                                           |
| no Wilson-line holonomy $\Rightarrow$ no Witten-sector tachyon at any radius (21/21)                                                                            | COMPUTATION | w33_pass11146_thermal_reading.py                                                                            |
| zero-radius vector lattice: even, rank 16, det 4, hyperbolic $\mathbb{Z}_2^2$ form, roots $A_{15}$ = the $U(16)$ lattice (21/21)                                | COMPUTATION | w33_pass11151_u16_lattice.py                                                                                |
| near cusp 0: potential = Fricke mirror of large volume to 0.3–0.5%, runs into the disk                                                                          | COMPUTATION | w33_pass11150_cusp0_potential.py                                                                            |
| First-generation regenerated RPV in the 23 $D$ -flat $\mathbb{Z}_6$ -I vacua $\geq 2 \times 10^{-3}$ (upper bounds); none below $10^{-10}$ <i>Machine</i> (§10) | COMPUTATION | w33_pass10980_rpv_flavour_upper_bound.py                                                                    |
| $\mathrm{ASp}(4, 3)$ , 4 199 040; four opcodes; length 19                                                                                                       | COMPUTATION | blueprint, Passes 2778/2789/2882<br>bt822_magic_census_matter_is_magic.py,<br>bt818_ovoid_nogo_theta_gap.py |
| 4 stabiliser + 36 magic rays; KS budget 36/40; $\alpha = 7$                                                                                                     | COMPUTATION |                                                                                                             |
| Steinberg memory $[[240, 81, 3]]_3$                                                                                                                             | COMPUTATION | bt861_code_register_is_steinberg.py                                                                         |
| Golay lane: 552 locations, threshold $5.32 \times 10^{-4}$                                                                                                      | COMPUTATION | w33_pass10947_five_front_execution.py                                                                       |
| cube-chart atlas; $\leq 8$ moves per digit                                                                                                                      | COMPUTATION | bt777_hypercube_chart_web.py, bt827_holonet_fractal_architecture.py                                         |
| single-photon sum gate, $0.92 \pm 0.01$                                                                                                                         | measured    | [25]                                                                                                        |

## B Glossary

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$A_2$  The root system of  $SU(3)$ : six vectors forming a regular hexagon.

**Albert algebra** The 27-dimensional algebra  $H_3(\mathbb{O})$  of  $3 \times 3$  Hermitian octonionic matrices; the only exceptional Euclidean Jordan algebra.

**Bell line** The line of  $W(3,3)$  formed by the unitary channels  $A \mapsto P_u A P_u^\dagger$ ; the “now” of Section 4.

**Bridge** A true mathematical statement together with a proposed physical reading.

**Cartan involution** An involution of a real Lie algebra splitting it into rotations ( $\mathfrak{k}$ ) and boosts ( $\mathfrak{p}$ ); it fixes the real form.

**Clifford group** Quantum operations mapping Pauli operators to Pauli operators; for two qutrits it acts on labels as  $Sp(4,3)$ .

**Context** A maximal set of simultaneously measurable Pauli operators; a line of  $W(3,3)$ .

**Contextuality** The impossibility of assigning pre-existing outcomes to all measurements consistently across contexts; a resource for quantum computation.

**CSS code** A quantum error-correcting code built from two classical codes;  $[[n,k,d]]_q$  means  $n$  physical qudits,  $k$  logical, distance  $d$ .

$E_8$  The largest exceptional simple Lie algebra: dimension 248, 240 roots.

$\mathbb{F}_3$  The field  $\{0,1,2\}$  with arithmetic modulo 3.

**Freudenthal triple system** A 56-dimensional space with a symplectic form and an  $E_7$ -invariant quartic; the charge space of certain supergravity black holes.

**Generalized quadrangle** A point–line geometry with no triangles in which a point off a line is collinear with exactly one of its points.

**Grading** A splitting of an algebra into pieces labelled by integers (or integers mod  $n$ ) that add under the bracket.

**Hesse clock** One of the four parallel classes of lines of  $AG(2,3)$ , read as three ticks.

**Heisenberg algebra** The algebra of positions, momenta and a central element; here  $56 + 1 = 57$  dimensions.

**Heterotic string** A closed string theory with gauge group  $E_8 \times E_8$  (or  $SO(32)$ ); on orbifolds it yields Standard-Model-like theories.

**Lagrangian** A maximal subspace on which the symplectic form vanishes; a line of  $W(3,3)$ .

**Magic state** A state that is not a stabiliser state; injecting it makes Clifford circuits universal.

**Majorana–Weyl spinor** A real, chiral spinor; in ten dimensions it has 16 components.

**Matter parity** A  $\mathbb{Z}_2$  symmetry,  $-1$  on matter fermions and  $+1$  on Higgs fields, that forbids fast proton decay in supersymmetric theories.

**Octonions**  $\mathbb{O}$  The eight-dimensional, non-associative normed division algebra.

**Orbifold** A space divided by a finite group; used to curl up extra dimensions.

**Pauli operator** For a qutrit,  $X^a Z^b$ ; for two qutrits, a tensor product of two.

**Peirce decomposition** The splitting of a Jordan algebra by the eigenvalues  $1, \frac{1}{2}, 0$  of multiplication by an idempotent.

**Qutrit** A quantum system with three perfectly distinguishable states.

**Real form** A real Lie algebra whose complexification is a given complex one;  $E_8$  has three.

**Root** An eigen-direction of the maximal commuting part of a Lie algebra.

**Signature** For a quadratic form, the numbers of positive and negative directions; for a real Lie algebra,  $\dim \mathfrak{p} - \dim \mathfrak{k}$ .

**Spread** A set of pairwise disjoint lines covering every point;  $W(3, 3)$  has 36.

**Steinberg module** A distinguished irreducible representation of a finite group of Lie type; for  $\mathrm{PSp}(4, 3)$  it is 81-dimensional and is  $H_1(W(3, 3))$ .

**Stabiliser state** A joint eigenstate of a maximal commuting set of Paulis.

**Strongly regular graph** A regular graph in which adjacent pairs have  $\lambda$ , and non-adjacent pairs  $\mu$ , common neighbours.

**Symplectic form** An alternating, nondegenerate bilinear form; here the commutation phase of Pauli operators.

**Tetracode** The ternary  $[4, 2, 3]$  code; here the evaluation code of the four clocks.

**Transvection** The symplectic map  $x \mapsto x + \langle x, v \rangle v$ .

**Tripotent** An element  $e$  of a Jordan triple system with  $\{e, e, e\} = 2e$  (normalisation of this paper).

**Witting polytope** A regular complex polytope in  $\mathbb{C}^4$  with 240 vertices, the  $E_8$  roots, and 40 diameters.

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